

Results

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1 Modules

$V_* \leftrightarrow$ Correspondence of finite-dimensional modules

V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8	V_9	V_{10}	V_{11}	V_{12}	V_{13}	V_{14}
V	V^*	S^2V	S^2V^*	W^2V	W^2V^*	S^3V	S^3V^*	W^3V	W^3V^*	S^4V	S^4V^*	W^4V	W^4V^*

2 Preliminaries

Proposition 2.0.1. Suppose that $\lambda \in X$, $\alpha_i \in \Phi_0^+$, $1 \leq i \leq k$, and $\beta, \gamma \in \Phi_1^+$. Let $w = s_{\alpha_k} s_{\alpha_{k-1}} \cdots s_{\alpha_1} \in \mathcal{W}$.

- (1) Suppose that $\langle \lambda, \alpha_1^\vee \rangle < 0$. Then $(P_\lambda : M_{s_{\alpha_1}\lambda}) > 0$.
- (2) Suppose that $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1} \lambda, \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w\lambda}) > 0$.
- (3) Suppose that $(\lambda, \beta) = 0$. Then $(P_\lambda : M_{\lambda+\beta}) > 0$.
- (4) Suppose that $(\lambda, \beta) = 0$ and $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1}(\lambda + \beta), \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w(\lambda+\beta)}) > 0$.
- (5) Suppose that $(\lambda, \beta) = (\lambda + \beta, \gamma) = 0$ and $\text{ht}(\beta) < \text{ht}(\gamma)$. Then $(P_\lambda : M_{\lambda+\beta+\gamma}) > 0$.
- (6) Suppose that $(\lambda, \beta) = (\lambda + \beta, \gamma) = 0$, $\text{ht}(\beta) < \text{ht}(\gamma)$, and $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1}(\lambda + \beta + \gamma), \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w(\lambda+\beta+\gamma)}) > 0$.

Proposition 2.0.2 (Gorelik). Any typical block in $\mathcal{O}$ is equivalent to a block in the BGG category $\mathcal{O}$ of $\mathfrak{g}_0$ -modules of integral weights.

Lemma 2.0.1. If $\lambda - \rho$ is the lowest weight in a standard filtration of a projective object P , then P_λ is a direct summand of P .

Proposition 2.0.3. If a projective P has a standard filtration given by $P_\lambda = \sum_\nu M_\nu$, the ν not necessarily distinct, then for any finite-dimensional representation N with weights μ , the standard filtration for $P \otimes N$ is given by $\sum_\nu \sum_\mu M_{\nu+\mu}$, where μ appears in the sum with multiplicity given by $\dim N^\mu$.

Theorem 2.1. [1] Let $\lambda \in X + \rho$ be such that $\lambda - \rho$ is a typical anti-dominant dot-singular weight. Let $\mathcal{W}^\lambda$ be a minimal set of left-coset representatives of $\mathcal{W}/\mathcal{W}_\lambda$, where $\mathcal{W}_\lambda = \{w \in \mathcal{W} \mid w\lambda = \lambda\}$. Then, if $\sigma \in \mathcal{W}^\lambda$,

$$P_{\sigma\lambda} = \sum_{\tau \geq \sigma, \tau \in \mathcal{W}^\lambda} M_{\tau\lambda}. \quad (2.2)$$

Theorem 2.2. We let P_λ denote the (unique) projective cover for L_λ for all $\lambda \in \mathfrak{h}^*$, that is the indecomposable projective such that $P_\lambda \rightarrow L_\lambda \rightarrow 0$. We recall the following facts about projectives.

- (1) All projectives have a standard filtration.
- (2) The category $\mathcal{O}$ has enough projectives.
- (3) The Verma modules M_μ which appear in a standard filtration of P_λ satisfy $\mu \geq \lambda$ in the Bruhat ordering, and M_λ appears with multiplicity 1.

2.1 Strategy

Let $\lambda \in \mathfrak{h}^*$ be such that $\lambda - \rho$ is an atypical weight. Our goal is to determine the standard filtration of the projective cover P_λ .

To this end, we choose a weight $\mu \in \mathfrak{h}^*$ and a finite-dimensional module W satisfying the following:

- The structure of P_μ is already known;
- There exists a weight ν of W (typically its highest weight) such that $\mu = \lambda + \nu$.

We may select μ so that $\mu - \rho$ is typical; in this case, the structure of P_μ follows from [Theorem 2.1](#). Alternatively, one may take μ with $\mu - \rho$ atypical, provided that $\text{Filt}(P_\mu)$ has been previously computed.

[Proposition 2.0.1](#) can be used to deduce the Verma modules which appear in a standard filtration of the projective $P_\mu \otimes W$, which must include M_λ .

We now project $P_\mu \otimes W$ onto the block corresponding to the linkage class of $\lambda - \rho$, denoted by $pr_\lambda(P_\mu \otimes W)$. The projection $pr_\lambda(P_\mu \otimes W)$ is defined precisely so that all highest weights of Verma modules appearing in its standard filtration belong to the linkage class of $\lambda - \rho$.

In all cases considered in this paper, unless stated otherwise, the weight λ is a minimal weight in the standard filtration of the projection $pr_\lambda(P_\mu \otimes W)$. Under these premises, [Lemma 2.0.1](#) guarantees that the indecomposable projective module P_λ must appear in the projection, constituting a direct summand.

Since P_λ is a summand, every composition factor in its standard filtration must also appear in that of $pr_\lambda(P_\mu \otimes W)$. Conversely, applying [Proposition 2.0.1](#) allows us to identify which Verma modules in $\text{Filt}(pr_\lambda(P_\mu \otimes W))$ belong to $\text{Filt}(P_\lambda)$.

If each Verma module appears with multiplicity one in $\text{Filt}(pr_\lambda(P_\mu \otimes W))$, and [Proposition 2.0.1](#) confirms that all of them lie in $\text{Filt}(P_\lambda)$, then it follows immediately that

$$pr_\lambda(P_\mu \otimes W) = P_\lambda.$$

However, this ideal situation does not always occur. In general, one aims to show that P_λ is the *only* indecomposable projective summand of $pr_\lambda(P_\mu \otimes W)$; equivalently, that $pr_\lambda(P_\mu \otimes W) = P_\lambda$. This is typically achieved by proving that no other indecomposable projective module P_β (with $\beta \neq \lambda$) can occur as a direct summand in the projection.

Consider a Verma module M_β in $\text{Filt}(pr_\lambda(P_\mu \otimes W))$. Its presence in $\text{Filt}(P_\lambda)$ cannot be determined solely by applying [Proposition 2.0.1](#). Such modules require individual analysis. Furthermore, if [Proposition 2.0.1](#) establishes that $(P_\lambda : M_\beta) > 0$, but M_β occurs with multiplicity $n > 1$ in $pr_\lambda(P_\mu \otimes W)$, then precisely $n - 1$ copies of M_β require separate determination beyond what [Proposition 2.0.1](#) provides.

We collect all such indeterminate Verma modules into a multiset denoted by U . Since P_λ is guaranteed to be a direct summand of $pr_\lambda(P_\mu \otimes W)$, our objective is to verify that all elements of U belong to $\text{Filt}(P_\lambda)$. To accomplish this, we employ the following methodology:

Suppose $pr_\lambda(P_\mu \otimes W)$ contains, besides P_λ , another indecomposable projective module P_β as a direct summand. Then the following conditions must hold:

- M_β must appear in multiset U ;
- Every Verma module in $\text{Filt}(P_\beta)$ must appear in U , and the multiplicity of each such Verma module in $\text{Filt}(P_\beta)$ must not exceed its multiplicity in U .

Consequently, the only possible indecomposable projective summands of $pr_\lambda(P_\mu \otimes W)$ other than P_λ are those P_β for which $M_\beta \in \text{Supp}(U)$. If we can definitively exclude all possible indecomposable projective as direct summands of $pr_\lambda(P_\mu \otimes W)$, then it follows that

$$pr_\lambda(P_\mu \otimes W) = P_\lambda$$

For such a candidate P_β , if there exists a Verma module M that either does not belong to U but appears in $\text{Filt}(P_\beta)$, or appears in $\text{Filt}(P_\beta)$ with multiplicity strictly greater than its multiplicity in U , then P_β cannot occur as a direct summand of $pr_\lambda(P_\mu \otimes W)$.

In most cases, it suffices to apply [Proposition 2.0.1](#) to identify certain necessary Verma modules in $\text{Filt}(P_\beta)$; the presence of even one such factor outside U already implies $\text{Filt}(P_\beta) \not\subseteq U$. When [Proposition 2.0.1](#) does not yield sufficient information, We instead use the known (or independently provable) complete structure of P_β , ensuring that no circularity is introduced.

However, in certain exceptional cases one indeed has $\text{Filt}(P_\beta) \subseteq U$ as multisets, so that filtration comparison alone cannot rule out P_β as a direct summand of $pr_\lambda(P_\mu \otimes W)$. To exclude this possibility, suppose for contradiction that P_β is a direct summand of $pr_\lambda(P_\mu \otimes W)$. Then there exists a module Q such that

$$pr_\lambda(P_\mu \otimes W) = Q \oplus P_\beta.$$

Since $pr_\lambda(P_\mu \otimes W)$ is a direct summand of the projective module $P_\mu \otimes V$, it is itself projective; hence both Q and P_β must be projective.

We will show that Q is *not* projective, thereby obtaining a contradiction. Specifically, it suffices to exhibit a weight ν and a finite-dimensional module V such that $pr_\nu(Q \otimes V)$ is *not* a direct sum of indecomposable projective modules—equivalently, $pr_\nu(Q \otimes V)$ fails to be projective.

This approach provides a rigorous framework for determining the complete structure of P_λ through elimination of alternative direct summands.

3 $\mathfrak{gl}(41)$

Theorem 3.1. *Let $a, b, c, d \in \mathbb{Z}$ and $\lambda = (a, b, c, d|a)$. The projective objects P_λ in $\mathfrak{B}_{b,c,d}$ have the following Verma flag formulae.*

Proof. The Cases are as follows:

(1) $a \neq \{b, c, d\}$

(1.1) $b < a < d < c, d > a + 1$

Let $\mu := \lambda + (1, 0, 0, 0|0)$.

$$\begin{aligned} \text{Pr}_\lambda(P_\mu \otimes V_2) &= M_{a,b,c,d|a} + M_{a+1,b,c,d|a+1} + M_{d,c,b,a|a} + M_{d,c,b,a+1|a+1} + 2M_{c,d,a,b|a} + 2M_{c,d,a+1,b|a+1} + M_{c,a+1,d,b|a+1} \\ &= \sum M_{a,b,c,d|a} + \sum M_{a+1,b,c,d|a+1} + \sum M_{c,a,d,b|a} + \sum M_{c,a+1,d,b|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,d,a,b|a}, M_{c,d,a+1,b|a+1}, M_{c,a,d,b|a}, M_{c,a+1,d,b|a+1}\}$.

From [table 1](#), [Proposition 2.0.1](#) alone is insufficient to rule out $P_{c,d,a,b|a}$ or $P_{c,a,d,b|a}$ as direct summands of $\text{pr}_\lambda(P_\lambda \otimes V_2)$.

We therefore invoke the complete standard filtrations of these projective modules. By [Theorem 3.3](#) and [Theorem 3.2](#) (whose proofs are independent of the present argument, thus avoiding circular reasoning), we have

$$\begin{aligned} P_{c,d,a,b|a} &= M_{c,d,a+1,b|a+1} + M_{c,d,a,b|a} \\ P_{c,a,d,b|a} &= M_{c,d,a+1,b|a+1} + M_{c,d,a,b|a} + M_{c,a+1,d,b|a+1} + M_{c,a,d,b|a} \\ &= \sum M_{c,a,d,b|a} + \sum M_{c,a+1,d,b|a+1} \end{aligned}$$

Since all Verma modules appearing in $\text{Filt}(P_{c,d,a,b|a})$ and $\text{Filt}(P_{c,a,d,b|a})$ belong to the multiset U , and their multiplicities do not exceed those in U , filtration comparison alone cannot exclude either $P_{c,d,a,b|a}$ or $P_{c,a,d,b|a}$ as direct summands of $\text{pr}_\lambda(P_\lambda \otimes V_2)$. We proceed by contradiction to eliminate both possibilities.

First, assume that $P_{c,d,a,b|a}$ is a direct summand of $\text{pr}_\lambda(P_\lambda \otimes V_2)$. Then there exists a projective module Q such that

$$\text{pr}_\lambda(P_\lambda \otimes V_2) = Q \oplus P_{c,d,a,b|a}$$

Let $\nu = (a + 2, b, c, d|a + 1)$. Under our standing assumptions on the parameters, ν is a typical weight. We divide the argument into two cases.

Case 1. $c > a + 2$.

A direct computation yields the standard filtration of $\text{pr}_\nu(Q \otimes V_1)$ as

$$\begin{aligned} \text{pr}_\nu(Q \otimes V_1) &= M_{a+2,b,c,d|a+1} + M_{d,c,b,a+2|a+1} + M_{c,d,a+2,b|a+1} + M_{c,a+2,b,d|a+1} + M_{a+2,c,d,b|a+1} + M_{c,b,d,a+2|a+1} \\ &= \sum M_{a+2,b,c,d|a+1} + M_{c,a+2,d,b|a+1} \end{aligned} \tag{1}$$

Observe that ν is the minimal highest weight occurring in this filtration. Since Q is projective, so is $\text{pr}_\nu(Q \otimes V_1)$. By [Lemma 2.0.1](#), this implies that P_ν must appear as a direct summand.

However, [Theorem 2.1](#) gives the complete standard filtration of P_ν as

$$\begin{aligned} P_\nu &= M_{a+2,b,c,d|a+1} + M_{d,c,b,a+2|a+1} + 2M_{c,d,a+2,b|a+1} + M_{c,a+2,b,d|a+1} + M_{a+2,c,d,b|a+1} + M_{c,b,d,a+2|a+1} + M_{c,a+2,d,b|a+1} \\ &= \sum M_{a+2,b,c,d|a+1} + \sum M_{c,a+2,d,b|a+1} \end{aligned} \tag{2}$$

Comparing [equation \(2\)](#) with [equation \(1\)](#), we see that the Verma module $M_{c,d,a+2,b|a+1}$ appears with multiplicity one in $\text{Filt}(\text{pr}_\nu(Q \otimes V_1))$, but with multiplicity two in $\text{Filt}(P_\nu)$. Consequently, P_ν cannot be a direct summand of $\text{pr}_\nu(Q \otimes V_1)$, contradicting [Lemma 2.0.1](#). Hence, $P_{c,d,a,b|a}$ is not a direct summand of $\text{pr}_\lambda(P_\lambda \otimes V_2)$.

Case 2. $d = a + 2$.

直接计算 $pr_\nu(Q \otimes V_1)$ 标准滤过如下

$$pr_\nu(Q \otimes V_1) = 2M_{a+2,b,c,a+2|a+1} + 2M_{a+2,c,b,a+2|a+1} + 3M_{c,a+2,a+2,b|a+1} + 2M_{c,a+2,b,a+2|a+1} + 2M_{a+2,c,a+2,b|a+1} \quad (3)$$

通过公式equation (3), 可以看出 ν 是 $\text{Filt}(pr_\nu(Q \otimes V_1))$ 中的极小权, 且 M_ν 重数是 2。这说明 $pr_\nu(Q \otimes V_1)$ 中有两个 P_ν 作为其分解的直和项。但是, Theorem 2.1 给出的 P_ν 的标准滤过是

$$\begin{aligned} P_\nu &= M_{a+2,b,c,a+2|a+1} + M_{a+2,c,b,a+2|a+1} + 2M_{c,a+2,a+2,b|a+1} + M_{c,a+2,b,a+2|a+1} + M_{a+2,c,a+2,b|a+1} + M_{c,b,a+2,a+2|a+1} \\ &= \sum M_{a+2,b,c,a+2|a+1} + M_{c,a+2,a+2,b|a+1} \end{aligned} \quad (4)$$

比较equation (3)和equation (4), 我们可知 $pr_\nu(Q \otimes V_1)$ 不可能有两个 P_ν 作为其分解的直和项, 矛盾。因此 $P_{c,a+2,a,b|a}$ 不是 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和项。

Next, assume that $P_{c,a,d,b|a}$ is a direct summand. Then there exists a projective module Q' such that

$$pr_\lambda(P_\lambda \otimes V_2) = Q' \oplus P_{c,a,d,b|a}$$

We also divide the argument into two cases.

Case 1. $d > a + 2$.

Using the same weight ν , we compute

$$\begin{aligned} pr_\nu(Q' \otimes V_1) &= M_{a+2,b,c,d|a+1} + M_{d,c,b,a+2|a+1} + M_{c,d,a+2,b|a+1} + M_{c,a+2,b,d|a+1} + M_{a+2,c,d,b|a+1} + M_{c,b,d,a+2|a+1} \\ &= \sum M_{a+2,b,c,d|a+1} \end{aligned} \quad (5)$$

Again, ν is the minimal highest weight, and $pr_\nu(Q' \otimes V_1)$ is projective, so P_ν should occur as a summand. However, $M_{c,a+2,d,b|a+1}$ has multiplicity one in equation (5) but multiplicity two in $\text{Filt}(P_\nu)$, making such an embedding impossible. This contradiction excludes $P_{c,a,d,b|a}$ as well.

Case 2. $d = a + 2$.

直接计算 $pr_\nu(Q' \otimes V_1)$ 标准滤过如下

$$pr_\nu(Q \otimes V_1) = 2M_{a+2,b,c,a+2|a+1} + 2M_{a+2,c,b,a+2|a+1} + 3M_{c,a+2,a+2,b|a+1} + 2M_{c,a+2,b,a+2|a+1} + 2M_{a+2,c,a+2,b|a+1} \quad (6)$$

通过公式equation (6), 可以看出 ν 是 $\text{Filt}(pr_\nu(Q' \otimes V_1))$ 中的极小权, 且 M_ν 重数是 2。这说明 $pr_\nu(Q \otimes V_1)$ 中有两个 P_ν 作为其分解的直和项。但是, 比较equation (6)和equation (4), 我们可知 $pr_\nu(Q' \otimes V_1)$ 不可能有两个 P_ν 作为其分解的直和项, 矛盾。因此 $P_{c,a,a+2,b|a}$ 不是 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和项。

In conclusion, neither $P_{c,d,a,b|a}$ nor $P_{c,a,d,b|a}$ can occur as a direct summand of $pr_\lambda(P_\lambda \otimes V_2)$.

Table 1: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{c,d,a,b a}$	$\emptyset$
$P_{c,d,a+1,b a+1}$	$M_{c,d,a+2,b a+2}$
$P_{c,a,d,b a}$	$\emptyset$
$P_{c,a+1,d,b a+1}$	$M_{c,a+2,d,b a+2}, M_{c,d,a+2,b a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2) \quad a < c < b < d$$

$$(1.2.1) \quad c > a + 1$$

Let $\mu := \lambda + (1, 0, 0, 0|0)$.

$$P_\mu = 2M_{b,d,a+1,c|a} + 2M_{d,b,c,a+1|a} + M_{b,a+1,c,d|a} + M_{d,c,a+1,b|a} + M_{a+1,b,d,c|a} + M_{c,d,b,a+1|a} + M_{c,b,a+1,d|a}$$

$$\begin{aligned} \text{Pr}_\lambda(P_\mu \otimes V_2) &= 2M_{b,d,a,c|a} + 2M_{b,d,a+1,c|a+1} + 2M_{d,b,c,a|a} + 2M_{d,b,c,a+1|a+1} + M_{b,a,c,d|a} + M_{b,a+1,c,d|a+1} \\ &= \sum M_{a,b,c,d|a} + \sum M_{b,d,a,c|a} + \sum M_{a+1,b,c,d|a+1} + \sum M_{b,d,a+1,c|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,a,c|a}, M_{b,d,a+1,c|a+1}, M_{d,b,c,a|a}, M_{d,b,c,a+1|a+1}, M_{b,a,c,d|a}, M_{b,a+1,c,d|a+1}, M_{a,b,c,d|a}, M_{a+1,b,c,d|a+1}\}$.

Table 2: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(\text{pr}_\lambda(P_\lambda \otimes V_2))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(\text{pr}_\lambda(P_\mu \otimes V_2))$ (via Prop. 2.1)
$P_{b,d,a,c a}$		$\emptyset$
$P_{b,d,a+1,c a+1}$	$M_{b,d,a+2,c a+2}, M_{d,b,a+2,c a+2}, M_{b,d,c,a+2 a+2}, M_{d,b,c,a+2 a+2}$	
$P_{d,b,c,a a}$		$\emptyset$
$P_{d,b,c,a+1 a+1}$		$M_{d,b,c,a+2 a+2}$
$P_{d,b,a,c a}$		$\emptyset$
$P_{d,b,a+1,c a+1}$		$M_{d,b,a+2,c a+2}, M_{d,b,c,a+2 a+2}$
$P_{b,d,c,a a}$		$\emptyset$
$P_{b,d,c,a+1 a+1}$		$M_{b,d,c,a+2 a+2}, M_{d,b,c,a+2 a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

根据 table 2, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,d,a,c|a}$ 或 $P_{d,b,c,a|a}$ 或 $P_{d,b,a,c|a}$ 或 $P_{b,d,c,a|a}$ 是否出现在 $\text{pr}_\lambda(P_\lambda \otimes V_2)$ 的直和分解中. 因此由 Theorem 3.3 和 Theorem 3.4, 我们引入它们完整的滤过结构如下, 这里并不引入循环论证.

$$\begin{aligned} P_{b,d,a,c|a} &= M_{b,d,a+1,c|a+1} + M_{b,d,a,c|a} + M_{d,b,c,a+1|a+1} + M_{d,b,c,a|a} + M_{d,b,a+1,c|a+1} + M_{d,b,a,c|a} + M_{b,d,c,a|a} \\ &= \sum M_{b,d,a,c|a} + \sum M_{b,d,a+1,c|a+1} \end{aligned} \tag{7}$$

$$\begin{aligned} P_{d,b,a,c|a} &= M_{d,b,c,a+1|a+1} + M_{d,b,c,a|a} + M_{d,b,a+1,c|a+1} + M_{d,b,a,c|a} \\ &= \sum M_{d,b,c,a|a} + \sum M_{d,b,a+1,c|a+1} \end{aligned} \tag{8}$$

$$P_{b,d,c,a|a} = M_{d,b,c,a+1|a+1} + M_{d,b,c,a|a} + M_{b,d,c,a+1|a+1} + M_{b,d,c,a|a} = \sum M_{b,d,c,a|a} + \sum M_{b,d,c,a+1|a+1} \tag{9}$$

$$P_{d,b,c,a|a} = M_{d,b,c,a+1|a+1} + M_{d,b,c,a|a} \tag{10}$$

可以看出 $\text{Filt}(P_{b,d,a,c|a})$ 或 $\text{Filt}(P_{d,b,c,a|a})$ 或 $\text{Filt}(P_{d,b,a,c|a})$ 或 $\text{Filt}(P_{b,d,c,a|a})$ 所有的 Verma 模都在多重集 U 中. 且其重数未超过 U 中对应项的重数. 因此因此仅通过滤过比较仍无法排除它们中某一个作为 $\text{pr}_\lambda(P_\mu \otimes V_2)$ 直和项的可能性. 我们寻求其他方法排除这些可能性.

事实上, 我们仅需证明 $M_{b,d,a,c|a}, M_{d,b,c,a|a}, M_{d,b,a,c|a}, M_{b,d,c,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, d|a-1)$, 我们计算 $\text{pr}_\nu((\text{pr}_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$\text{pr}_\nu((\text{pr}_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,d,a,c|a-1} + 2M_{b,d,c,a|a-1} + 2M_{d,b,a,c|a-1} + 2M_{d,b,c,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, d|a) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, d, a, c|a-1)$ 只能由 $(b, d, a, c|a) + (0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,c|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,c,a|a}, M_{d,b,a,c|a}, M_{b,d,c,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

$$(1.2.2) \quad c = a + 1$$

Let $\mu := \lambda + (1, 0, 0, 0|0)$.

$$P_\mu = 2M_{b,d,a+1,a+1|a} + M_{b,a+1,a+1,d|a} + M_{a+1,b,d,a+1|a} + M_{a+1,d,a+1,b|a} + M_{d,a+1,b,a+1|a} + 2M_{d,b,a+1,a+1|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,d,a,a+1|a} + 2M_{b,d,a+1,a|a} + 2M_{b,d,a+1,a+1|a+1} + M_{b,a,a+1,d|a} + M_{b,a+1,a,d|a} + M_{b,a+1,d,a|a} \\ &= \sum M_{a,b,a+1,d|a} + \sum M_{b,d,a,a+1|a} + \sum M_{a+1,b,a+1,d|a+1} + \sum M_{b,d,a+1,a+1|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,a,a+1|a}, M_{b,d,a+1,a|a}, M_{b,d,a+1,a+1|a+1}\}$. 根据 table 3, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,d,a,c|a}$ 或 $P_{d,b,a,c|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,c|a}, M_{d,b,a,c|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, d|a-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,d,a,c|a-1} + 2M_{b,d,c,a|a-1} + 2M_{d,b,a,c|a-1} + 2M_{d,b,c,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, d|a) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, d, a, c|a-1)$ 只能由 $(b, d, a, c|a) + (0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,c|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,c|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

Table 3: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{b,d,a,a+1 a}$	$\emptyset$
$P_{b,d,a+1,a a}$	$M_{b,d,a+2,a+1 a+2}, M_{d,b,a+2,a+1 a+2}$
$P_{b,d,a+1,a+1 a+1}$	$M_{b,d,a+2,a+1 a+2}, M_{b,d,a+1,a+2 a+2}, M_{d,b,a+2,a+1 a+2}, M_{d,b,a+1,a+2 a+2}$
$P_{d,b,a,a+1 a}$	$\emptyset$
$P_{d,b,a+1,a a}$	$M_{d,b,a+2,a+1 a+2}$
$P_{d,b,a+1,a+1 a+1}$	$M_{d,b,a+2,a+1 a+2}, M_{d,b,a+1,a+2 a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.3) \quad a < c < b = d$$

$$(1.3.1) \quad c > a + 1$$

Let $\mu := \lambda + (1, 0, 0, 0|0)$.

$$P_\mu = 2M_{b,b,a+1,c|a} + M_{b,a+1,c,b|a} + M_{a+1,b,b,c|a} + M_{c,b,a+1,b|a} + M_{b,c,b,a+1|a} + 2M_{b,b,c,a+1|a} + M_{a+1,b,c,b|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,b,a,c|a} + 2M_{b,b,a+1,c|a+1} + M_{b,a,c,b|a} + M_{b,a+1,c,b|a+1} + M_{a,b,b,c|a} + M_{a+1,b,b,c|a+1} + \\ &= \sum M_{a,b,c,b|a} + \sum M_{b,b,a,c|a} + \sum M_{a+1,b,c,b|a+1} + \sum M_{b,b,a+1,c|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b,a,c|a}, M_{b,b,a+1,c|a+1}, M_{b,b,c,a|a}, M_{b,b,c,a+1|a+1}\}$. 根据 table 4, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,b,c,a|a}$ 或 $P_{b,b,a,c|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b,c,a|a}, M_{b,b,a,c|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, b|a-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,b,a,c|a-1} + 2M_{b,b,c,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, b|a) + (0, 0, 0, 0| - 1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, b, a, c|a-1)$ 只能由 $(b, b, a, c|a) + (0, 0, 0, 0| - 1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,b,a,c|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{b,b,c,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

Table 4: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{b,b,a,c a}$		$\emptyset$
$P_{b,b,a+1,c a+1}$		$M_{b,b,a+2,c a+2}, M_{b,b,c,a+2 a+2}$
$P_{b,b,c,a a}$		$\emptyset$
$P_{b,b,c,a+1 a+1}$		$M_{b,b,c,a+2 a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.3.2) \quad c = a + 1$$

Let $\mu := \lambda + (1, 0, 0, 0|0)$.

$$P_\mu = 2M_{b,b,a+1,a+1|a} + M_{b,a+1,a+1,b|a} + M_{a+1,b,b,a+1|a} + M_{a+1,b,a+1,b|a} + M_{b,a+1,b,a+1|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,b,a,a+1|a} + 2M_{b,b,a+1,a|a} + 2M_{b,b,a+1,a+1|a+1} + M_{b,a,a+1,b|a} + M_{b,a+1,a,b|a} + M_{b,a+1,b,a|a} \\ &= \sum M_{a,b,a+1,b|a} + \sum M_{b,b,a,a+1|a} + \sum M_{a+1,b,a+1,b|a+1} + M_{b,b,a+1,a+1|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b,a,a+1|a}, M_{b,b,a+1,a|a}, M_{b,b,a+1,a+1|a+1}\}$. 根据 table 5, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,b,a,c|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b,a,c|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, b|a-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,b,a,c|a-1} + 2M_{b,b,c,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, b|a) + (0, 0, 0, 0| - 1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, b, a, c|a-1)$ 只能由 $(b, b, a, c|a) + (0, 0, 0, 0| - 1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,b,a,c|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

$$(1.4) \quad \text{Others} \quad \text{Let } \mu := \lambda + (1, 0, 0, 0|0) = (a+1, b, c, d|a).$$

$$Pr_\lambda(P_\mu \otimes V_2) = \sum M_{a,b,c,d|a} + \sum M_{a+1,b,c,d|a+1}$$

All items belong to P_λ by Proposition 2.0.1.

Table 5: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{b,b,a,a+1 a}$	$\emptyset$
$P_{b,b,a+1,a a}$	$M_{b,b,a+2,a+1 a+2}$
$P_{b,b,a+1,a+1 a+1}$	$M_{b,b,a+2,a+1 a+2}, M_{b,b,a+1,a+2 a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(2) $b = a, a \neq \{c, d\}$

Let $\mu := \lambda + (1, 1, 0, 0|0) = (a + 1, a + 1, c, d|a)$.

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{a,a,c,d|a} + \sum M_{a,a+1,c,d|a+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(3) $c = a, d \neq a$

(3.1) $c = a, d \neq a, b < a$

Let $\mu := \lambda + (1, 0, 1, 0|0) = (a + 1, b, a + 1, d|a)$.

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{a,b,a,d|a} + \sum M_{a,b,a+1,d|a+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(3.2) $c = a, d \neq a, b = a$

Let $\mu := \lambda + (1, 1, 1, 0|0) = (a + 1, a + 1, a + 1, d|a)$.

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{a,b,a,d|a} + \sum M_{a,b,a+1,d|a+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(3.3) $c = a, d \neq a, b > a$

(3.3.1) $b \neq a + 1, d < b$

Let $\mu := \lambda + (1, 1, 1, 0|0) = (a + 1, b + 1, a + 1, d|a)$.

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{a,b,a,d|a} + \sum M_{a,b,a+1,d|a+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(3.3.2) $b = a + 1, d < b$

Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{a+1,a+2,a+1,d|a} + M_{a+2,a+1,a+1,d|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{a,a+1,a,d|a} + M_{a,a+1,a+1,d|a+1} + 2M_{a+1,a+1,a,d|a+1} + M_{a,a+2,a+1,d|a+2} + M_{a+1,a+2,a,d|a+2} \\ &= \sum M_{a,a+1,a,d|a} + \sum M_{a,a+1,a+1,d|a+1} + \sum M_{a,a+2,a+1,d|a+2} + M_{a+1,a+1,a,d|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+1,a,d|a+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 6](#).

Table 6: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a+1,a+1,a,d a+1}$	$M_{a+2,a+1,a,d a+2}, M_{a+1,a+2,a,d a+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.3.3) \quad b \neq a+1, d > b$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|1).$$

$$P_\mu = 2M_{b,d,a,a|a+1} + M_{b,a,a,d|a+1} + M_{a,b,d,a|a+1} + M_{a,d,a,b|a+1} + M_{d,a,b,a|a+1} + 2M_{d,b,a,a|a+1} + M_{a,b,a,d|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b,d,a+1,a|a+1} + 2M_{b,d,a,a+1|a+1} + 2M_{b,d,a,a|a} + M_{b,a+1,a,d|a+1} + M_{b,a,a+1,d|a+1} + M_{b,a,a+1,d|a} \\ &= \sum M_{a,b,a,d|a} + \sum M_{b,d,a,a|a} + \sum M_{a,b,a+1,d|a+1} + \sum M_{b,d,a,a+1|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,a+1,a|a+1}, M_{b,d,a,a+1|a+1}, M_{b,d,a,a|a}\}$. 根据 table 7, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,d,a,a|a}$ 或 $P_{d,b,a,a|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,a|a}, M_{d,b,a,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a, d|a-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,d,a,a|a-1} + 2M_{d,b,a,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, a, d|a) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, d, a, a|a-1)$ 只能由 $(b, d, a, c|a) + (0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,a|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 7: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b,d,a+1,a a+1}$	$M_{b,d,a+2,a a+2}, M_{d,b,a+2,a a+2}$	
$P_{b,d,a,a+1 a+1}$	$M_{b,d,a,a+2 a+2}, M_{d,b,a,a+2 a+2}, M_{b,d,a+2,a a+2}, M_{d,b,a+2,a a+2}$	
$P_{b,d,a,a a}$	$\emptyset$	
$P_{d,b,a+1,a a+1}$	$M_{d,b,a+2,a a+2}$	
$P_{d,b,a,a+1 a+1}$	$M_{d,b,a,a+2 a+2}, M_{d,b,a+2,a a+2}$	
$P_{d,b,a,a a}$	$\emptyset$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.3.4) \quad b \neq a+1, d = b$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|1).$$

$$P_\mu = 2M_{b,b,a,a|a+1} + M_{b,a,a,b|a+1} + M_{a,b,b,a|a+1} + M_{a,b,a,b|a+1} + M_{b,a,b,a|a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b,b,a+1,a|a+1} + 2M_{b,b,a,a+1|a+1} + 2M_{b,b,a,a|a} + M_{b,a+1,a,b|a+1} + M_{b,a,a+1,b|a+1} + M_{b,a,a+1,b|a} \\ &= \sum M_{a,b,a,b|a} + M_{b,b,a,a|a} + \sum M_{a,b,a+1,b|a+1} + \sum M_{b,b,a,a+1|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b,a+1,a|a+1}, M_{b,b,a,a+1|a+1}, M_{b,b,a,a|a}\}$. 根据 table 8, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,b,a,a|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b,a,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a, b|a-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,b,a,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, a, b|a) + (0, 0, 0, 0| - 1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, b, a, a|a-1)$ 只能由 $(b, b, a, c|a) + (0, 0, 0, 0| - 1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,b,a,a|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 8: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b,b,a+1,a a+1}$		$M_{b,b,a+2,a a+2}$
$P_{b,b,a,a+1 a+1}$		$M_{b,b,a,a+2 a+2}, M_{b,b,a+2,a a+2}$
$P_{b,b,a,a a}$		$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.3.5) \quad b = a + 1, d > b$$

Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = 2M_{a+2,d,a+1,a+1|a} + M_{a+2,a+1,a+1,d|a} + M_{a+1,a+2,d,a+1|a} + M_{a+1,d,a+1,a+2|a} + M_{d,a+1,a+2,a+1|a} + \dots$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= 2M_{a+1,d,a,a|a} + 3M_{a+1,d,a,a+1|a+1} + 3M_{a+1,d,a+1,a|a+1} + 2M_{a+2,d,a,a+1|a+2} + 2M_{a+2,d,a+1,a|a+2} \\ &= \sum M_{a,a+1,a,d|a} + \sum M_{a+1,d,a,a|a} + \sum M_{a,a+1,a+1,d|a+1} + \sum M_{a+1,a+1,a,d|a+1} + \dots \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,d,a,a|a}, 2M_{a+1,d,a,a+1|a+1}, 2M_{a+1,d,a+1,a|a+1}\}$. 根据 table 9, 仅仅由 Proposition 2.0.1 不足以判断 $P_{a+1,d,a,a|a}$ 或 $P_{d,a+1,a,a|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_{10})$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{a+1,d,a,a|a}, M_{d,a+1,a,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, a+1, a, d|a-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_{10})) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_{10})) \otimes V_1) = P_\nu = 2M_{a+1,d,a,a|a-1} + 2M_{d,a+1,a,a|a-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_{10})) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, a+1, a, d|a) + (0, 0, 0, 0| - 1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(a+1, d, a, a|a-1)$ 只能由 $(a+1, d, a, c|a) + (0, 0, 0, 0| - 1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{a+1,d,a,a|a}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,a+1,a,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$.

$$(3.3.6) \quad b = a + 1, d = b$$

Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{a+1,a+2,a+1,a+1|a} + M_{a+2,a+1,a+1,a+1|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{a,a+1,a,a+1|a} + M_{a,a+1,a+1,a|a} + 2M_{a+1,a+1,a,a|a} + M_{a,a+1,a+1,a+1|a+1} + 2M_{a+1,a+1,a,a+1|a+1} \\ &= \sum M_{a,a+1,a,a+1|a} + \sum M_{a,a+1,a+1,a+1|a+1} + \sum M_{a,a+2,a+1,a+1|a+2} + M_{a+1,a+1,a,a|a} + \dots \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+1,a,a|a}, M_{a+1,a+1,a,a+1|a+1}, M_{a+1,a+1,a,a+1|a+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by table 10.

Table 9: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a+1,d,a,a a}$	$\emptyset$
$P_{a+1,d,a,a+1 a+1}$	$M_{a+1,d,a,a+2 a+2}, M_{d,a+1,a,a+2 a+2}, M_{a+1,d,a+2,a a+2}, M_{d,a+1,a+2,a a+2}$
$P_{a+1,d,a+1,a a+1}$	$M_{a+1,d,a+2,a a+2}, M_{d,a+1,a+2,a a+2}$
$P_{a+2,d,a,a+1 a+2}$	$M_{a+3,d,a,a+1 a+3}, M_{d,a+3,a,a+1 a+3}, M_{a+3,d,a+1,a a+3}, M_{d,a+3,a+1,a a+3}$
$P_{a+2,d,a+1,a a+2}$	$M_{a+3,d,a+1,a a+3}, M_{d,a+3,a+1,a a+3}$
$P_{d,a+1,a+1,a a+1}$	$M_{d,a+1,a+2,a a+2}$
$P_{d,a+1,a,a a}$	$\emptyset$
$P_{d,a+1,a,a+1 a+1}$	$M_{d,a+1,a,a+2 a+2}, M_{d,a+1,a+2,a a+2}$
$P_{d,a+2,a,a+1 a+2}$	$M_{d,a+3,a,a+1 a+3}, M_{d,a+3,a+1,a a+3}$
$P_{d,a+2,a+1,a a+2}$	$M_{d,a+3,a+1,a a+3}$
$P_{a+1,a+1,a,d a+1}$	$M_{a+2,a+1,a,d a+2}, M_{a+1,a+2,a,d a+2}, M_{d,a+1,a,a+2 a+2}, M_{a+2,a+1,d,a a+2},$ $M_{d,a+1,a+2,a a+2}, M_{a+1,d,a,a+2 a+2}, M_{a+1,a+2,d,a a+2}, M_{a+1,d,a+2,a a+2}$
$P_{a+1,a+1,d,a a+1}$	$M_{a+2,a+1,d,a a+2}, M_{a+1,a+2,d,a a+2}, M_{d,a+1,a+2,a a+2}, M_{a+1,d,a+2,a a+2}$
$P_{d,a,a+1,a+1 a+1}$	$M_{d,a,a+2,a+1 a+2}, M_{d,a,a+1,a+2 a+2}, M_{d,a+1,a+2,a a+2}, M_{d,a+1,a,a+2 a+2}$
$P_{a,d,a+1,a+1 a+1}$	$M_{a,d,a+2,a+1 a+2}, M_{a,d,a+1,a+2 a+2}, M_{d,a,a+2,a+1 a+2}, M_{a+1,d,a+2,a a+2},$ $M_{d,a+1,a+2,a a+2}, M_{d,a,a+1,a+2 a+2}, M_{a+1,d,a,a+2 a+2}, M_{d,a+1,a,a+2 a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 10: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a+1,a+1,a,a a}$	$M_{a+2,a+1,a+1,a a+2}, M_{a+2,a+1,a,a+1 a+2}, M_{a+1,a+2,a+1,a a+2}, M_{a+1,a+2,a,a+1 a+2}$
$P_{a+1,a+1,a,a+1 a+1}$	$M_{a+2,a+1,a,a+1 a+2}, M_{a+1,a+2,a,a+1 a+2}, M_{a+1,a+1,a,a+2 a+2}, M_{a+2,a+1,a+1,a a+2},$ $M_{a+1,a+2,a+1,a a+2}, M_{a+1,a+1,a+2,a a+2}$
$P_{a+1,a+1,a+1,a a+1}$	$M_{a+2,a+1,a+1,a a+2}, M_{a+1,a+2,a+1,a a+2}, M_{a+1,a+1,a+2,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(4) $d = a$

(4.1) $b \neq a + 1, c \neq a + 1$

(4.1.1) $b < a, c > a$

Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$P_\mu = M_{a,b,c,a|a+1} + M_{c,a,b,a|a+1} + M_{a,c,a,b|a+1} + M_{a,c,b,a|a+1} + 2M_{c,a,a,b|a+1} + M_{c,b,a,a|a+1} + M_{a,a,c,b|a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+1,b,c,a|a+1} + M_{a,b,c,a+1|a+1} + M_{a,b,c,a|a} + M_{c,a+1,b,a|a+1} + M_{c,a,b,a+1|a+1} + M_{c,a,b,a} \\ &= \sum M_{a,b,c,a|a} + \sum M_{a,b,c,a+1|a+1} + M_{c,a,a,b|a} + \sum M_{c,a,a+1,b|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,a+1,a,b|a+1}, M_{c,a,a+1,b|a+1}, M_{c,a,a,b|a}\}$. 根据 table 11, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,a,a,b|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中. 为了排除这种可能, 我们分两种情况进行判断.

情况 1: $c > 2$

事实上, 我们仅需证明 $M_{c,a,a,b|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, a + 2|a + 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1) = P_\nu = 2M_{c,a+2,a,b|a+1} + 2M_{c,a,a+2,b|a+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, a + 1|a + 1) + (0, 0, 0, 1|0)$ 得到. 而前面计算结果可知 $\text{Filt}(P_\lambda)$ 恰好有一个 $M_{a,b,c,a+1|a+1} > 0$, 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$.

又由于 $(c, a + 2, a, b|a + 1)$ 只能由 $(c, a + 1, a, b|a + 1) + (0, 1, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c,a+1,a,b|a+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{c,a,a+1,b|a+1}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2. 这时, 未确定的 Verma 模只剩下一个 $M_{c,a,a,b|a}$, 它自己当然不是一个不可分分解投射. 因此 $M_{c,a,a,b|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也应该是 2.

情况 2: $c = 2$

为了排除这种可能, 我们仅需证明 $M_{b,b,a,a|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a + 3, a + 2|a + 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_5)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_5) = P_\nu = 2M_{a+3,a+2,a,b|a+1} + 2M_{a+3,a,a+2,b|a+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_5)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, a + 2, a + 1|a + 1) + (0, 0, 1, 1|0)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_5) = P_\nu$. 又由于 $(a + 3, a + 2, a, b|a + 1)$ 只能由 $(a + 2, a + 1, a, b|a + 1) + (1, 1, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{a+2,a+1,a,b|a+1}$, 否则 $pr_\nu(P_\lambda \otimes V_5)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{a+2,a,a+1,b|a+1}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2. 这时, 未确定的 Verma 模只剩下一个 $M_{a+2,a,a,b|a}$, 它自己当然不是一个不可分分解投射. 因此 $M_{a+2,a,a,b|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也应该是 2.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 11: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{c,a+1,a,b a+1}$		$M_{c,a+2,a,b a+2}$
$P_{c,a,a+1,b a+1}$		$M_{c,a,a+2,b a+2}, M_{c,a+2,a,b a+2}$
$P_{c,a,a,b a}$		$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(4.1.2) *Others*

Let $\mu := \lambda + (0, 0, 0, 0|1) = (a, b, c, a|a+1)$.

$$Pr_\lambda(P_\mu \otimes V_1) = \sum M_{a,b,c,a|a} + \sum M_{a,b,c,a+1|a+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(4.2) $b = a + 1, c \neq a + 1$

(4.2.1) $c > a + 2$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{a+2,c,a+1,a+1|a} + M_{a+1,a+2,c,a+1|a} + M_{c,a+1,a+2,a+1|a} + M_{c,a+2,a+1,a+1|a} + M_{a+2,a+1,c,a+1|a} + M_{a+1,a+2,c,a+1|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{a+1,c,a,a|a} + M_{a+1,c,a,a+1|a+1} + 2M_{a+1,c,a+1,a|a+1} + M_{a+2,c,a,a+1|a+2} + M_{a+2,c,a+1,a|a+2} \\ &= \sum M_{a,a+1,c,a|a} + \sum M_{a,a+1,c,a+1|a+1} + \sum M_{a,a+2,c,a+1|a+2} + \sum M_{a+1,a+1,c,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,a+1,a+1,a|a+1}, M_{a+1,a+1,c,a|a+1}, M_{a+1,a+1,c,a+1|a+1}\}$.
We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 12](#).

Table 12: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c,a+1,a+1,a a+1}$	$M_{c,a+2,a+1,a a+2}, M_{c,a+1,a+2,a a+2}$
$P_{a+1,a+1,c,a a+1}$	$M_{a+2,a+1,c,a a+2}, M_{a+1,a+2,c,a a+2}, M_{c,a+1,a+2,a a+2}, M_{a+2,c,a+1,a a+2},$ $M_{c,a+2,a+1,a a+2}, M_{a+1,c,a+2,a a+2}$
$P_{a+1,c,a+1,a a+1}$	$M_{a+2,c,a+1,a a+2}, M_{a+1,c,a+2,a a+2}, M_{c,a+2,a+1,a a+2}, M_{c,a+1,a+2,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(4.2.2) $c = a + 2$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{a+2,a+2,a+1,a+1|a} + M_{a+1,a+2,a+2,a+1|a} + M_{a+2,a+1,a+2,a+1|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{a+1,a+2,a,a|a} + M_{a+2,a+1,a,a|a} + M_{a+1,a+2,a,a+1|a+1} + 2M_{a+1,a+2,a+1,a|a+1} + M_{a+2,a+1,a+2,a|a+1} \\ &= \sum M_{a,a+1,a+2,a|a} + \sum M_{a,a+1,a+2,a+1|a+1} + \sum M_{a,a+2,a+2,a+1|a+2} + \sum M_{a+1,a+1,a+2,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+2,a+1,a|a+1}, M_{a+1,a+1,a+2,a|a+1}, M_{a+1,a+1,a+2,a+1|a+1}\}$.
We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 13](#).

(4.2.3) $c < a$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{a+2,a+1,c,a+1|a} + M_{a+1,a+2,a+1,c|a} + M_{a+1,a+2,c,a+1|a} + M_{a+2,a+1,a+1,c|a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{a+1,a,c,a|a} + M_{a+1,a,c,a+1|a+1} + 2M_{a+1,a+1,c,a|a+1} + M_{a+2,a,c,a+1|a+2} + M_{a+2,a+1,c,a|a+2} \\ &= \sum M_{a,a+1,c,a|a} + \sum M_{a,a+1,c,a+1|a+1} + \sum M_{a+1,a+1,c,a|a+1} + \sum M_{a,a+2,c,a+1|a+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+1,c,a|a+1}, M_{a+1,a+1,a,c|a+1}\}$.
We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 14](#).

Table 13: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a+1,a+2,a+1,a a+1}$	$M_{a+2,a+2,a+1,a a+2}, M_{a+1,a+2,a+2,a a+2}, M_{a+1,a+3,a+2,a a+3}, M_{a+2,a+1,a+2,a a+2},$ $M_{a+3,a+1,a+2,a a+3}, M_{a+2,a+3,a+1,a a+3}, M_{a+3,a+2,a+1,a a+3}$
$P_{a+1,a+1,a+2,a a+1}$	$M_{a+2,a+1,a+2,a a+2}, M_{a+1,a+2,a+2,a a+2}, M_{a+2,a+2,a+1,a a+2}$
$P_{a+2,a+1,a+1,a a+1}$	$M_{a+3,a+2,a+1,a a+3}, M_{a+3,a+1,a+2,a a+3}, M_{a+2,a+2,a+1,a a+2}, M_{a+2,a+1,a+2,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 14: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a+1,a+1,c,a a+1}$	$M_{a+2,a+1,c,a a+2}, M_{a+1,a+2,c,a a+2}, M_{a+2,a+1,a,c a+2}, M_{a+1,a+2,a,c a+2}$
$P_{a+1,a+1,a,c a+1}$	$M_{a+2,a+1,a,c a+2}, M_{a+1,a+2,a,c a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.2.4) \quad c = a$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|2).$$

$$P_\mu = M_{a,a+1,a,a|a+2} + M_{a+1,a,a,a|a+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{a+1,a+2,a,a|a+2} + M_{a,a+2,a+1,a|a+2} + M_{a,a+2,a,a+1|a+2} + 2M_{a+1,a+1,a,a|a+1} + M_{a,a+1,a+2,a|a+2} \\ &= \sum M_{a,a+1,a,a|a} + \sum M_{a,a+1,a,a+1|a+1} + \sum M_{a,a+2,a,a+1|a+2} + M_{a+1,a+1,a,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+1,a,a|a+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_5) = P_\lambda$ by table 15.

Table 15: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_5))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_5))$ (via Prop. ~ 1)
$P_{a+1,a+1,a,a a+1}$	$M_{a+2,a+1,a,a a+2}, M_{a+1,a+2,a,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.3) \quad b = a + 1, c = a + 1$$

我们提前使用 Theorem 3.4 中 $P_{a,a+1,a+1,a|a}$ 结果, 注意这并没有引入循环论证. Let $\mu := \lambda + (1, 0, 0, 0|0)$.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a+2,a,a+1,a+1|a+2} + M_{a+2,a+1,a,a+1|a+2} + M_{a+2,a+1,a+1,a|a+2} + M_{a,a+2,a+1,a+1|a+2} + M_{a+1,a+2,a,a+1|a+2} \\ &= \sum M_{a,a+1,a+1,a|a} + \sum M_{a,a+1,a+1,a+1|a+1} + \sum M_{a,a+1,a+2,a+1|a+2} + M_{a+1,a+1,a+1,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+1,a+1,a|a+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$ by table 16.

$$(4.4) \quad c = a + 1, b \neq a + 1$$

Table 16: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{a+1,a+1,a+1,a a+1}$	$M_{a+2,a+1,a+1,a a+2}, M_{a+1,a+2,a+1,a a+2}, M_{a+1,a+1,a+2,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.4.1) \quad b > a + 2$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|2).$$

$$P_\mu = M_{a+1,b,a,a|a+2} + M_{b,a,a+1,a|a+2} + M_{b,a+1,a,a|a+2} + M_{a,b,a+1,a|a+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{a+2,b,a+1,a|a+2} + M_{a+2,b,a,a+1|a+2} + 2M_{a+1,b,a+1,a|a+1} + M_{a+1,b,a,a+1|a+1} + M_{a+1,b,a} \\ &= \sum M_{a,b,a+1,a|a} + \sum M_{a,b,a+1,a+1|a+1} + \sum M_{a,b,a+2,a+1|a+2} + \sum M_{a+1,b,a+1,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,a+1,a+1,a|a+1}, M_{a+1,b,a+1,a|a+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_5) = P_\lambda$ by table 17.

Table 17: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_5))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_5))$ (via Prop. ~ 1)
$P_{b,a+1,a+1,a a+1}$	$M_{b,a+2,a+1,a a+2}, M_{b,a+1,a+2,a a+2}$
$P_{a+1,b,a+1,a a+1}$	$M_{a+2,b,a+1,a a+2}, M_{a+1,b,a+2,a a+2}, M_{b,a+2,a+1,a a+2}, M_{b,a+1,a+2,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.4.2) \quad b = a$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|2).$$

$$P_\mu = M_{a,a+1,a,a|a+2} + M_{a,a,a+1,a|a+2} + M_{a+1,a,a,a|a+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{a+1,a+2,a,a|a+2} + M_{a,a+2,a+1,a|a+2} + M_{a,a+2,a,a+1|a+2} + 2M_{a+1,a+1,a,a|a+1} + 2M_{a,a+1} \\ &= \sum M_{a,a,a+1,a|a} + \sum M_{a,a,a+1,a+1|a+1} + \sum M_{a,a,a+2,a+1|a+2} + \sum M_{a,a+1,a+1,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,a+1,a+1,a|a+1}, M_{a+1,a+1,a,a|a+1}, M_{a+1,a+1,a,a+1|a+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_5) = P_\lambda$ by table 18.

Table 18: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_5))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_5))$ (via Prop. ~ 1)
$P_{a,a+1,a+1,a a+1}$	$M_{a+2,a+1,a,a a+2}, M_{a+1,a+2,a,a a+2}, M_{a,a+2,a+1,a a+2}, M_{a,a+1,a+2,a a+2},$ $M_{a+2,a,a+1,a a+2}, M_{a+1,a,a+2,a a+2}$
$P_{a+1,a+1,a,a a+1}$	$M_{a+2,a+1,a,a a+2}, M_{a+1,a+2,a,a a+2}$
$P_{a+1,a,a+1,a a+1}$	$M_{a+2,a,a+1,a a+2}, M_{a+1,a,a+2,a a+2}, M_{a+2,a+1,a,a a+2}, M_{a+1,a+2,a,a a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(4.4.3) $b < a$

Let $\mu := \lambda + (0, 0, 0, 0|2)$.

$$P_\mu = M_{a,b,a+1,a|a+2} + M_{a+1,a,b,a|a+2} + M_{a,a+1,a,b|a+2} + M_{a,a+1,b,a|a+2} + 2M_{a+1,a,a,b|a+2} + M_{a+1,b,a,a|a+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{a+1,b,a+2,a|a+2} + M_{a,b,a+2,a+1|a+2} + 2M_{a+1,b,a+1,a|a+1} + M_{a,b,a+1,a+1|a+1} + M_{a,b,a+1,a|a+1} \\ &= \sum M_{a,b,a+1,a|a} + \sum M_{a,b,a+1,a+1|a+1} + \sum M_{a,b,a+2,a+1|a+2} + M_{a+1,a,a,b|a} + \sum M_{a+1,a,b,a|a} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+1,a+1,b,a|a+1}, 2M_{a+1,a+1,a,b|a+1}, M_{a+1,a+1,a+1,b|a+1}\}$. 根据 table 19, 仅仅由 Proposition 2.0.1 不足以判断 $P_{a+1,a,a,b|a}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_5)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{a+1,a,a,b|a}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a+3, a+1|a+2)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_5)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_5)) \otimes V_1) = P_\nu = 2M_{a+3,a+1,a,b|a+2} + 2M_{a+3,a,a+1,b|a+2} \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_5)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, a+2, a+1|a+2) + (0, 0, 1, 0|0)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(a+3, a+1, a, b|a+2)$ 只能由 $(a+2, a+1, a, b|a+2) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{a+2,a+1,a,b|a+2}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $\text{Filt}(P_\lambda)$ 中 $M_{a+2,a,a+1,b|a+2}$ 的重数也必然是 2.

直接由 Proposition 2.0.1 得到, $M_{a+2,a+1,a,b|a+2}, M_{a+2,a,a+1,b|a+2} \in \text{Filt}(P_{a+1,a,a,b|a})$, 但是由于 P_λ 和 $pr_\lambda(P_\lambda \otimes V_5)$ 中 $M_{a+2,a+1,a,b|a+2}, M_{a+2,a,a+1,b|a+2}$ 重数都是 2. 因此 $pr_\lambda(P_\lambda \otimes V_5)$ 不可能再有不可分解直和项 $P_{a+1,a,a,b|a}$ 了.

综上 $Pr_\lambda(P_\mu \otimes V_5) = P_\lambda$.

Table 19: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_5))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_5))$ (via Prop. ~ 1)
$P_{a+1,a+1,b,a a+1}$	$M_{a+2,a+1,b,a a+2}, M_{a+1,a+2,b,a a+2}, M_{a+1,a+2,a,b a+2}$
$2M_{a+1,a+1,a,b a+1}$	$M_{a+1,a+2,a,b a+2}$
$P_{a+2,a+1,a,b a+2}$	$M_{a+3,a+1,a,b a+3}$
$P_{a+2,a,a+1,b a+2}$	$M_{a+3,a,a+1,b a+3}, M_{a+3,a+1,a,b a+3}$
$2M_{a+1,a,a+1,b a+1}$	$M_{a+1,a,a+2,b a+2}, M_{a+1,a+2,a,b a+2}$
$P_{a+1,a,a,b a}$	$\emptyset$
$P_{a+1,b,a+1,a a+1}$	$M_{a+2,b,a+1,a a+2}, M_{a+1,b,a+2,a a+2}, M_{a+2,a+1,b,a a+2}, M_{a+1,a+2,b,a a+2},$ $M_{a+1,a,a+2,b a+2}, M_{a+1,a+2,a,b a+2}$
$P_{a,a+1,a+1,b a+1}$	$M_{a,a+2,a+1,b a+2}, M_{a,a+1,a+2,b a+2}, M_{a+1,a+2,a,b a+2}, M_{a+1,a,a+2,b a+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(4.4.4) $b = a+2$

我们提前使用 Theorem 3.3 中 $P_{a,a+2,a+1,a|a+1}$ 结果, 注意这并没有引入循环论证. Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{a+1,a+2,a+1,a|a+1} + M_{a+1,a+2,a,a+1|a+1} + M_{a+1,a+2,a,a|a} + 2M_{a+2,a+2,a+1,a|a+2} + 2M_{a+2,a+2,a,a|a+2} \\ &= \sum M_{a,a+2,a+1,a|a} + \sum M_{a,a+2,a+1,a+1|a+1} + \sum M_{a+1,a+2,a+1,a|a+1} + \sum M_{a,a+2,a+1,a|a+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a+2,a+3,a+1,a|a+3}, M_{a+2,a+3,a,a+1|a+3}\}$. 根据 table 20, 仅仅由 Proposition 2.0.1 不足以判断 $P_{a+2,a+2,a+1,a|a+2}$ 或 $P_{a+2,a+2,a,a+1|2}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中. 根据前文的结论, 我们可以有

$$\begin{aligned} P_{a+2,a+2,a+1,a|a+2} &= M_{a+3,a+2,a+1,a|a+3} + M_{a+2,a+3,a+1,a|a+3} + M_{a+2,a+2,a+1,a|a+2} \\ &= M_{a+2,a+2,a+1,a|a+2} + \sum M_{a+2,a+3,a+1,a|a+3} \end{aligned} \quad (11)$$

$$\begin{aligned} P_{a+2,a+2,a,a+1|a+2} &= M_{a+3,a+2,a,a+1|a+3} + M_{a+2,a+3,a,a+1|a+3} + M_{a+2,a+2,a,a+1|a+2} + M_{a+3,a+2,a+1,a|a+3} \\ &= \sum M_{a+2,a+2,a,a+1|a+2} + \sum M_{a+2,a+3,a,a+1|a+3} \end{aligned} \quad (12)$$

由于 $\text{Filt}(P_{a+2,a+2,a+1,a|a+2}) \in U$ 和 $\text{Filt}(P_{a+2,a+2,a,a+1|a+2}) \subseteq U$, 因此我们进行反证, 分别排除它们作为 $pr_\lambda(P_\lambda \otimes V_1)$ 直和项的可能

首先, 假设 $P_{a+2,a+2,a+1,a|a+2}$ 是 $pr_\lambda(P_\lambda \otimes V_1)$ 的一个直和项, 那么, 存在一个投射模 Q 使得

$$pr_\lambda(P_\lambda \otimes V_1) = Q + P_{a+2,a+2,a+1,a|a+2}$$

令 $\nu = (a, a+4, a+2, a+1|a+3)$, 那么 ν 是典型的权. 我们就有

$$pr_\nu(Q \otimes V_1) = M_{a+2,a+4,a,a+1|a+3} + M_{a+4,a+1,a+2,a|a+3} + M_{a+4,a,a+2,a+1|a+3} + M_{a+4,a+2,a,a+1|a+3} + M_{a+4,a+2,a+1,a|a+3} \quad (13)$$

观察到 equation (13) 中, 权 ν 在滤过中是极小的. Q 是投射模, 因此 $pr_\nu(Q \otimes V_1)$ 也是投射模. 根据 Lemma 2.0.1, 表明 P_ν 必然是 $pr_\nu(Q \otimes V_1)$ 的一个不可分解的直和项. 但是 Theorem 2.1 给出 P_ν 的完整滤过的结构如下

$$\begin{aligned} P_\nu &= M_{a+2,a+4,a+1,a|a+3} + M_{a+2,a+4,a,a+1|a+3} + M_{a+4,a+1,a+2,a|a+3} + M_{a+4,a,a+2,a+1|a+3} + M_{a+4,a+2,a,a+1|a+3} \\ &= \sum M_{a,a+4,a+2,a+1|a+3} \end{aligned} \quad (14)$$

比较 equation (13) 和 equation (14), 我们发现 P_ν 不可能是 $pr_\nu(Q \otimes V_1)$ 的直和项. 矛盾, 因此排除了 $P_{a+2,a+2,a+1,a|a+2}$ 作为 $pr_\lambda(P_\lambda \otimes V_1)$ 直和项的可能. 同样使用 ν 和 V_1 , 也可以排除 $P_{a+2,a+2,a,a+1|a+2}$ 作为 $pr_\lambda(P_\lambda \otimes V_1)$ 直和项的可能.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

□

Table 20: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)	
$P_{a+2,a+3,a+1,a a+3}$	$M_{a+2,a+4,a+1,a a+4}, M_{a+4,a+2,a+1,a a+4}$	
$P_{a+2,a+3,a,a+1 a+3}$	$M_{a+2,a+4,a,a+1 a+4}, M_{a+4,a+2,a,a+1 a+4}, M_{a+2,a+4,a+1,a a+4}, M_{a+4,a+2,a+1,a a+4}$	
$P_{a+3,a+1,a+2,a a+3}$	$M_{a+4,a+1,a+2,a a+4}, M_{a+4,a+2,a+1,a a+4}$	
$P_{a+3,a,a+2,a+1 a+3}$	$M_{a+4,a,a+2,a+1 a+4}, M_{a+4,a+2,a,a+1 a+4}, M_{a+4,a+1,a+2,a a+4}, M_{a+4,a+2,a+1,a a+4}$	
$P_{a+2,a+1,a+1,a a+1}$	$M_{a+2,a+1,a+2,a a+2}$	
$P_{a+2,a+2,a+1,a a+2}$	$\emptyset$	
$P_{a+2,a+2,a,a+1 a+2}$	$\emptyset$	
$P_{a+3,a+2,a+1,a a+3}$	$M_{a+4,a+2,a+1,a a+4}$	
$P_{a+3,a+2,a,a+1 a+3}$	$M_{a+4,a+2,a,a+1 a+4}, M_{a+4,a+2,a+1,a a+4}$	
$P_{a+1,a+2,a+1,a a+1}$	$M_{a+1,a+2,a+2,a a+2}, M_{a+2,a+1,a+2,a a+2}$	
$P_{a+1,a+3,a+2,a a+3}$	$M_{a+1,a+4,a+2,a a+4}, M_{a+4,a+1,a+2,a a+4}, M_{a+2,a+4,a+1,a a+4}, M_{a+4,a+2,a+1,a a+4}$	
$P_{a,a+3,a+2,a+1 a+3}$	$M_{a,a+4,a+2,a+1 a+4}, M_{a+4,a,a+2,a+1 a+4}, M_{a+2,a+4,a,a+1 a+4}, M_{a+1,a+4,a+2,a a+4},$ $M_{a+4,a+2,a,a+1 a+4}, M_{a+4,a+1,a+2,a a+4}, M_{a+2,a+4,a+1,a a+4}, M_{a+4,a+2,a+1,a a+4}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Theorem 3.2. Let $a, b, c, d \in \mathbb{Z}$ and $\lambda = (a, b, c, d|b)$ with $a \neq b$. The projective objects P_λ in $\mathfrak{B}_{a,c,d}$ have the following Verma flag formulae.

Proof. The Cases are as follows:

(1) $a \neq b + 1$

(1.1) $b \neq \{c, d\}$

(1.1.1) $a < c < b < d, d > b + 1$

Let $\mu := \lambda + (0, 1, 0, 0|0)$.

$$P_\mu = 2M_{b+1,d,a,c|b} + 2M_{d,b+1,c,a|b} + M_{b+1,a,c,d|b} + M_{d,c,a,b+1|b} + M_{a,b+1,d,c|b} + M_{c,d,b+1,a|b} + M_{c,b+1,a,d|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,d,a,c|b} + 2M_{b+1,d,a,c|b+1} + 2M_{d,b,c,a|b} + 2M_{d,b+1,c,a|b+1} + M_{b,a,c,d|b} + M_{b+1,a,c,d|b+1} \\ &= \sum M_{a,b,c,d|b} + \sum M_{b,d,a,c|b} + \sum M_{a,b+1,c,d|b+1} + \sum M_{b+1,d,a,c|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,a,c|b}, M_{b+1,d,a,c|b+1}, M_{d,b,c,a|b}, M_{d,b+1,c,a|b+1}, M_{b,a,c,d|b}, M_{b+1,a,c,d|b+1}\}$. 根据 table 21, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,d,a,c|b}$ 或 $P_{d,b,c,a|b}$ 或 $P_{b,a,c,d|b}$ 或 $P_{b+1,d,a,c|b+1}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中. 我们寻求其他方法, 排除它们中某一个作为 $pr_\lambda(P_\mu \otimes V_2)$ 直和项的可能性. 事实上, 我们仅需证明 $M_{b,d,a,c|b}, M_{d,b,c,a|b}, M_{d,b,a,c|b}, M_{b,d,c,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

情况 1: $c < b - 1$

令 $\nu = (a, b, c, d|b - 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,d,a,c|b-1} + 2M_{b,d,c,a|b-1} + 2M_{d,b,a,c|b-1} + 2M_{d,b,c,a|b-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, d|b) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, d, a, c|b - 1)$ 只能由 $(b, d, a, c|b) + (0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,c|b}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,c,a|b}, M_{d,b,a,c|b}, M_{b,d,c,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

情况 1: $c = b - 1$

令 $\nu = (a, b, b - 1, d|b + 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2) = 2P_\nu = 2M_\nu + 4M_{b,d,a,c|b+1} + 4M_{b,d,c,a|b+1} + 4M_{d,b,a,c|b+1} + 4M_{d,b,c,a|b+1}$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(a, b, c, d|b) + (0, 0, 0, 0|1)$ 和 $(a, b + 1, c, d|b + 1) + (0, -1, 0, 0|0)$ 得到. 由 Proposition 2.0.1 可知, $M_{a,b,c,d|b}$ 和 $M_{a,b+1,c,d|b+1}$ 都在 $\text{Filt}(P_\lambda)$ 中, 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$.

又由于 $(b, d, a, c|b + 1)$ 只能由 $(b, d, a, c|b) + (0, 0, 0, 0|1)$ 和 $(b + 1, d, a, c|b + 1) + (-1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,c|b}$ 和两个 $M_{b+1,d,a,c|b+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,c,a|b}, M_{d,b,a,c|b}, M_{b,d,c,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综和情况 1, 2, $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

(1.1.2) $a < c < b < d, d = b + 1$

Let $\mu := \lambda + (0, 1, 0, 0|0)$.

$$P_\mu = 2M_{b+1,b+1,a,c|b} + M_{b+1,a,c,b+1|b} + M_{a,b+1,b+1,c|b} + M_{c,b+1,a,b+1|b} + M_{b+1,c,b+1,a|b} + 2M_{b+1,b+1,c,a|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,b+1,a,c|b} + 2M_{b+1,b,a,c|b} + 2M_{b+1,b+1,a,c|b+1} + M_{b,a,c,b+1|b} + M_{b+1,a,c,b|b} + M_{b+1,a,c,b+1|b+1} \\ &= \sum M_{a,b,c,b+1|b} + \sum M_{b,b+1,a,c|b} + \sum M_{a,b+1,c,b+1|b+1} + \sum M_{b+1,b+1,a,c|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b+1,a,c|b}, M_{b+1,b,a,c|b}, M_{b+1,b+1,a,c|b}, M_{b+1,b+1,c,a|b+1}, M_{b,a,c,b+1|b}, M_{b+1,a,c,b|b}, M_{b+1,a,c,b+1|b+1}\}$. 根据 table 22, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,b+1,a,c|b}$ 或 $P_{b+1,b,a,c|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b+1,a,c|b}, M_{b+1,b,a,c|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

Table 21: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{b,d,a,c b}$	$\emptyset$
$P_{b+1,d,a,c b+1}$	$M_{b+2,d,a,c b+2}, M_{d,b+2,a,c b+2}, M_{b+2,d,c,a b+2}, M_{d,b+2,c,a b+2}$
$P_{d,b,c,a b}$	$\emptyset$
$P_{d,b+1,c,a b+1}$	$M_{d,b+2,c,a b+2}$
$P_{d,b,a,c b}$	$\emptyset$
$P_{d,b+1,a,c b+1}$	$M_{d,b+2,a,c b+2}, M_{d,b+2,c,a b+2}$
$P_{b,d,c,a b}$	$\emptyset$
$P_{b+1,d,c,a b+1}$	$M_{b+2,d,c,a b+2}, M_{d,b+2,c,a b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

令 $\nu = (a, b+1, c, b+1|b+2)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2) = P_\nu = 2M_{b+1,b+1,a,c|b+2} + 2M_{b+1,b+1,c,a|b+2} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$ 极小并且重数为 1, 且 ν 只能由 $(a, b+1, c, b+1|b+1) + (0, 0, 0, 0|1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_2) = P_\nu$. 又由于 $(b+1, b+1, a, c|b+2)$ 只能由 $(b+1, b+1, a, c|b+1) + (0, 0, 0, 0|1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b+1,b+1,a,c|b+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{b+1,b+1,c,a|b+1}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{b,b+1,a,c|b}, M_{b+1,b,a,c|b}, M_{b,b+1,c,a|b}, M_{b+1,b,c,a|b}\}$. 我们发现, 但是 Proposition 2.0.1 表明 $M_{b+1,b+1,a,c|b+1} \in \text{Filt}(P_{b,b+1,a,c|b})$ 和 $M_{b+1,b+1,c,a|b+1} \in \text{Filt}(P_{b,b+1,c,a|b})$. 这说明 U' 不能包含 $\text{Filt}(P_{b,b+1,a,c|b})$ 和 $\text{Filt}(P_{b,b+1,c,a|b})$, 进一步说明除了 $P_\lambda, Pr_\lambda(P_\mu \otimes V_2)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

$$(1.1.3) \quad a = c < b < d, d > b+1$$

Let $\mu := \lambda + (0, 1, 0, 0|0)$.

$$P_\mu = 2M_{b+1,d,a,a|b} + M_{b+1,a,a,d|b} + M_{a,b+1,d,a|b} + M_{a,d,a,b+1|b} + M_{d,a,b+1,a|b} + 2M_{d,b+1,a,a|b} + M_{a,b+1,a,d|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,d,a,a|b} + 2M_{b+1,d,a,a|b+1} + M_{b,a,a,d|b} + M_{b+1,a,a,d|b+1} + M_{a,b,d,a|b} + M_{a,b+1,d,a|b+1} \\ &= \sum M_{a,b,a,d|b} + \sum M_{b,d,a,a|b} + \sum M_{a,b+1,a,d|b+1} + \sum M_{b+1,d,a,a|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,a,a|b}, M_{b+1,d,a,a|b+1}, M_{d,b,a,a|b}, M_{d,b+1,a,a|b+1}\}$. 根据 table 23, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,d,a,a|b}$ 或 $P_{d,b,a,a|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,a|b}, M_{d,b,a,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

情况 1: $a < b-1$

令 $\nu = (a, b, a, d|b-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{b,d,a,a|b-1} + 2M_{d,b,a,a|b-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, a, d|b-1) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b, d, a, a|b-1)$ 只能由 $(b, d, a, a|b) +$

Table 22: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
	$P_{b,b+1,a,c b}$	$\emptyset$
	$P_{b+1,b,a,c b}$	$M_{b+2,b+1,a,c b+2}, M_{b+2,b+1,c,a b+2}$
	$P_{b+1,b+1,a,c b+1}$	$M_{b+2,b+1,a,c b+2}, M_{b+1,b+2,a,c b+2}, M_{b+2,b+1,c,a b+2}, M_{b+1,b+2,c,a b+2}$
	$P_{b,b+1,c,a b}$	$\emptyset$
	$P_{b+1,b,c,a b}$	$M_{b+2,b+1,c,a b+2}$
	$P_{b+1,b+1,c,a b+1}$	$M_{b+2,b+1,c,a b+2}, M_{b+1,b+2,c,a b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$(0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,a|b}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

情况 1: $a = b - 1$

令 $\nu = (b - 1, b, b - 1, d|b + 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2) = 2P_\nu = 2M_\nu + 4M_{b,d,b-1,b-1|b+1} + 4M_{d,b,b-1,b-1|b+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(b - 1, b, b - 1, d|b) + (0, 0, 0, 0|1)$ 和 $(b - 1, b + 1, b - 1, d|b + 1) + (0, -1, 0, 0|0)$ 得到. 由 Proposition 2.0.1 可知, $M_{b-1,b,b-1,d|b}$ 和 $M_{b-1,b+1,b-1,d|b+1}$ 都在 $\text{Filt}(P_\lambda)$ 中, 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$.

又由于 $(b, d, b - 1, b - 1|b + 1)$ 只能由 $(b, d, b - 1, b - 1|b) + (0, 0, 0, 0|1)$ 和 $(b + 1, d, b - 1, b - 1|b + 1) + (-1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,b-1,b-1|b}$ 和两个 $M_{b+1,d,b-1,b-1|b+1}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,b-1,b-1|b}$, $M_{d,b+1,b-1,b-1|b+1}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综和情况 1, 2, $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

Table 23: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
	$P_{b,d,a,a b}$	$\emptyset$
	$P_{b+1,d,a,a b+1}$	$M_{b+2,d,a,a b+2}, M_{d,b+2,a,a b+2}$
	$P_{d,b,a,a b}$	$\emptyset$
	$P_{d,b+1,a,a b+1}$	$M_{d,b+2,a,a b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.1.4) \quad a = c < b < d, d = b + 1$$

Let $\mu := \lambda + (0, 1, 0, 0|0)$.

$$P_\mu = 2M_{b+1,b+1,a,a|b} + M_{b+1,a,a,b+1|b} + M_{a,b+1,b+1,a|b} + M_{a,b+1,a,b+1|b} + M_{b+1,a,b+1,a|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= 2M_{b,b+1,a,a|b} + 2M_{b+1,b,a,a|b} + 2M_{b+1,b+1,a,a|b+1} + M_{b,a,a,b+1|b} + M_{b+1,a,a,b|b} + M_{b+1,a,b+1|b+1} \\ &= \sum M_{a,b,a,b+1|b} + \sum M_{b,b+1,a,a|b} + \sum M_{a,b+1,a,b+1|b+1} + M_{b+1,b+1,a,a|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b+1,a,a|b}, M_{b+1,b,a,a|b}, M_{b+1,b+1,a,a|b+1}, M_{b,a,a,b+1|b}, M_{b+1,a,a,b|b}, M_{b+1,a,b+1|b+1}\}$. 根据 table 24, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,b+1,a,a|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b+1,a,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (b-1, b, b-1, b+1|b+2)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2) = P_\nu = 2M_{b+1,b+1,b-1,b-1|b+2} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_2)$ 极小并且重数为 1, 且 ν 只能由 $(b-1, b, b-1, b+1|b+1) + (0, 0, 0, 0|1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_2) = P_\nu$. 又由于 $(b+1, b+1, b-1, b-1|b+2)$ 只能由 $(b+1, b+1, b-1, b-1|b+1) + (0, 0, 0, 0|1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b+1,b+1,b-1,b-1|b+1}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的 $\text{Filt}(P_\nu)$.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{b,b+1,a,a|b}, M_{b+1,b,a,a|b}, M_{b+1,b+1,a,a|b+1}, M_{b,a,a,b+1|b}, M_{b+1,a,a,b|b}, M_{b+1,a,b+1|b+1}\}$. 我们发现, 但是 Proposition 2.0.1 表明 $M_{b+1,b+1,b-1,b-1|b+1} \in \text{Filt}(P_{b,b+1,b-1,b-1|b})$. 这说明 U' 不能包含 $\text{Filt}(P_{b,b+1,b-1,b-1|b})$, 进一步说明除了 $P_\lambda, Pr_\lambda(P_\mu \otimes V_2)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

Table 24: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{b,b+1,a,a b}$	$\emptyset$
$P_{b+1,b,a,a b}$	$M_{b+2,b+1,a,a b+2}$
$P_{b+1,b+1,a,a b+1}$	$M_{b+2,b+1,a,a b+2}, M_{b+1,b+2,a,a b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.1.5) \quad b < a < d < c$$

$$\text{Let } \mu := \lambda + (0, 1, 0, 0|0).$$

$$P_\mu = M_{a,b+1,c,d|b} + M_{d,c,b+1,a|b} + 2M_{c,d,a,b+1|b} + M_{c,a,b+1,d|b} + M_{a,c,d,b+1|b} + M_{c,b+1,d,a|b} + M_{d,a,c,b+1|b} + \dots$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b,c,d|b} + M_{a,b+1,c,d|b+1} + M_{d,c,b,a|b} + M_{d,c,b+1,a|b+1} + 2M_{c,d,a,b|b} + 2M_{c,d,a,b+1|b+1} + \dots \\ &= \sum M_{a,b,c,d|b} + \sum M_{a,b+1,c,d|b+1} + \sum M_{c,a,d,b|b} + \sum M_{c,a,d,b+1|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,d,a,b|b}, M_{c,d,a,b+1|b+1}, M_{c,a,d,b|b}, M_{c,a,d,b+1|b+1}, M_{c,b+1,d,a|b}, M_{d,a,c,b+1|b}, M_{d,c,b+1,a|b+1}, M_{d,c,b,a|b}, M_{a,c,d,b+1|b}, M_{a,b+1,c,d|b+1}, M_{a,b,c,d|b}\}$. 根据 table 25, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,d,a,b|b}$ 和 $P_{c,a,d,b|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c,d,a,b|b}$ 和 $M_{c,a,d,b|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, d|b-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{c,d,a,b|b-1} + 2M_{c,a,d,b|b-1} \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, d|b) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c, d, a, b|b-1)$ 只能由 $(c, d, a, b|b) + (0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c,d,a,b|b}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $\text{Filt}(P_\lambda)$ 中 $M_{c,a,d,b|b}$ 重数也应该是 2.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

Table 25: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{c,d,a,b b}$		$\emptyset$
$P_{c,d,a,b+1 b+1}$		$M_{c,d,a,b+2 b+2}$
$P_{c,a,d,b b}$		$\emptyset$
$P_{c,a,d,b+1 b+1}$		$M_{c,a,d,b+2 b+2}, M_{c,d,a,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.1.6) $b < a = d < c$
Let $\mu := \lambda + (0, 1, 0, 0|0)$.

$$P_\mu = M_{a,b+1,c,a|b} + M_{c,a,b+1,a|b} + M_{a,c,a,b+1|b} + M_{a,c,b+1,a|b} + 2M_{c,a,a,b+1|b} + M_{c,b+1,a,a|b} + M_{a,a,c,b+1|b}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b,c,a|b} + M_{a,b+1,c,a|b+1} + M_{c,a,b,a|b} + M_{c,a,b+1,a|b+1} + M_{a,c,a,b|b} + M_{a,c,a,b+1|b+1} + M_{a,c,a,b+1|b}$$

$$= \sum M_{a,b,c,a|b} + \sum M_{a,b+1,c,a|b+1} + M_{c,a,a,b|b} + M_{c,a,a,b+1|b+1}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,a,a,b|b}, M_{c,a,a,b+1|b+1}\}$.
根据 table 26, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,a,a,b|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_2)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c,a,a,b|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, a|b-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1) = P_\nu = 2M_{c,a,a,b|b-1} \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_2)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, a|b) + (0, 0, 0, 0| -1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c, a, a, b|b-1)$ 只能由 $(c, a, a, b|b) + (0, 0, 0, 0| -1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c,a,a,b|b}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

Table 26: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)
$P_{c,a,a,b b}$		$\emptyset$
$P_{c,a,a,b+1 b+1}$		$M_{c,a,a,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.1.7) *Others*
Let $\mu := \lambda + (0, 1, 0, 0|0) = (a, b+1, c, d|b)$.

$$Pr_\lambda(P_\mu \otimes V_2) = \sum M_{a,b,c,d|b} + \sum M_{a,b+1,c,d|b+1}$$

All items belong to P_λ by Proposition 2.0.1.

$$(1.2) \quad c = b, d \neq b$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 0|0) = (a, b+1, b+1, d|b).$$

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{a,b,b,d|b} + \sum M_{a,b,b+1,d|b+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(1.3) \quad c = b, d = b$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 1|0) = (a, b+1, b+1, b+1|b).$$

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{a,b,b,b|b} + \sum M_{a,b,b,b+1|b+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(1.4) \quad d = b, c \neq b, c \neq b+1$$

$$(1.4.1) \quad a < c < b$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|1).$$

$$P_\mu = 2M_{b,b,a,c|b+1} + M_{b,a,c,b|b+1} + M_{a,b,b,c|b+1} + M_{c,b,a,b|b+1} + M_{b,c,b,a|b+1} + 2M_{b,b,c,a|b+1} + M_{a,b,c,b|b+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b+1,b,a,c|b+1} + 2M_{b,b+1,a,c|b+1} + 2M_{b,b,a,c|b} + M_{b+1,a,c,b|b+1} + M_{b,a,c,b+1|b+1} + M_{b,a,c,b|b} \\ &= \sum M_{a,b,c,b|b} + \sum M_{b,b,a,c|b} + \sum M_{a,b,c,b+1|b+1} + \sum M_{b,b+1,a,c|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+1,b,a,c|b+1}, M_{b,b+1,a,c|b+1}, M_{b,b,a,c|b}\}$. 根据 [table 27](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{b,b,a,c|b}$ 和 $P_{b,b,c,a|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b,a,c|b}$ 和 $M_{b,b,c,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, b+2|b+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 [Theorem 2.1](#) 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1) = P_\nu = 2M_{b+2,b,a,c|b+1} + 2M_{b,b+2,a,c|b+1} + 2M_{b+2,b,c,a|b+1} + 2M_{b+2,b,a,c|b+1}.$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c, b+1|b+1) + (0, 0, 0, 1|0)$ 得到. 而根据 [Proposition 2.0.1](#), $M_{a,b,c,b+1|b+1} \in \text{Filt}(P_\lambda)$, 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b+2, b, a, c|b+1)$ 只能由 $(b+1, b, a, c|b+1) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b+1,b,a,c|b+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{b,b+1,a,c|b+1}, M_{b+1,b,c,a|b+1}, M_{b+1,b,a,c|b+1}$ 在 P_λ 重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{b,b,a,c|b}, M_{b,b,c,a|b}\}$. [Theorem 3.1](#) 说明 U' 不能包含其他不可分解投射模, 进一步说明除了 $P_\lambda, Pr_\lambda(P_\mu \otimes V_1)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

$$(1.4.2) \quad a = c < b$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|1).$$

$$P_\mu = 2M_{b,b,a,a|b+1} + M_{b,a,a,b|b+1} + M_{a,b,b,a|b+1} + M_{a,b,a,b|b+1} + M_{b,a,b,a|b+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b+1,b,a,a|b+1} + 2M_{b,b+1,a,a|b+1} + 2M_{b,b,a,a|b} + M_{b+1,a,a,b|b+1} + M_{b,a,a,b+1|b+1} + M_{b,a,a,b|b} \\ &= \sum M_{a,b,a,b|b} + M_{b,b,a,a|b} + \sum M_{a,b,a,b+1|b+1} + \sum M_{b,b+1,a,a|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+1,b,a,a|b+1}, M_{b,b+1,a,a|b+1}, M_{b,b,a,a|b}\}$. We have $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$ by [table 28](#).

根据 [table 28](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{b,b,a,a|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b,a,a|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a, b+2|b+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 [Theorem 2.1](#) 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1) = P_\nu = 2M_{b+2,b,a,a|b+1} + 2M_{b,b+2,a,a|b+1} + \dots$$

Table 27: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b+1,b,a,c b+1}$	$M_{b+2,b,a,c b+2}, M_{b+2,b,c,a b+2}$
$P_{b,b+1,a,c b+1}$	$M_{b,b+2,a,c b+2}, M_{b+2,b,a,c b+2}, M_{b,b+2,c,a b+2}, M_{b+2,b,c,a b+2}$
$P_{b,b,a,c b}$	$\emptyset$
$P_{b+1,b,c,a b+1}$	$M_{b+2,b,c,a b+2}$
$P_{b,b+1,c,a b+1}$	$M_{b,b+2,c,a b+2}, M_{b+2,b,c,a b+2}$
$P_{b,b,c,a b}$	$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, a, b+1|b+1) + (0, 0, 0, 1|0)$ 得到. 而根据 [Proposition 2.0.1](#), $M_{a,b,a,b+1|b+1} \in \text{Filt}(P_\lambda)$, 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(b+2, b, a, a|b+1)$ 只能由 $(b+1, b, a, a|b+1) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b+1,b,a,a|b+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{b,b+1,a,a|b+1}$ 在 P_λ 重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{b,b,a,a|b}\}$. [Theorem 3.1](#)说明 U' 不能包含其他不可分解投射模, 进一步说明除了 $P_\lambda, Pr_\lambda(P_\mu \otimes V_1)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 28: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b+1,b,a,a b+1}$	$M_{b+2,b,a,a b+2}$
$P_{b,b+1,a,a b+1}$	$M_{b,b+2,a,a b+2}, M_{b+2,b,a,a b+2}$
$P_{b,b,a,a b}$	$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.4.3) *Others* Let $\mu := \lambda + (0, 1, 0, 1|0) = (a, b+1, c, b+1|b)$.

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{a,b,c,b|b} + \sum M_{a,b,c,b+1|b+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(1.5) $d = b, c = b+1$

(1.5.1) $a < b$

Let $\mu := \lambda + (0, 1, 1, 1|0)$.

$$P_\mu = M_{b+1,a,b+2,b+1|b} + M_{b+2,b+1,a,b+1|b} + M_{b+1,b+2,b+1,a|b} + M_{a,b+2,b+1,b+1|b} + M_{a,b+1,b+2,b+1|b} + M_{b+1,b+2,b+1,b|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{b,a,b+1,b|b} + M_{b,a,b+1,b+1|b+1} + 2M_{b+1,a,b+1,b|b+1} + M_{b,a,b+2,b+1|b+2} + M_{b+1,a,b+2,b|b+2} \\ &= \sum M_{a,b,b+1,b|b} + \sum M_{a,b,b+1,b+1|b+1} + \sum M_{a,b+1,b+1,b|b+1} + \sum M_{a,b,b+2,b+1|b+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,b+1,b+1,b|b+1}, M_{b+1,b+1,a,b|b+1}, M_{b+1,b+1,b+1,b|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 29](#).

Table 29: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a,b+1,b+1,b b+1}$	$M_{a,b+2,b+1,b b+2}, M_{a,b+1,b+2,b b+2}, M_{b+2,a,b+1,b b+2}, M_{b+1,b+2,a,b b+2},$ $M_{b,b+2,b+1,a b+2}, M_{b+2,b+1,a,b b+2}, M_{b+2,b,b+1,a b+2}, M_{b+1,b+2,b,a b+2},$ $M_{b+2,b+1,b,a b+2}, M_{b+1,a,b+2,b b+2}, M_{b,b+1,b+2,a b+2}, M_{b+1,b,b+2,a b+2}$
$P_{b+1,b+1,a,b b+1}$	$M_{b+2,b+1,a,b b+2}, M_{b+1,b+2,a,b b+2}, M_{b+2,b+1,b,a b+2}, M_{b+1,b+2,b,a b+2}$
$P_{b+1,b+1,b,a b+1}$	$M_{b+2,b+1,b,a b+2}, M_{b+1,b+2,b,a b+2}$
$P_{b+1,a,b+1,b b+1}$	$M_{b+2,a,b+1,b b+2}, M_{b+1,a,b+2,b b+2}, M_{b+2,b+1,a,b b+2}, M_{b+2,b,b+1,a b+2},$ $M_{b+2,b+1,b,a b+2}, M_{b+1,b+2,a,b b+2}, M_{b+1,b,b+2,a b+2}, M_{b+1,b+2,b,a b+2}$
$P_{b,b+1,b+1,a b+1}$	$M_{b,b+2,b+1,a b+2}, M_{b,b+1,b+2,a b+2}, M_{b+2,b,b+1,a b+2}, M_{b+1,b+2,b,a b+2},$ $M_{b+2,b+1,b,a b+2}, M_{b+1,b,b+2,a b+2}$
$P_{b+1,b,b+1,a b+1}$	$M_{b+2,b,b+1,a b+2}, M_{b+1,b,b+2,a b+2}, M_{b+2,b+1,b,a b+2}, M_{b+1,b+2,b,a b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.5.2) \quad a > b + 2$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 1|0).$$

$$P_\mu = M_{a,b+1,b+2,b+1|b} + M_{a,b+2,b+1,b+1|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{a,b,b+1,b|b} + M_{a,b,b+1,b+1|b+1} + 2M_{a,b+1,b+1,b|b+1} + M_{a,b,b+2,b+1|b+2} + M_{a,b+1,b+2,b|b+2} \\ &= \sum M_{a,b,b+1,b|b} + \sum M_{a,b,b+1,b+1|b+1} + \sum M_{a,b,b+2,b+1|b+2} + M_{a,b+1,b+1,b|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,b+1,b+1,b|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 30](#).

Table 30: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{a,b+1,b+1,b b+1}$	$M_{a,b+2,b+1,b b+2}, M_{a,b+1,b+2,b b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.5.3) \quad a = b + 2$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|3).$$

$$P_\mu = M_{b+2,b,b+1,b|b+3} + M_{b+2,b+1,b,b|b+3}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_9) &= M_{b+3,b+1,b+2,b|b+3} + M_{b+3,b,b+2,b+1|b+3} + M_{b+2,b+1,b+2,b|b+2} + M_{b+2,b,b+2,b+1|b+2} + 2M_{b+2,b,b+1,b|b+2} \\ &= \sum M_{b+2,b,b+1,b|b} + \sum M_{b+2,b,b+1,b+1|b+1} + \sum M_{b+2,b,b+2,b+1|b+2} + \sum M_{b+2,b+1,b+1,b|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+3,b+1,b+2,b|b+3}, M_{b+3,b,b+2,b+1|b+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_9) = P_\lambda$ by [table 31](#).

Table 31: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_9))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_9))$ (via Prop. ~ 1)
$P_{b+3,b+1,b+2,b b+3}$	$M_{b+4,b+1,b+2,b b+4}, M_{b+4,b+2,b+1,b b+4}$
$P_{b+3,b,b+2,b+1 b+3}$	$M_{b+4,b,b+2,b+1 b+4}, M_{b+4,b+2,b,b+1 b+4}, M_{b+4,b+1,b+2,b b+4}, M_{b+4,b+2,b+1,b b+4}$
$P_{b+3,b+2,b+1,b b+3}$	$M_{b+4,b+2,b+1,b b+4}$
$P_{b+3,b+2,b,b+1 b+3}$	$M_{b+4,b+2,b,b+1 b+4}, M_{b+4,b+2,b+1,b b+4}$
$P_{b+2,b+1,b+1,b b+1}$	$M_{b+2,b+2,b+1,b b+2}, M_{b+2,b+1,b+2,b b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(2) $a = b + 1$

(2.1) $c \neq b, d \neq b$

(2.1.1) $c < b, d > b + 2$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{b+2,d,c,b+1|b} + M_{d,b+2,b+1,c|b} + M_{d,b+1,c,b+2|b} + M_{b+2,b+1,d,c|b} + M_{d,b+2,c,b+1|b} + M_{b+2,d,b+1,c|b} + \dots$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,d,c,b|b} + M_{b+1,d,c,b+1|b+1} + M_{b+2,d,c,b+1|b+2} + M_{d,b+1,b,c|b} + 2M_{d,b+1,b+1,c|b+1} + M_{d,b+1,b+1,c|b+2} \\ &= \sum M_{b+1,b,c,d|b} + \sum M_{b+1,b+1,c,d|b+1} + \sum M_{d,b+1,c,b+1|b+1} + \sum M_{b+2,b+1,c,d|b+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d,b+1,c,b+1|b+1}, M_{d,b+1,b+1,c|b+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 32.

Table 32: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d,b+1,c,b+1 b+1}$	$M_{d,b+2,c,b+1 b+2}, M_{d,b+1,c,b+2 b+2}, M_{d,b+2,b+1,c b+2}, M_{d,b+1,b+2,c b+2}$
$P_{d,b+1,b+1,c b+1}$	$M_{d,b+2,b+1,c b+2}, M_{d,b+1,b+2,c b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(2.1.2) $c < b, d = b + 2$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{b+2,b+2,c,b+1|b} + M_{b+2,b+1,b+2,c|b} + M_{b+2,b+2,b+1,c|b} + M_{b+2,b+1,c,b+2|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b+2,c,b|b} + M_{b+2,b+1,c,b|b} + M_{b+1,b+2,c,b+1|b+1} + 2M_{b+2,b+1,c,b+1|b+1} + M_{b+2,b+2,c,b+1|b+2} \\ &= \sum M_{b+1,b,c,b+2|b} + \sum M_{b+1,b+1,c,b+2|b+1} + \sum M_{b+2,b+1,c,b+1|b+1} + \sum M_{b+2,b+1,c,b+1|b+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,c|b+1}, M_{b+2,b+1,c,b+1|b+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 33.

(2.1.3) $c = b + 1, d = b + 2$

Table 33: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,c b+1}$	$M_{b+2,b+2,b+1,c b+2}, M_{b+2,b+1,b+2,c b+2}, M_{b+3,b+2,b+1,c b+3}, M_{b+3,b+1,b+2,c b+3}$
$P_{b+2,b+1,c,b+1 b+1}$	$M_{b+2,b+2,c,b+1 b+2}, M_{b+2,b+1,c,b+2 b+2}, M_{b+3,b+2,c,b+1 b+3}, M_{b+3,b+1,c,b+2 b+3},$ $M_{b+2,b+2,b+1,c b+2}, M_{b+2,b+1,b+2,c b+2}, M_{b+3,b+2,b+1,c b+3}, M_{b+3,b+1,b+2,c b+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

直接使用 $P_{b+1,b,b+1,b+2|b}$ 的结果, 它包含在本情况下item (2.1.15)Others 的情况中。Let $\mu := \lambda + (0, 0, 0, -1|0)$.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{b+2,b,b+1,b+1|b} + M_{b+1,b,b+2,b+1|b} + M_{b+1,b,b+1,b+2|b} + M_{b+2,b+1,b,b+1|b} + M_{b+1,b+2,b,b+1|b} \\ &= \sum M_{b+1,b,b+1,b+2|b} + \sum M_{b+1,b+1,b+1,b+2|b+1} + M_{b+2,b+1,b+1,b+1|b+1} + \sum M_{b+2,b+1,b+1,b+1|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,b+1|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$ by table 34.

Table 34: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,b+1 b+1}$	$M_{b+2,b+2,b+1,b+1 b+2}, M_{b+2,b+1,b+2,b+1 b+2}, M_{b+2,b+1,b+1,b+2 b+2}, M_{b+3,b+2,b+1,b+1 b+3},$ $M_{b+3,b+1,b+2,b+1 b+3}, M_{b+3,b+1,b+1,b+2 b+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.1.4) \quad c = b + 1, d > b + 2$$

结合上一种情况item (2.1.3), 我们进行归纳 Let $\mu := \lambda + (0, 0, 0, -1|0)$.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{d,b,b+1,b+1|b} + M_{b+1,b,d,b+1|b} + M_{b+1,b,b+1,d|b} + M_{d,b+1,b,b+1|b} + M_{b+1,d,b,b+1|b} + M_{b+1,b+1,d,b+1|b} \\ &= \sum M_{b+1,b,b+1,d|b} + \sum M_{b+1,b+1,b+1,d|b+1} + \sum M_{b+2,b+1,b+1,d|b+2} + M_{d,b+1,b+1,b+1|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d,b+1,b+1,b+1|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$ by table 35.

Table 35: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{d,b+1,b+1,b+1 b+1}$	$M_{d,b+2,b+1,b+1 b+2}, M_{d,b+1,b+2,b+1 b+2}, M_{d,b+1,b+1,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.1.5) \quad d > c > b + 2$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$\begin{aligned} P_\mu &= M_{b+2,b+1,d,c|b} + M_{c,d,b+1,b+2|b} + M_{d,c,b+2,b+1|b} + M_{c,b+1,b+2,d|b} + M_{d,b+2,b+1,c|b} + M_{b+2,d,c,b+1|b} + M_{b+1,b+2,d,c|b} \\ Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b,d,c|b} + M_{b+1,b+1,d,c|b+1} + M_{b+2,b+1,d,c|b+2} + M_{c,d,b,b+1|b} + 2M_{c,d,b+1,b+1|b+1} + M_{b+1,b+1,d,c|b+1} \\ &= \sum M_{b+1,b,c,d|b} + \sum M_{b+1,b+1,c,d|b+1} + \sum M_{b+2,b+1,c,d|b+2} + \sum M_{c,b+1,b+1,d|b+1} \end{aligned}$$

Table 36: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d,c,b+1,b+1 b+1}$	$M_{d,c,b+2,b+1 b+2}, M_{d,c,b+1,b+2 b+2}$
$P_{c,d,b+1,b+1 b+1}$	$M_{c,d,b+2,b+1 b+2}, M_{c,d,b+1,b+2 b+2}, M_{d,c,b+2,b+1 b+2}, M_{d,c,b+1,b+2 b+2}$
$P_{c,b+1,d,b+1 b+1}$	$M_{c,b+2,d,b+1 b+2}, M_{c,b+1,d,b+2 b+2}, M_{d,b+2,c,b+1 b+2}, M_{c,d,b+2,b+1 b+2},$ $M_{d,c,b+2,b+1 b+2}, M_{d,b+1,c,b+2 b+2}, M_{c,d,b+1,b+2 b+2}, M_{d,c,b+1,b+2 b+2}$
$P_{d,b+1,c,b+1 b+1}$	$M_{d,b+2,c,b+1 b+2}, M_{d,b+1,c,b+2 b+2}, M_{d,c,b+2,b+1 b+2}, M_{d,c,b+1,b+2 b+2}$
$P_{c,b+1,b+1,d b+1}$	$M_{c,b+2,b+1,d b+2}, M_{c,b+1,b+2,d b+2}, M_{d,b+2,b+1,c b+2}, M_{c,d,b+1,b+2 b+2},$ $M_{c,b+2,d,b+1 b+2}, M_{d,c,b+1,b+2 b+2}, M_{d,b+2,c,b+1 b+2}, M_{c,d,b+2,b+1 b+2},$ $M_{d,c,b+2,b+1 b+2}, M_{d,b+1,b+2,c b+2}, M_{c,b+1,d,b+2 b+2}, M_{d,b+1,c,b+2 b+2}$
$P_{d,b+1,b+1,c b+1}$	$M_{d,b+2,b+1,c b+2}, M_{d,b+1,b+2,c b+2}, M_{d,c,b+1,b+2 b+2}, M_{d,b+2,c,b+1 b+2},$ $M_{d,c,b+2,b+1 b+2}, M_{d,b+1,c,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Similarly, the set of indeterminate composition factors is $U = \{M_{d,c,b+1,b+1|b+1}, M_{c,d,b+1,b+1|b+1}, M_{c,b+1,d,b+1|b+1},$
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 36](#).

$$(2.1.6) \quad d > c = b + 2$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{b+2,b+1,d,b+2|b} + M_{d,b+2,b+1,b+2|b} + M_{b+2,d,b+2,b+1|b} + M_{b+2,b+2,b+1,d|b} + M_{b+2,b+1,b+2,d|b} + M_{b+2,b+1,b+1,d|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b,d,b+2|b} + M_{b+2,b,d,b+1|b} + M_{b+1,b+1,d,b+2|b+1} + 2M_{b+2,b+1,d,b+1|b+1} + M_{b+2,b+1,d,b+1|b+1} \\ &= \sum M_{b+1,b,b+2,d|b} + \sum M_{b+1,b+1,b+2,d|b+1} + \sum M_{b+2,b+1,b+1,d|b+1} + \sum M_{b+2,b+1,b+1,d|b} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,d|b+1}, M_{b+2,d,b+1,b+1|b+1}, M_{c,b+1,b+1,d|b+1},$
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 37](#).

$$(2.1.7) \quad d = c > b + 2$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{c,c,b+1,b+2|b} + M_{c,b+1,b+2,c|b} + M_{b+2,c,b+1,c|b} + M_{c,b+2,c,b+1|b} + M_{b+2,b+1,c,c|b} + M_{c,c,b+2,b+1|b} + M_{c,b+1,b+2,c|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c,c,b,b+1|b} + 2M_{c,c,b+1,b+1|b+1} + M_{c,c,b+1,b+2|b+2} + M_{c,b,b+1,c|b} + 2M_{c,b+1,b+1,c|b+1} + M_{c,b+1,b+1,c|b+1} \\ &= \sum M_{b+1,b,c,c|b} + \sum M_{b+1,b+1,c,c|b+1} + \sum M_{b+2,b+1,c,c|b+2} + \sum M_{c,b+1,b+1,c|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,c,b+1,b+1|b+1}, M_{c,b+1,c,b+1|b+1}, M_{c,b+1,b+1,c|b+1},$
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 38](#).

$$(2.1.8) \quad d = c = b + 2$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{b+2,b+2,b+1,b+2|b} + M_{b+2,b+1,b+2,b+2|b} + M_{b+2,b+2,b+2,b+1|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b+2,b,b+2|b} + M_{b+2,b+1,b,b+2|b} + M_{b+2,b+2,b,b+1|b} + M_{b+1,b+2,b+1,b+2|b+1} + 2M_{b+2,b+1,b+1,b+2|b+1} \\ &= \sum M_{b+1,b,b+2,b+2|b} + \sum M_{b+1,b+1,b+2,b+2|b+1} + \sum M_{b+2,b+1,b+1,b+2|b+1} + \sum M_{b+2,b+1,b+1,b+2|b} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,b+2|b+1}, M_{b+2,b+1,b+2,b+1|b+1}, M_{b+2,b+1,b+2,b+1|b},$
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 39](#).

Table 37: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,d b+1}$	$M_{b+2,b+2,b+1,d b+2}, M_{b+2,b+1,b+2,d b+2}, M_{b+3,b+2,b+1,d b+3}, M_{b+3,b+1,b+2,d b+3},$ $M_{d,b+2,b+1,b+2 b+2}, M_{b+2,d,b+1,b+2 b+2}, M_{b+2,b+2,d,b+1 b+2}, M_{d,b+2,b+2,b+1 b+2},$ $M_{b+2,d,b+2,b+1 b+2}, M_{d,b+1,b+2,b+2 b+2}, M_{b+2,b+1,d,b+2 b+2}, M_{d,b+2,b+1,b+3 b+3},$ $M_{b+3,d,b+1,b+2 b+3}, M_{b+3,b+2,d,b+1 b+3}, M_{d,b+3,b+1,b+2 b+3}, M_{d,b+2,b+3,b+1 b+3},$ $M_{b+3,d,b+2,b+1 b+3}, M_{d,b+3,b+2,b+1 b+3}, M_{d,b+1,b+2,b+3 b+3}, M_{b+3,b+1,d,b+2 b+3},$ $M_{d,b+1,b+3,b+2 b+3}$
$P_{b+2,d,b+1,b+1 b+1}$	$M_{b+2,d,b+2,b+1 b+2}, M_{b+2,d,b+1,b+2 b+2}, M_{b+3,d,b+2,b+1 b+3}, M_{b+3,d,b+1,b+2 b+3},$ $M_{d,b+2,b+2,b+1 b+2}, M_{d,b+2,b+1,b+2 b+2}, M_{d,b+3,b+2,b+1 b+3}, M_{d,b+3,b+1,b+2 b+3}$
$P_{d,b+2,b+1,b+1 b+1}$	$M_{d,b+2,b+2,b+1 b+2}, M_{d,b+2,b+1,b+2 b+2}, M_{d,b+3,b+2,b+1 b+3}, M_{d,b+3,b+1,b+2 b+3}$
$P_{d,b+1,b+1,b+2 b+1}$	$M_{d,b+2,b+1,b+2 b+2}, M_{d,b+1,b+2,b+2 b+2}, M_{d,b+2,b+2,b+1 b+2}$
$P_{d,b+1,b+2,b+1 b+1}$	$M_{d,b+2,b+2,b+1 b+2}, M_{d,b+1,b+2,b+2 b+2}, M_{d,b+1,b+3,b+2 b+3}, M_{d,b+2,b+1,b+2 b+2},$ $M_{d,b+3,b+1,b+2 b+3}, M_{d,b+2,b+3,b+1 b+3}, M_{d,b+3,b+2,b+1 b+3}$
$P_{b+2,b+1,d,b+1 b+1}$	$M_{b+2,b+2,d,b+1 b+2}, M_{b+2,b+1,d,b+2 b+2}, M_{b+3,b+2,d,b+1 b+3}, M_{b+3,b+1,d,b+2 b+3},$ $M_{d,b+2,b+2,b+1 b+2}, M_{b+2,d,b+2,b+1 b+2}, M_{d,b+1,b+2,b+2 b+2}, M_{b+2,d,b+1,b+2 b+2},$ $M_{d,b+2,b+1,b+2 b+2}, M_{d,b+2,b+3,b+1 b+3}, M_{b+3,d,b+2,b+1 b+3}, M_{d,b+3,b+2,b+1 b+3},$ $M_{d,b+1,b+3,b+2 b+3}, M_{b+3,d,b+1,b+2 b+3}, M_{d,b+3,b+1,b+2 b+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 38: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c,c,b+1,b+1 b+1}$	$M_{c,c,b+2,b+1 b+2}, M_{c,c,b+1,b+2 b+2}$
$P_{c,b+1,c,b+1 b+1}$	$M_{c,b+2,c,b+1 b+2}, M_{c,b+1,c,b+2 b+2}, M_{c,c,b+2,b+1 b+2}, M_{c,c,b+1,b+2 b+2}$
$P_{c,b+1,b+1,c b+1}$	$M_{c,b+2,b+1,c b+2}, M_{c,b+1,b+2,c b+2}, M_{c,c,b+1,b+2 b+2}, M_{c,b+2,c,b+1 b+2},$ $M_{c,c,b+2,b+1 b+2}, M_{c,b+1,c,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 39: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,b+2 b+1}$	$M_{b+2,b+2,b+1,b+2 b+2}, M_{b+2,b+1,b+2,b+2 b+2}, M_{b+3,b+2,b+1,b+2 b+3}, M_{b+3,b+1,b+2,b+2 b+3},$ $M_{b+2,b+2,b+2,b+1 b+2}, M_{b+3,b+2,b+2,b+1 b+3}$
$P_{b+2,b+1,b+2,b+1 b+1}$	$M_{b+2,b+2,b+2,b+1 b+2}, M_{b+2,b+1,b+2,b+2 b+2}, M_{b+3,b+2,b+2,b+1 b+3}, M_{b+3,b+1,b+2,b+2 b+3},$ $M_{b+2,b+1,b+3,b+2 b+3}, M_{b+2,b+2,b+1,b+2 b+2}, M_{b+3,b+2,b+1,b+2 b+3}, M_{b+2,b+3,b+1,b+2 b+3},$ $M_{b+2,b+2,b+3,b+1 b+3}, M_{b+2,b+3,b+2,b+1 b+3}$
$P_{b+2,b+2,b+1,b+1 b+1}$	$M_{b+2,b+2,b+2,b+1 b+2}, M_{b+2,b+2,b+1,b+2 b+2}, M_{b+3,b+2,b+2,b+1 b+3}, M_{b+2,b+3,b+2,b+1 b+3},$ $M_{b+3,b+2,b+1,b+2 b+3}, M_{b+2,b+3,b+1,b+2 b+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.1.9) \quad b+2 < d < c$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$\begin{aligned} P_\mu &= M_{b+2,b+1,c,d|b} + M_{d,c,b+1,b+2|b} + 2M_{c,d,b+2,b+1|b} + M_{c,b+2,b+1,d|b} + M_{b+2,c,d,b+1|b} + M_{c,b+1,d,b+2|b} + \dots \\ pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b,c,d|b} + M_{b+1,b+1,c,d|b+1} + M_{b+2,b+1,c,d|b+2} + M_{d,c,b,b+1|b} + 2M_{d,c,b+1,b+1|b+1} + M_{b+2,b+1,c,d|b+1} + \dots \\ &= \sum M_{b+1,b,c,d|b} + \sum M_{b+1,b+1,c,d|b+1} + \sum M_{b+2,b+1,c,d|b+2} + \sum M_{d,b+1,c,b+1|b+1} + \dots \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,d,b+1,b|b}, 2M_{c,d,b+1,b+1|b+1}, M_{c,d,b+1,b+1|b+1}, \dots\}$. 根据 table 40, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,d,b+1,b|b}$ 或 $P_{c,b+1,d,b|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_6)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c,d,b+1,b|b}, M_{c,b+1,d,b|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可. 令 $\nu = (b+1, b, c, d|b-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1) = P_\nu = 2M_{c,d,b+1,b|b-1} + 2M_{c,b+1,d,b|b-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(b+1, b, c, d|b) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c, d, b+1, b|b-1)$ 只能由 $(c, d, b+1, b|b) + (0, 0, 0, 0|-1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c,d,b+1,b|b}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{c,b+1,d,b|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2. 综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

$$(2.1.10) \quad b+2 = d < c$$

Let $\mu := \lambda + (0, 0, 1, 1|0)$.

$$\begin{aligned} P_\mu &= M_{b+2,b+1,c,b+2|b} + M_{c,b+2,b+1,b+2|b} + M_{b+2,c,b+2,b+1|b} + M_{b+2,c,b+1,b+2|b} + 2M_{c,b+2,b+2,b+1|b} + M_{c,b+1,b+2,b+1|b} + \dots \\ pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b,c,b+2|b} + M_{b+2,b,c,b+1|b} + M_{b+1,b+1,c,b+2|b+1} + 2M_{b+2,b+1,c,b+1|b+1} + M_{b+2,b+1,c,b+1|b+1} + \dots \\ &= \sum M_{b+1,b,c,b+2|b} + \sum M_{b+1,b+1,c,b+2|b+1} + \sum M_{c,b+1,b+2,b|b} + \sum M_{c,b+1,b+1,b+2|b+1} + \dots \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,c,b+1,b+1|b+1}, 2M_{c,b+2,b+1,b+1|b+1}, \dots\}$. 根据 table 41, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,b+2,b+1,b|b}$ 或 $P_{c,b+1,b+2,b|b}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_6)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c,b+2,b+1,b|b}, M_{c,b+1,b+2,b|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (b+1, b, c, b+2|b-1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1) = P_\nu = 2M_{c,b+2,b+1,b|b-1} + 2M_{c,b+1,b+2,b|b-1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(b+1, b, c, b+2|b) + (0, 0, 0, 0|-1)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c, b+2, b+1, b|b-1)$ 只

Table 40: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c,d,b+1,b b}$	$\emptyset$
$M_{c,d,b+1,b+1 b+1}$	$M_{c,d,b+1,b+2 b+2}$
$P_{c,d,b+2,b+1 b+2}$	$M_{c,d,b+3,b+1 b+3}$
$P_{d,c,b+1,b+1 b+1}$	$M_{d,c,b+2,b+1 b+2}, M_{d,c,b+1,b+2 b+2}, M_{c,d,b+1,b+2 b+2}$
$P_{d,b+1,c,b+1 b+1}$	$M_{d,b+2,c,b+1 b+2}, M_{d,b+1,c,b+2 b+2}, M_{d,c,b+2,b+1 b+2}, M_{c,b+1,d,b+2 b+2},$ $M_{d,c,b+1,b+2 b+2}, M_{c,d,b+1,b+2 b+2}$
$M_{c,b+1,d,b+1 b+1}$	$M_{c,b+1,d,b+2 b+2}, M_{c,d,b+1,b+2 b+2}$
$P_{c,b+1,d,b b}$	$\emptyset$
$P_{c,b+2,d,b+1 b+2}$	$M_{c,b+3,d,b+1 b+3}, M_{c,d,b+3,b+1 b+3}$
$P_{c,b+1,b+1,d b+1}$	$M_{c,b+2,b+1,d b+2}, M_{c,b+1,b+2,d b+2}, M_{c,d,b+1,b+2 b+2}, M_{c,b+1,d,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

能由 $(c, b+2, b+1, b|b) + (0, 0, 0, 0| - 1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c,b+2,b+1,b|b}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{c,b+1,b+2,b|b}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$.

$$(2.1.11) \quad c > b+2, d = b+1$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{b+2,c,b+1,b+1|b} + M_{c,b+1,b+2,b+1|b} + M_{c,b+2,b+1,b+1|b} + M_{b+2,b+1,c,b+1|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,c,b,b+1|b} + M_{b+1,c,b+1,b|b} + M_{b+1,c,b+1,b+1|b+1} + M_{b+2,c,b+1,b+1|b+2} + M_{c,b,b+1,b+1|b+1} \\ &= \sum M_{b+1,b,c,b+1|b} + M_{c,b+1,b+1,b|b} + \sum M_{b+1,b+1,c,b+1|b+1} + M_{c,b+1,b+1,b+1|b+1} + \sum \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,b+1,b+1,b|b}, M_{c,b+1,b+1,b+1|b+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 42.

$$(2.1.12) \quad c = b+2, d = b+1$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{b+2,b+2,b+1,b+1|b} + M_{b+2,b+1,b+2,b+1|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b+2,b,b+1|b} + M_{b+1,b+2,b+1,b|b} + M_{b+2,b+1,b,b+1|b} + 2M_{b+2,b+1,b+1,b|b} + M_{b+1,b+2,b+1,b|b} \\ &= \sum M_{b+1,b,b+2,b+1|b} + M_{b+2,b+1,b+1,b|b} + \sum M_{b+1,b+1,b+2,b+1|b+1} + M_{b+2,b+1,b+1,b+1|b+1} + \sum \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,b|b}, M_{b+2,b+1,b+1,b+1|b+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 43.

$$(2.1.13) \quad c > b+2, d < b$$

Let $\mu := \lambda + (1, 1, 0, 0|0)$.

$$P_\mu = M_{c,b+2,b+1,d|b} + M_{b+2,b+1,c,d|b} + M_{b+2,c,b+1,d|b} + M_{c,b+1,b+2,d|b}$$

Table 41: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b+2,c,b+1,b+1 b+1}$	$M_{b+2,c,b+2,b+1 b+2}, M_{b+2,c,b+1,b+2 b+2}, M_{b+3,c,b+2,b+1 b+3}, M_{b+3,c,b+1,b+2 b+3},$ $M_{c,b+2,b+1,b+2 b+2}, M_{c,b+3,b+2,b+1 b+3}, M_{c,b+3,b+1,b+2 b+3}$
$2M_{c,b+2,b+1,b+1 b+1}$	$M_{c,b+2,b+1,b+2 b+2}, M_{c,b+3,b+2,b+1 b+3}, M_{c,b+3,b+1,b+2 b+3}$
$P_{c,b+1,b+2,b b}$	$\emptyset$
$P_{c,b+2,b+1,b b}$	$\emptyset$
$2M_{c,b+1,b+2,b+1 b+1}$	$M_{c,b+1,b+2,b+2 b+2}, M_{c,b+1,b+3,b+2 b+3}, M_{c,b+2,b+1,b+2 b+2}, M_{c,b+3,b+1,b+2 b+3},$ $M_{c,b+2,b+3,b+1 b+3}, M_{c,b+3,b+2,b+1 b+3}$
$P_{c,b+2,b+2,b+1 b+2}$	$M_{c,b+3,b+2,b+1 b+3}, M_{c,b+2,b+3,b+1 b+3}$
$P_{c,b+1,b+1,b+2 b+1}$	$M_{c,b+2,b+1,b+2 b+2}, M_{c,b+1,b+2,b+2 b+2}$
$P_{b+2,b+1,c,b+1 b+1}$	$M_{b+2,b+2,c,b+1 b+2}, M_{b+2,b+1,c,b+2 b+2}, M_{b+3,b+2,c,b+1 b+3}, M_{b+3,b+1,c,b+2 b+3},$ $M_{b+2,c,b+2,b+1 b+2}, M_{c,b+1,b+2,b+2 b+2}, M_{b+2,c,b+1,b+2 b+2}, M_{c,b+2,b+1,b+2 b+2},$ $M_{c,b+2,b+3,b+1 b+3}, M_{b+3,c,b+2,b+1 b+3}, M_{c,b+3,b+2,b+1 b+3}, M_{c,b+1,b+3,b+2 b+3},$ $M_{b+3,c,b+1,b+2 b+3}, M_{c,b+3,b+1,b+2 b+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 42: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c,b+1,b+1,b b}$	$M_{c,b+2,b+1,b+1 b+2}, M_{c,b+1,b+2,b+1 b+2}$
$P_{c,b+1,b+1,b+1 b+1}$	$M_{c,b+2,b+1,b+1 b+2}, M_{c,b+1,b+2,b+1 b+2}, M_{c,b+1,b+1,b+2 b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 43: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,b b}$	$M_{b+2,b+2,b+1,b+1 b+2}, M_{b+2,b+1,b+2,b+1 b+2}$
$P_{b+2,b+1,b+1,b+1 b+1}$	$M_{b+2,b+2,b+1,b+1 b+2}, M_{b+2,b+1,b+2,b+1 b+2}, M_{b+2,b+1,b+1,b+2 b+2}, M_{b+3,b+2,b+1,b+1 b+3},$ $M_{b+3,b+1,b+2,b+1 b+3}, M_{b+3,b+1,b+1,b+2 b+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$\begin{aligned}
Pr_\lambda(P_\mu \otimes V_6) &= M_{c,b+1,b,d|b} + 2M_{c,b+1,b+1,d|b+1} + M_{c,b+2,b+1,d|b+2} + M_{b+1,b,c,d|b} + M_{b+1,b+1,c,d|b+1} + M_{b+1,b+2,c,d|b+2} \\
&= \sum M_{b+1,b,c,d|b} + \sum M_{b+1,b+1,c,d|b+1} + \sum M_{b+2,b+1,c,d|b+2} + M_{c,b+1,b+1,d|b+1}
\end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,b+1,b+1,d|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 44](#).

Table 44: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c,b+1,b+1,d b+1}$	$M_{c,b+2,b+1,d b+2}, M_{c,b+1,b+2,d b+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$\begin{aligned}
(2.1.14) \quad &c = b + 2, d < b \\
&\text{Let } \mu := \lambda + (1, 1, 0, 0|0).
\end{aligned}$$

$$P_\mu = M_{b+2,b+1,b+2,d|b} + M_{b+2,b+2,b+1,d|b}$$

$$\begin{aligned}
Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b,b+2,d|b} + M_{b+2,b,b+1,d|b} + M_{b+1,b+1,b+2,d|b+1} + 2M_{b+2,b+1,b+1,d|b+1} + M_{b+2,b+1,b+2,d|b+2} \\
&= \sum M_{b+1,b,b+2,d|b} + \sum M_{b+1,b+1,b+2,d|b+1} + M_{b+2,b+1,b+1,d|b+1} + \sum M_{b+2,b+1,b+2,d|b+2}
\end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,d|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 45](#).

Table 45: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,d b+1}$	$M_{b+2,b+2,b+1,d b+2}, M_{b+2,b+1,b+2,d b+2}, M_{b+3,b+2,b+1,d b+3}, M_{b+3,b+1,b+2,d b+3}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$\begin{aligned}
(2.1.15) \quad &\text{Others} \\
&\text{Let } \mu := \lambda + (1, 1, 0, 0|0) = (b + 2, b + 1, c, d|b).
\end{aligned}$$

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{b+1,b,c,d|b} + \sum M_{b+1,b+1,c,d|b+1} + \sum M_{b+2,b+1,c,d|b+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$\begin{aligned}
(2.2) \quad &c = b, d = b \\
&\text{Let } \mu := \lambda + (0, 0, 0, 0|2) = (b + 1, b, b, b|b + 2).
\end{aligned}$$

$$Pr_\lambda(P_\mu \otimes V_5) = M_{b+1,b,b,b|b} + \sum M_{b+1,b,b,b+1|b+1} + \sum M_{b+2,b,b,b+1|b+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2.3) \quad c = b, d \neq b$$

$$\begin{aligned}
(2.3.1) \quad &d \leq b + 1 \\
&\text{Let } \mu := \lambda + (1, 1, 1, 0|0) = (b + 2, b + 1, b + 1, d|b).
\end{aligned}$$

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{b+1,b,b,d|b} + \sum M_{b+1,b,b+1,d|b+1} + \sum M_{b+2,b,b+1,d|b+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2.3.2) \quad d > b + 1$$

$$\text{Let } \mu := \lambda + (1, 1, 1, 0|0).$$

$$P_\mu = M_{b+2,d,b+1,b+1|b} + M_{b+2,b+1,b+1,d|b} + M_{d,b+1,b+2,b+1|b} + M_{d,b+2,b+1,b+1|b} + M_{b+2,b+1,d,b+1|b} + M_{d,b+1,b+2,b+1|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{b+1,d,b,b|b} + M_{b+1,d,b,b+1|b+1} + M_{b+1,d,b+1,b|b+1} + M_{b+2,d,b,b+1|b+2} + M_{b+2,d,b+1,b|b+2} \\ &= \sum M_{b+1,b,b,d|b} + \sum M_{b+1,b,b+1,d|b+1} + \sum M_{b+2,b,b+1,d|b+2} + \sum M_{d,b,b+1,b+1|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d,b+1,b+1,b|b+1}, M_{d,b,b+1,b+1|b+1}, M_{d,b+1,b+1,b+1|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 46](#).

Table 46: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d,b+1,b+1,b b+1}$	$M_{d,b+2,b+1,b b+2}, M_{d,b+1,b+2,b b+2}$
$P_{d,b,b+1,b+1 b+1}$	$M_{d,b,b+2,b+1 b+2}, M_{d,b,b+1,b+2 b+2}, M_{d,b+2,b,b+1 b+2}, M_{d,b+1,b+2,b b+2},$ $M_{d,b+2,b+1,b b+2}, M_{d,b+1,b,b+2 b+2}$
$P_{d,b+1,b,b+1 b+1}$	$M_{d,b+2,b,b+1 b+2}, M_{d,b+1,b,b+2 b+2}, M_{d,b+2,b+1,b b+2}, M_{d,b+1,b+2,b b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.4) \quad c \neq b, d = b$$

$$(2.4.1) \quad c < b$$

$$\text{Let } \mu := \lambda + (1, 1, 0, 1|0).$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{b+1,b,c,b|b} + M_{b+1,b,c,b+1|b+1} + M_{b+1,b+1,c,b|b+1} + M_{b+2,b,c,b+1|b+2} + M_{b+2,b+1,c,b|b+2} \\ &= \sum M_{b+1,b,c,b|b} + \sum M_{b+1,b,c,b+1|b+1} + \sum M_{b+2,b,c,b+1|b+2} \end{aligned}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2.4.2) \quad c = b + 1$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|2).$$

$$P_\mu = M_{b+1,b+1,b,b|b+2} + M_{b+1,b,b+1,b|b+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= 2M_{b+2,b+1,b+1,b|b+2} + M_{b+2,b+1,b,b+1|b+2} + M_{b+1,b+2,b+1,b|b+2} + M_{b+1,b+2,b,b+1|b+2} + 2M_{b+1,b+1,b+1,b|b+2} \\ &= \sum M_{b+1,b,b+1,b|b} + \sum M_{b+1,b,b+1,b+1|b+1} + \sum M_{b+1,b,b+2,b+1|b+2} + \sum M_{b+1,b+1,b+1,b|b+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b+2,b+1,b+1,b|b+2}, M_{b+1,b+1,b+1,b|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_5) = P_\lambda$ by [table 47](#).

Table 47: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_5))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_5))$ (via Prop. ~ 1)
$P_{b+2,b+1,b+1,b b+2}$	$M_{b+3,b+1,b+1,b b+3}$
$P_{b+1,b+1,b+1,b b+1}$	$M_{b+1,b+2,b+1,b b+2}, M_{b+1,b+1,b+2,b b+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(2.4.3) $c > b + 1$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{b+2,c,b+1,b+1|b} + M_{c,b+1,b+2,b+1|b} + M_{c,b+2,b+1,b+1|b} + M_{b+2,b+1,c,b+1|b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{b+1,c,b,b|b} + M_{b+1,c,b,b+1|b+1} + M_{b+1,c,b+1,b|b+1} + M_{b+2,c,b,b+1|b+2} + M_{b+2,c,b+1,b|b+2} \\ &= \sum M_{b+1,b,c,b|b} + \sum M_{b+1,b,c,b+1|b+1} + M_{c,b+1,b+1,b|b+1} + \sum M_{b+2,b,c,b+1|b+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,b+1,b+1,b|b+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 48](#).

Table 48: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c,b+1,b+1,b b+1}$	$M_{c,b+2,b+1,b b+2}, M_{c,b+1,b+2,b b+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

□

Theorem 3.3. Let $a, b, c, d \in \mathbb{Z}$ and $\lambda = (a, b, c, d|c)$ with $c \notin \{a, b\}$. The projective objects P_λ in $\mathfrak{B}_{a,b,d}$ have the following Verma flag formulae.

Proof. The Cases are as follows:

(1) $d \neq c$

(1.1) $a \neq c+1, b \neq c+1$

(1.1.1) $a < c, c+1 < b < d$

Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$P_\mu = 2M_{b,d,a,c|c+1} + 2M_{d,b,c,a|c+1} + M_{b,a,c,d|c+1} + M_{d,c,a,b|c+1} + M_{a,b,d,c|c+1} + M_{c,d,b,a|c+1} + M_{c,b,a,d|c+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b,d,a,c+1|c+1} + 2M_{b,d,a,c|c} + 2M_{d,b,c+1,a|c+1} + 2M_{d,b,c,a|c} + M_{b,a,c+1,d|c+1} + M_{b,a,c,d|c} \\ &= \sum M_{a,b,c,d|c} + \sum M_{b,d,a,c|c} + \sum M_{a,b,c+1,d|c+1} + \sum M_{b,d,a,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,a,c+1|c+1}, M_{b,d,a,c|c}, M_{d,b,c+1,a|c+1}, M_{d,b,c,a|c}\}$. 根据 table 49, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,d,a,c|c}$ 或 $P_{d,b,a,c|c}$ 或 $P_{b,d,c,a|c}$ 或 $P_{d,b,c,a|c}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,c|c}, M_{d,b,a,c|c}, M_{b,d,c,a|c}, M_{d,b,c,a|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, d|c+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2) = 2P_\nu = 4M_{b,d,a,c|c+1} + 4M_{d,b,a,c|c+1} + 4M_{b,d,c,a|c+1} + 4M_{d,b,c,a|c+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(a, b, c, d|c) + (0, 0, 0, 0|1)$ 和 $(a, b, c+1, d|c+1) + (0, 0, -1, 0|0)$ 得到. 由 Proposition 2.0.1, $M_{(a,b,c+1,d|c+1)} \subset \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$. 又由于 $(b, d, a, c|c+1)$ 只能由 $(b, d, a, c|c) + (0, 0, 0, 0|1)$ 和 $(b, d, a, c+1|c+1) + (0, 0, 0, -1|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,c|c}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,c|c}, M_{b,d,c,a|c}, M_{d,b,c,a|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 49: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b,d,a,c+1 c+1}$	$M_{b,d,a,c+2 c+2}, M_{d,b,a,c+2 c+2}, M_{b,d,c+2,a c+2}, M_{d,b,c+2,a c+2}$
$P_{b,d,a,c c}$	$\emptyset$
$P_{d,b,c+1,a c+1}$	$M_{d,b,c+2,a c+2}$
$P_{d,b,c,a c}$	$\emptyset$
$P_{d,b,a,c+1 c+1}$	$M_{d,b,a,c+2 c+2}, M_{d,b,c+2,a c+2}$
$P_{d,b,a,c c}$	$\emptyset$
$P_{b,d,c+1,a c+1}$	$M_{b,d,c+2,a c+2}, M_{d,b,c+2,a c+2}$
$P_{b,d,c,a c}$	$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.1.2) $a < c, c + 1 < b = d$
 Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$\begin{aligned} P_\mu &= 2M_{b,b,a,c|c+1} + M_{b,a,c,b|c+1} + M_{a,b,b,c|c+1} + M_{c,b,a,b|c+1} + M_{b,c,b,a|c+1} + 2M_{b,b,c,a|c+1} + M_{a,b,c,b|c+1} \\ Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b,b,a,c+1|c+1} + 2M_{b,b,a,c|c} + M_{b,a,c+1,b|c+1} + M_{b,a,c,b|c} + M_{a,b,b,c+1|c+1} + M_{a,b,b,c|c} + \\ &= \sum M_{a,b,c,b|c} + \sum M_{b,b,a,c|c} + \sum M_{a,b,c+1,b|c+1} + \sum M_{b,b,a,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b,a,c+1|c+1}, M_{b,b,a,c|c}, M_{b,b,c+1,a|c+1}\}$. 根据 table 50, 仅仅由 Proposition 2.0.1 不足以判断 $P_{b,b,a,c|c}$ 或 $P_{b,b,c,a|c}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,b,a,c|c}, M_{b,b,c,a|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可. 令 $\nu = (a, b, c, b|c + 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2) = 2P_\nu = 4M_{b,b,a,c|c+1} + 4M_{b,b,c,a|c+1} \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(a, b, c, b|c) + (0, 0, 0, 0|1)$ 和 $(a, b, c+1, b|c+1) + (0, 0, -1, 0|0)$ 得到. 由 Proposition 2.0.1, $M_{(a,b,c+1,b|c+1)} \subset \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$. 又由于 $(b, b, a, c|c+1)$ 只能由 $(b, b, a, c|c) + (0, 0, 0, 0|1)$ 和 $(b, b, a, c+1|c+1) + (0, 0, 0, -1|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,b,a,c|c}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{b,b,c,a|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 50: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b,b,a,c+1 c+1}$		$M_{b,b,a,c+2 c+2}, M_{b,b,c+2,a c+2}$
$P_{b,b,a,c c}$		$\emptyset$
$P_{b,b,c+1,a c+1}$		$M_{b,b,c+2,a c+2}$
$P_{b,b,c,a c}$		$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.1.3) $b < a < d < c$
 Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$\begin{aligned} P_\mu &= M_{a,b,c,d|c+1} + M_{d,c,b,a|c+1} + 2M_{c,d,a,b|c+1} + M_{c,a,b,d|c+1} + M_{a,c,d,b|c+1} + M_{c,b,d,a|c+1} + M_{d,a,c,b|c+1} \\ Pr_\lambda(P_\mu \otimes V_1) &= M_{a,b,c+1,d|c+1} + M_{a,b,c,d|c} + M_{d,c+1,b,a|c+1} + M_{d,c,b,a|c} + 2M_{c+1,d,a,b|c+1} + 2M_{c,d,a,b|c} + \\ &= \sum M_{a,b,c,d|c} + \sum M_{a,b,c+1,d|c+1} + \sum M_{c,a,d,b|c} + \sum M_{c+1,a,d,b|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,d,a,b|c+1}, M_{c,d,a,b|c}, M_{c+1,a,d,b|c+1}\}$. 根据 table 51, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,d,a,b|c}$ 或 $P_{c,a,d,b|c}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c,d,a,b|c}, M_{c,a,d,b|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可. 令 $\nu = (a, b, c+2, d|c+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1) = P_\nu = 2M_{c+2,d,a,b|c+1} + 2M_{c+2,a,d,b|c+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c+1, d|c+1) + (0, 0, 1, 0|0)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c+2, d, a, b|c+1)$ 只能由 $(c+2, d, a, b|c+1) + (0, 0, 0, 0|0)$ 得到.

$1, d, a, b|c+1) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c+1,d,a,b|c+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{c+1,a,d,b|c+1}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{c,d,a,b|c}, M_{c,a,d,b|c}\}$. 我们发现, 但是 [Proposition 2.0.1](#) 表明 $M_{c+1,d,a,b|c+1} \in \text{Filt}(P_{c,d,a,b|c})$ 和 $M_{c+1,a,d,b|c+1} \in \text{Filt}(P_{c,a,d,b|c})$. 这说明 U' 不能包含 $\text{Filt}(P_{c,a,d,b|c})$ 和 $\text{Filt}(P_{c,d,a,b|c})$, 进一步说明除了 P_λ , $Pr_\lambda(P_\mu \otimes V_1)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 51: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{c+1,d,a,b c+1}$		$M_{c+2,d,a,b c+2}$
$P_{c,d,a,b c}$		$\emptyset$
$P_{c+1,a,d,b c+1}$		$M_{c+2,a,d,b c+2}, M_{c+2,d,a,b c+2}$
$P_{c,a,d,b c}$		$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.1.4) \quad b < a = d < c$$

Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$P_\mu = M_{a,b,c,a|c+1} + M_{c,a,b,a|c+1} + M_{a,c,a,b|c+1} + M_{a,c,b,a|c+1} + 2M_{c,a,a,b|c+1} + M_{c,b,a,a|c+1} + M_{a,a,c,b|c+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a,b,c+1,a|c+1} + M_{a,b,c,a|c} + M_{c+1,a,b,a|c+1} + M_{c,a,b,a|c} + M_{a,c+1,a,b|c+1} + M_{a,c,a,b|c} + M_{a,b,c+1,a|c} \\ &= \sum M_{a,b,c,a|c} + \sum M_{a,b,c+1,a|c+1} + M_{c,a,a,b|c} + M_{c+1,a,a,b|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,a,a,b|c+1}, M_{c,a,a,b|c}\}$.

根据 [table 52](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{c,a,a,b|c}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c,a,a,b|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c+2, a|c+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1)$, 结合 [Theorem 2.1](#) 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_1) = P_\nu = 2M_{c+2,a,a,b|c+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, c+1, a|c+1) + (0, 0, 1, 0|0)$ 得到. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c+2, a, a, b|c+1)$ 只能由 $(c+1, a, a, b|c+1) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c+1,a,a,b|c+1}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的 $\text{Filt}(P_\nu)$.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{c,a,a,b|c}\}$. 我们发现, 但是 [Proposition 2.0.1](#) 表明 $M_{c+1,a,a,b|c+1} \in \text{Filt}(P_{c,a,a,b|c})$. 这说明 U' 不能包含 $\text{Filt}(P_{c,a,a,b|c})$, 进一步说明除了 P_λ , $Pr_\lambda(P_\mu \otimes V_1)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

$$(1.1.5) \quad \text{Others not}\{(a < c, c+1 < b \leq d), (a < c, b < a \leq d < c)\}$$

$$(1.2) \quad a \neq c+1, b = c+1$$

$$(1.2.1) \quad a > c+2, d \leq c+1$$

Let $\mu := \lambda + (0, 1, 1, 0|0) = (a, b+1, c+1, c+1|c)$.

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{a,c+1,c,d|c} + M_{a,c+1,c+1,d|c+1} + M_{a,c+2,c+1,d|c+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

Table 52: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{c+1,a,a,b c+1}$		$M_{c+2,a,a,b c+2}$
$P_{c,a,a,b c}$		$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.2) \quad a > c + 2, d > c + 1$$

$$(1.2.2.1) \quad d > a > c + 2$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 0|0).$$

$$P_\mu = M_{a,d,c+1,c+2|c} + M_{d,a,c+2,c+1|c} + M_{d,c+2,c+1,a|c} + M_{a,c+2,d,c+1|c} + M_{d,a,c+1,c+2|c} + M_{a,d,c+2,c+1|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{a,d,c,c+1|c} + 2M_{a,d,c+1,c+1|c+1} + M_{a,d,c+1,c+2|c+2} + M_{d,a,c+1,c|c} + 2M_{d,a,c+1,c+1|c} \\ &= \sum M_{a,c+1,c,d|c} + \sum M_{a,c+1,c+1,d|c+1} + \sum M_{a,d,c+1,c+1|c+1} + \sum M_{a,c+2,c+1,d|c} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d,a,c+1,c+1|c+1}, M_{a,d,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 53](#).

Table 53: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d,a,c+1,c+1 c+1}$		$M_{d,a,c+2,c+1 c+2}, M_{d,a,c+1,c+2 c+2}$
$P_{a,d,c+1,c+1 c+1}$	$M_{a,d,c+2,c+1 c+2}, M_{a,d,c+1,c+2 c+2}, M_{d,a,c+2,c+1 c+2}, M_{d,a,c+1,c+2 c+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.2.2) \quad d = a > c + 2$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 0|0).$$

$$P_\mu = M_{a,a,c+1,c+2|c} + M_{a,c+2,a,c+1|c} + M_{a,a,c+2,c+1|c} + M_{a,c+2,c+1,a|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{a,a,c,c+1|c} + 2M_{a,a,c+1,c+1|c+1} + M_{a,a,c+1,c+2|c+2} + M_{a,c+1,a,c|c} + M_{a,c+1,a,c+1|c} \\ &= \sum M_{a,c+1,c,a|c} + \sum M_{a,c+1,c+1,a|c+1} + M_{a,a,c+1,c+1|c+1} + \sum M_{a,c+2,c+1,a|c+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,a,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 54](#).

Table 54: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{a,a,c+1,c+1 c+1}$	$M_{a,a,c+2,c+1 c+2}, M_{a,a,c+1,c+2 c+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.2.3) \quad a > d > c + 2$$

Let $\mu := \lambda + (0, 1, 1, 0|0)$.

$$P_\mu = M_{a,d,c+2,c+1|c} + M_{a,c+2,c+1,d|c} + M_{a,d,c+1,c+2|c} + M_{a,c+2,d,c+1|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{a,d,c+1,c|c} + 2M_{a,d,c+1,c+1|c+1} + M_{a,d,c+2,c+1|c+2} + M_{a,c+1,c,d|c} + M_{a,c+1,c+1,d|c+1} \\ &= \sum M_{a,c+1,c,d|c} + \sum M_{a,c+1,c+1,d|c+1} + \sum M_{a,c+2,c+1,d|c+2} + M_{a,d,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,d,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 55](#).

Table 55: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{a,d,c+1,c+1 c+1}$	$M_{a,d,c+2,c+1 c+2}$	$M_{a,d,c+1,c+2 c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.2.4) \quad a > d = c + 2$$

Let $\mu := \lambda + (0, 1, 1, 0|0)$.

$$P_\mu = M_{a,c+2,c+1,c+2|c} + M_{a,c+2,c+2,c+1|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{a,c+1,c,c+2|c} + M_{a,c+2,c,c+1|c} + M_{a,c+1,c+1,c+2|c+1} + 2M_{a,c+2,c+1,c+1|c+1} + M_{a,c+2,c+1,c+1|c+1} \\ &= \sum M_{a,c+1,c,c+2|c} + \sum M_{a,c+1,c+1,c+2|c+1} + M_{a,c+2,c+1,c+1|c+1} + \sum M_{a,c+2,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,c+2,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 56](#).

Table 56: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{a,c+2,c+1,c+1 c+1}$	$M_{a,c+2,c+2,c+1 c+2}$	$M_{a,c+2,c+1,c+2 c+2}, M_{a,c+3,c+2,c+1 c+3}, M_{a,c+3,c+1,c+2 c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.3) \quad a = c + 2$$

$$(1.2.3.1) \quad a = c + 2, d > c + 3 \text{ Let } \mu := \lambda + (1, 1, 1, 0|0).$$

$$P_\mu = M_{c+3,d,c+1,c+2|c} + M_{d,c+3,c+2,c+1|c} + M_{d,c+2,c+1,c+3|c} + M_{c+3,c+2,d,c+1|c} + M_{d,c+3,c+1,c+2|c} + M_{c+3,c+2,c+1,c+1|c+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{c+2,d,c,c+1|c} + 2M_{c+2,d,c+1,c+1|c+1} + M_{c+2,d,c+1,c+2|c+2} + M_{c+3,d,c+1,c+2|c+3} + M_{c+3,d,c+1,c+2|c+3} \\ &= \sum M_{c+2,c+1,c,d|c} + \sum M_{c+2,c+1,c+1,d|c+1} + \sum M_{c+2,d,c+1,c+1|c+1} + \sum M_{c+2,d,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+3,d,c+1,c+2|c+3}, M_{d,c+3,c+2,c+1|c+3}, M_{c+3,c+2,c+1,c+1|c+1}\}$. 根据 [table 57](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{d,c+2,c+1,c+1|c+1}$ 或 $P_{d,c+2,c+1,c+2|c+2}$ 或 $P_{d,c+2,c+2,c+1|c+2}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_{10})$ 的直和分解中. [table 57](#)也说明, 除了 P_λ 和上述三种可能的不可约直和分解项, $pr_\lambda(P_\lambda \otimes V_{10})$, 不可能有其他不可约的直和分解项. 又由于 Bruhat 序下, U 中 $(c+3, c+2, c+1, d|c+3)$ 并不大于 $(d, c+2, c+2, c+1|c+2)$, $(d, c+2, c+2, c+1|c+2)$, $(d, c+2, c+2, c+1|c+2)$ 任一项. 因此由 [Theorem 2.2](#) 可以确定的是 $M_{c+3,c+2,c+1,d|c+3}$ 不会出现在 $\text{Filt}(P_{d,c+2,c+1,c+1|c+1})$

或 $\text{Filt}(P_{d,c+2,c+1,c+2|c+2})$ 或 $\text{Filt}(P_{d,c+2,c+2,c+1|c+2})$, 进一步可以确定 $M_{c+3,c+2,c+1,d|c+3} \in \text{Filt}(P_\lambda)$.

Proposition 2.0.1 可以知道 $M_{d,c+2,c+2,c+1|c+2} \in \text{Filt}(P_\lambda)$ 和 $M_{d,c+2,c+1,c+2|c+2} \in \text{Filt}(P_\lambda)$, 我们先证明 $M_{d,c+2,c+2,c+1|c+2}$ 和 $M_{d,c+2,c+1,c+2|c+2}$ 在 $\text{Filt}(P_\lambda)$ 中的重数皆是 2.

令 $\nu = (c+2, c+2, c+1, d|c+3)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_{10})) \otimes V_2)$, 结合 **Theorem 2.1** 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_{10})) \otimes V_2) = 2P_\nu = 2\nu + 4M_{d,c+2,c+2,c+1|c+3} + 4M_{d,c+2,c+1,c+2|c+3} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_{10})) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(c+2, c+2, c+1, d|c+2) + (0, 0, 0, 0|1)$ 和 $(c+3, c+2, c+1, d|c+3) + (-1, 0, 0, 0|0)$ 得到. 上文和 **Proposition 2.0.1**, 已经分别确定 $M_{c+3,c+2,c+1,d|c+3}, M_{c+2,c+2,c+1,d|c+2} \in \text{Filt}(P_\lambda)$, 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$. 又由于 $(d, c+2, c+2, c+1|c+3)$ 只能由 $(d, c+2, c+2, c+1|c+2) + (0, 0, 0, 0|1)$ 和 $(d, c+3, c+2, c+1|c+3) + (0, -1, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中 $M_{d,c+2,c+2,c+1|c+2}$ 和 $M_{d,c+3,c+2,c+1|c+3}$ 的重数都是 2, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理 $M_{d,c+2,c+1,c+2|c+2}$ 和 $M_{d,c+3,c+1,c+2|c+3}$ 的重数也都是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{c+3,d,c+1,c+2|c+3}, M_{d,c+3,c+2,c+1|c+3}, M_{d,c+2,c+1,c+2|c+2}, M_{d,c+2,c+1,c+1|c+1}\}$. 我们发现, **Proposition 2.0.1** 表明 $M_{d,c+2,c+2,c+1|c+2} \in \text{Filt}(P_{d,c+1,c+1,c+1|c+1})$. 这说明 U' 不能包含 $\text{Filt}(P_{d,c+1,c+1,c+1|c+1})$, 进一步说明除了 $P_\lambda, Pr_\lambda(P_\mu \otimes V_{10})$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$.

Table 57: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+3,d,c+1,c+2 c+3}$	$M_{c+4,d,c+1,c+2 c+4}, M_{d,c+4,c+1,c+2 c+4}, M_{c+4,d,c+2,c+1 c+4}, M_{d,c+4,c+2,c+1 c+4}$
$P_{d,c+3,c+2,c+1 c+3}$	$M_{d,c+4,c+2,c+1 c+4}$
$P_{d,c+2,c+1,c+3 c+3}$	$M_{d,c+2,c+1,c+4 c+4}, M_{d,c+4,c+1,c+2 c+4}, M_{d,c+2,c+4,c+1 c+4}, M_{d,c+4,c+2,c+1 c+4}$
$P_{c+3,c+2,d,c+1 c+3}$	$M_{c+4,c+2,d,c+1 c+4}, M_{d,c+2,c+4,c+1 c+4}, M_{c+4,d,c+2,c+1 c+4}, M_{d,c+4,c+2,c+1 c+4}$
$P_{d,c+2,c+1,c+1 c+1}$	$\emptyset$
$P_{d,c+2,c+1,c+2 c+2}$	$\emptyset$
$P_{d,c+3,c+1,c+2 c+3}$	$M_{d,c+4,c+1,c+2 c+4}, M_{d,c+4,c+2,c+1 c+4}$
$P_{c+2,d,c+1,c+1 c+1}$	$M_{c+2,d,c+2,c+1 c+2}, M_{c+2,d,c+1,c+2 c+2}$
$P_{c+3,d,c+2,c+1 c+3}$	$M_{c+4,d,c+2,c+1 c+4}, M_{d,c+4,c+2,c+1 c+4}$
$P_{d,c+2,c+2,c+1 c+2}$	$\emptyset$
$P_{d,c+2,c+3,c+1 c+3}$	$M_{d,c+2,c+4,c+1 c+4}, M_{d,c+4,c+2,c+1 c+4}$
$P_{c+3,c+2,c+1,d c+3}$	$M_{c+4,c+2,c+1,d c+4}, M_{d,c+2,c+1,c+4 c+4}, M_{c+4,d,c+1,c+2 c+4}, M_{c+4,c+2,d,c+1 c+4},$ $M_{d,c+4,c+1,c+2 c+4}, M_{d,c+2,c+4,c+1 c+4}, M_{c+4,d,c+2,c+1 c+4}, M_{d,c+4,c+2,c+1 c+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.2.3.2) $a = c + 2, d = c + 3$ Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{c+3,c+3,c+1,c+2|c} + M_{c+3,c+2,c+3,c+1|c} + M_{c+3,c+3,c+2,c+1|c} + M_{c+3,c+2,c+1,c+3|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{c+2,c+3,c,c+1|c} + M_{c+3,c+2,c,c+1|c} + 2M_{c+2,c+3,c+1,c+1|c+1} + 2M_{c+3,c+2,c+1,c+1|c+1} \\ &= \sum M_{c+2,c+1,c,c+3|c} + \sum M_{c+2,c+1,c+1,c+3|c+1} + \sum M_{c+2,c+3,c+1,c+1|c+1} + \sum \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+3,c+3,c+1,c+2|c+3}, M_{c+3,c+2,c+1,c+3|c+3}, M_{c+3,c+2,c+1,c+1|c+1}\}$ 仅仅在 table 58 中, 并不能直接通过 Proposition 2.0.1 排除 $P_{c+3,c+2,c+1,c+1|c+1}$ 出现在 $Pr_\lambda(P_\mu \otimes V_{10})$ 的直和分解中的可能, 但是我们直接使用下文中 $P_{c+3,c+2,c+1,c+1|c+1}$ 的结构 (注意这里并不产生循环论证), 发现 $M_{c+4,c+3,c+2,c+1|c+4} \in \text{Filt}(P_{c+3,c+2,c+1,c+1|c+1}) \setminus U$, 排除了 $P_{c+3,c+2,c+1,c+1|c+1}$ 出现在 $Pr_\lambda(P_\mu \otimes V_{10})$ 的直和分解中的可能. 综上, $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$.

Table 58: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+3,c+3,c+1,c+2 c+3}$	$M_{c+4,c+3,c+1,c+2 c+4}, M_{c+3,c+4,c+1,c+2 c+4}, M_{c+4,c+3,c+2,c+1 c+4}, M_{c+3,c+4,c+2,c+1 c+4}$
$P_{c+3,c+2,c+3,c+1 c+3}$	$M_{c+4,c+2,c+3,c+1 c+4}, M_{c+3,c+2,c+4,c+1 c+4}, M_{c+4,c+3,c+2,c+1 c+4}, M_{c+3,c+4,c+2,c+1 c+4}$
$P_{c+2,c+3,c+1,c+1 c+1}$	$M_{c+2,c+3,c+2,c+1 c+2}, M_{c+2,c+3,c+1,c+2 c+2}$
$P_{c+3,c+2,c+1,c+1 c+1}$	$\emptyset$
$P_{c+3,c+2,c+2,c+1 c+2}$	$M_{c+4,c+3,c+2,c+1 c+4}, M_{c+4,c+2,c+3,c+1 c+4}$
$P_{c+3,c+3,c+2,c+1 c+3}$	$M_{c+4,c+3,c+2,c+1 c+4}, M_{c+3,c+4,c+2,c+1 c+4}$
$P_{c+3,c+2,c+1,c+2 c+2}$	$M_{c+4,c+3,c+1,c+2 c+4}, M_{c+4,c+2,c+1,c+3 c+4}, M_{c+4,c+3,c+2,c+1 c+4}, M_{c+4,c+2,c+3,c+1 c+4}$
$P_{c+3,c+2,c+1,c+3 c+3}$	$M_{c+4,c+2,c+1,c+3 c+4}, M_{c+3,c+2,c+1,c+4 c+4}, M_{c+4,c+3,c+1,c+2 c+4}, M_{c+4,c+2,c+3,c+1 c+4},$ $M_{c+4,c+3,c+2,c+1 c+4}, M_{c+3,c+4,c+1,c+2 c+4}, M_{c+3,c+2,c+4,c+1 c+4}, M_{c+3,c+4,c+2,c+1 c+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.2.3.3) $a = c + 2, d = c + 2$ Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{c+3,c+2,c+1,c+2|c} + M_{c+3,c+2,c+2,c+1|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{c+2,c+1,c,c+2|c} + M_{c+2,c+2,c,c+1|c} + M_{c+2,c+1,c+1,c+2|c+1} + 2M_{c+2,c+2,c+1,c+1|c+1} \\ &= \sum M_{c+2,c+1,c,c+2|c} + \sum M_{c+2,c+1,c+1,c+2|c+1} + M_{c+2,c+2,c+1,c+1|c+1} + \sum M_{c+2,c+2,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+3,c+2,c+1,c+2|c+3}, M_{c+2,c+2,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by table 59.

(1.2.3.4) $a = c + 2, d = c + 1$ Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{c+3,c+2,c+1,c+1|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{c+2,c+1,c,c+1|c} + M_{c+2,c+1,c+1,c|c} + M_{c+2,c+1,c+1,c+1|c+1} + M_{c+2,c+2,c+1,c+1|c+1} \\ &= \sum M_{c+2,c+1,c,c+1|c} + M_{c+2,c+1,c+1,c+1|c+1} + M_{c+2,c+2,c+1,c+1|c+1} + M_{c+3,c+2,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+3,c+2,c+1,c+1|c+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by table 60.

Table 59: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+3,c+2,c+1,c+2 c+3}$	$M_{c+4,c+2,c+1,c+2 c+4}, M_{c+4,c+2,c+2,c+1 c+4}$
$P_{c+2,c+2,c+1,c+1 c+1}$	$M_{c+2,c+2,c+2,c+1 c+2}, M_{c+2,c+2,c+1,c+2 c+2}, M_{c+2,c+3,c+2,c+1 c+3}, M_{c+2,c+3,c+1,c+2 c+3}$
$P_{c+3,c+2,c+2,c+1 c+3}$	$M_{c+4,c+2,c+2,c+1 c+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 60: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+3,c+2,c+1,c+1 c+3}$	$M_{c+4,c+2,c+1,c+1 c+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.3.5) \quad a = c + 2, d < c$$

$$\text{Let } \mu := \lambda + (1, 1, 1, 0|0).$$

$$P_\mu = M_{c+3,c+2,c+1,d|c}$$

$$Pr_\lambda(P_\mu \otimes V_{10}) = M_{c+2,c+1,c,d|c} + M_{c+2,c+1,c+1,d|c+1} + M_{c+2,c+2,c+1,d|c+2} + M_{c+3,c+2,c+1,d|c+3}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+3,c+2,c+1,d|c+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 61](#).

Table 61: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+3,c+2,c+1,d c+3}$	$M_{c+4,c+2,c+1,d c+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.2.4) \quad a < c$$

$$(1.2.4.1) \quad d < a < c$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 0|0).$$

$$P_\mu = M_{c+2,c+1,a,d|c} + M_{a,c+2,c+1,d|c} + M_{c+1,c+2,a,d|c} + M_{c+2,a,c+1,d|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c+1,c,a,d|c} + 2M_{c+1,c+1,a,d|c+1} + M_{c+2,c+1,a,d|c+2} + M_{a,c+1,c,d|c} + M_{a,c+1,c+1,d|c+1} \\ &= \sum M_{a,c+1,c,d|c} + \sum M_{a,c+1,c+1,d|c+1} + \sum M_{a,c+2,c+1,d|c+2} + M_{c+1,c+1,a,d|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+1,a,d|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 62](#).

$$(1.2.4.2) \quad a < d < c$$

$$\text{Let } \mu := \lambda + (0, 1, 1, 0|0).$$

$$P_\mu = M_{c+1,c+2,a,d|c} + M_{c+2,c+1,d,a|c} + M_{d,c+2,c+1,a|c} + M_{c+2,a,c+1,d|c} + M_{c+2,c+1,a,d|c} + M_{c+1,c+2,d,a|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c,c+1,a,d|c} + 2M_{c+1,c+1,a,d|c+1} + M_{c+1,c+2,a,d|c+2} + M_{c+1,c,d,a|c} + 2M_{c+1,c+1,d,a|c} \\ &= \sum M_{a,c+1,c,d|c} + \sum M_{a,c+1,c+1,d|c+1} + \sum M_{c+1,c+1,a,d|c+1} + \sum M_{a,c+2,c+1,d|c} \end{aligned}$$

Table 62: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+1,c+1,a,d c+1}$	$M_{c+2,c+1,a,d c+2}, M_{c+1,c+2,a,d c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+1,a,d|c+1}, M_{c+1,c+1,d,a|c+1}\}$.
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 63](#).

Table 63: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+1,c+1,a,d c+1}$	$M_{c+2,c+1,a,d c+2}, M_{c+1,c+2,a,d c+2}, M_{c+2,c+1,d,a c+2}, M_{c+1,c+2,d,a c+2}$
$P_{c+1,c+1,d,a c+1}$	$M_{c+2,c+1,d,a c+2}, M_{c+1,c+2,d,a c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.2.4.3) $a = d < c$

Let $\mu := \lambda + (0, 1, 1, 0|0)$.

$$P_\mu = M_{c+1,c+2,a,a|c} + M_{c+2,a,c+1,a|c} + M_{c+2,c+1,a,a|c} + M_{a,c+2,c+1,a|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c,c+1,a,a|c} + 2M_{c+1,c+1,a,a|c+1} + M_{c+1,c+2,a,a|c+2} + M_{c+1,a,c,a|c} + M_{c+1,a,c+1,a|c} \\ &= \sum M_{a,c+1,c,a|c} + \sum M_{a,c+1,c+1,a|c+1} + \sum M_{c+1,c+1,a,a|c+1} + \sum M_{a,c+2,c+1,a|c} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+1,a,a|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 64](#).

Table 64: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+1,c+1,a,a c+1}$	$M_{c+2,c+1,a,a c+2}, M_{c+1,c+2,a,a c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.2.4.4) $a < c, d = c + 1$

Let $\mu := \lambda + (0, 1, 1, 0|0)$.

$$P_\mu = M_{c+2,c+1,a,c+1|c} + M_{c+1,c+2,c+1,a|c} + M_{a,c+2,c+1,c+1|c} + M_{c+1,c+2,a,c+1|c} + M_{c+2,c+1,c+1,a|c} + M_{c+1,c+2,c+1,a|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c+1,c,a,c+1|c} + 2M_{c+1,c+1,a,c|c} + 2M_{c+1,c+1,a,c+1|c+1} + M_{c+2,c+1,a,c+1|c+2} + M_{c,c+1,a,c+1|c} \\ &= \sum M_{a,c+1,c,c+1|c} + \sum M_{a,c+1,c+1,c+1|c+1} + \sum M_{a,c+2,c+1,c+1|c+2} + \sum M_{c+1,c,c+1|c} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+1,a,c|c}, M_{c+1,c+1,a,c+1|c+1}\}$.
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 65](#).

(1.2.4.5) $a < c, d = c + 2$

Let $\mu := \lambda + (0, 1, 1, 0|0)$.

$$P_\mu = 2M_{c+2,c+2,a,c+1|c} + M_{c+2,a,c+1,c+2|c} + M_{a,c+2,c+2,c+1|c} + M_{c+1,c+2,a,c+2|c} + M_{c+2,c+1,c+2,a|c} + M_{c+1,c+2,c+1,a|c}$$

Table 65: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+1,c+1,a,c c}$	$M_{c+2,c+1,a,c+1 c+2}, M_{c+1,c+2,a,c+1 c+2}, M_{c+2,c+1,c+1,a c+2}, M_{c+1,c+2,c+1,a c+2}$
$P_{c+1,c+1,a,c+1 c+1}$	$M_{c+2,c+1,a,c+1 c+2}, M_{c+1,c+2,a,c+1 c+2}, M_{c+1,c+1,a,c+2 c+2}, M_{c+2,c+1,c+1,a c+2},$ $M_{c+1,c+2,c+1,a c+2}, M_{c+1,c+1,c+2,a c+2}$
$P_{c+1,c+1,c,a c}$	$M_{c+2,c+1,c+1,a c+2}, M_{c+1,c+2,c+1,a c+2}$
$P_{c+1,c+1,c+1,a c+1}$	$M_{c+2,c+1,c+1,a c+2}, M_{c+1,c+2,c+1,a c+2}, M_{c+1,c+1,c+2,a c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= 2M_{c+1,c+2,a,c|c} + 2M_{c+2,c+1,a,c|c} + 3M_{c+1,c+2,a,c+1|c+1} + 3M_{c+2,c+1,a,c+1|c+1} + 2M_{c+1,c+2,c+1,a|c+2} \\ &= \sum M_{a,c+1,c,c+2|c} + \sum M_{c+1,c+2,a,c|c} + \sum M_{a,c+1,c+1,c+2|c+1} + \sum M_{a,c+2,c+1,c|c} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+2,a,c|c}, M_{c+2,c+1,a,c|c}, 2M_{c+1,c+2,c+1,a|c+2}, 2M_{c+2,c+1,c+1,a|c+2}\}$. 根据 table 66, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c+1,c+2,a,c|c}$ 或 $P_{c+2,c+1,a,c|c}$ 或 $P_{c+1,c+2,c+1,a|c+2}$ 或 $P_{c+2,c+1,c+1,a|c+2}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_6)$ 的直和分解中.

我们使用 Theorem 3.4, 有 $M_{c+3,c+2,c+1,a|c+3}, M_{c+3,c+2,a,c+1|c+3} \in \text{Filt}(P_{c+2,c+1,a,c,c})$. 以及根据上文 item (1.2.3.5) 结论, $M_{c+3,c+2,a,c+1|c+3} \in \text{Filt}(P_{c+2,c+1,c,a,c})$. 这表明 U 中不可能包含一个完整的 $\text{Filt}(P_{c+2,c+1,a,c,c})$ 或 $\text{Filt}(P_{c+2,c+1,c,a,c})$.

接下来我们仅需排除 $P_{c+1,c+2,a,c|c}$ 或 $P_{c+1,c+2,c+1,a|c+2}$ 出现在 $pr_\lambda(P_\lambda \otimes V_6)$ 的直和分解中可能, 事实上, 我们仅需证明 $M_{c+1,c+2,a,c|c}, M_{c+1,c+2,c+1,a|c+2}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可. 令 $\nu = (a, c+2, c+1, c+2|c+3)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_2)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_2) = P_\nu = 2M_{c+2,c+2,a,c+1|c+3} + 4M_{c+2,c+2,c+1,a|c+3} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_2)$ 极小并且重数为 1, 且 ν 只能由 $(a, c+2, c+1, c+2|c+2) + (0, 0, 0, 0|1)$ 得到. 由 Proposition 2.0.1, $M_{(a,c+2,c+1,c+2|c+2)} \in \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_2) = P_\nu$. 又由于 $(c+2, c+2, a, c+1|c+3)$ 只能由 $(c+2, c+2, a, c+1|c+2) + (0, 0, 0, 0|1)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c+2,c+2,a,c+1|c+2}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{c+2,c+2,c+1,a|c+2}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{c+1,c+2,a,c|c}, M_{c+2,c+1,a,c|c}, 2M_{c+1,c+2,a,c+1|c+1}, 2M_{c+2,c+1,a,c+1|c+1}\}$. 我们发现, 但是 Proposition 2.0.1 表明 $M_{c+2,c+2,c+1,a|c+2} \in \text{Filt}(P_{c+1,c+2,c,a|c})$ 和 $M_{c+2,c+2,a,c+1|c+2} \in \text{Filt}(P_{c+1,c+2,a,c|c})$. 这说明 U' 不能包含 $\text{Filt}(P_{c+1,c+2,c,a|c})$ 和 $\text{Filt}(P_{c+1,c+2,a,c|c})$, 进一步说明除了 $P_\lambda, Pr_\lambda(P_\mu \otimes V_6)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

$$(1.2.4.6) \quad a < c, d > c+2$$

Let $\mu := \lambda + (0, 1, 1, 0|0)$.

$$P_\mu = 2M_{c+2,d,a,c+1|c} + 2M_{d,c+2,c+1,a|c} + M_{c+2,a,c+1,d|c} + M_{d,c+1,a,c+2|c} + M_{a,c+2,d,c+1|c} + M_{c+1,d,c+2,a|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= 2M_{c+1,d,a,c|c} + 3M_{c+1,d,a,c+1|c+1} + 2M_{c+2,d,a,c+1|c+2} + 2M_{d,c+1,c,a|c} + 3M_{d,c+1,c+1,a|c+2} \\ &= \sum M_{a,c+1,c,d|c} + \sum M_{c+1,d,a,c|c} + \sum M_{a,c+1,c+1,d|c+1} + \sum M_{a,c+2,c+1,d|c+2} + \dots \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,d,a,c|c}, M_{c+1,d,a,c+1|c+1}, M_{c+2,d,a,c+1|c+2}, M_{d,c+1,c,a|c}, 2M_{d,c+1,c+1,a|c+2}, 2M_{c+1,d,c+1,a|c+2}\}$. 根据 table 67, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c+1,d,a,c|c}$ 或 $P_{d,c+1,a,c|c}$ 或 $P_{c+1,d,c,a|c}$ 或 $P_{d,c+1,c,a|c}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_6)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{c+1,d,a,c|c}, M_{d,c+1,a,c|c}, M_{c+1,d,c,a|c}, M_{d,c+1,c,a|c}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

Table 66: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
	$P_{c+1,c+2,a,c c}$	$\emptyset$
	$P_{c+2,c+1,a,c c}$	$\emptyset$
$2M_{c+1,c+2,a,c+1 c+1}$		$M_{c+1,c+2,a,c+2 c+2}, M_{c+1,c+3,a,c+2 c+3}, M_{c+2,c+1,a,c+2 c+2}, M_{c+1,c+2,c+2,a c+2},$ $M_{c+2,c+1,c+2,a c+2}, M_{c+3,c+1,a,c+2 c+3}, M_{c+2,c+3,a,c+1 c+3}, M_{c+1,c+3,c+2,a c+3},$ $M_{c+3,c+2,a,c+1 c+3}, M_{c+3,c+1,c+2,a c+3}, M_{c+2,c+3,c+1,a c+3}, M_{c+3,c+2,c+1,a c+3}$
$2M_{c+2,c+1,a,c+1 c+1}$		$M_{c+2,c+1,a,c+2 c+2}, M_{c+3,c+2,a,c+1 c+3}, M_{c+3,c+1,a,c+2 c+3}, M_{c+2,c+1,c+2,a c+2},$ $M_{c+3,c+2,c+1,a c+3}, M_{c+3,c+1,c+2,a c+3}$
$P_{c+2,c+2,a,c+1 c+2}$		$M_{c+3,c+2,a,c+1 c+3}, M_{c+2,c+3,a,c+1 c+3}, M_{c+3,c+2,c+1,a c+3}, M_{c+2,c+3,c+1,a c+3}$
$2M_{c+2,c+1,c+1,a c+1}$		$M_{c+2,c+1,c+2,a c+2}, M_{c+3,c+2,c+1,a c+3}, M_{c+3,c+1,c+2,a c+3}$
	$P_{c+1,c+2,c,a c}$	$\emptyset$
	$P_{c+2,c+1,c,a c}$	$\emptyset$
$2M_{c+1,c+2,c+1,a c+1}$		$M_{c+1,c+2,c+2,a c+2}, M_{c+1,c+3,c+2,a c+3}, M_{c+2,c+1,c+2,a c+2}, M_{c+3,c+1,c+2,a c+3},$ $M_{c+2,c+3,c+1,a c+3}, M_{c+3,c+2,c+1,a c+3}$
$P_{c+2,c+2,c+1,a c+2}$		$M_{c+3,c+2,c+1,a c+3}, M_{c+2,c+3,c+1,a c+3}$
$P_{a,c+2,c+1,c+1 c+1}$		$M_{a,c+2,c+2,c+1 c+2}, M_{a,c+2,c+1,c+2 c+2}, M_{a,c+3,c+2,c+1 c+3}, M_{a,c+3,c+1,c+2 c+3},$ $M_{c+2,a,c+2,c+1 c+2}, M_{c+1,c+2,c+2,a c+2}, M_{c+2,c+1,c+2,a c+2}, M_{c+2,a,c+1,c+2 c+2},$ $M_{c+1,c+2,a,c+2 c+2}, M_{c+2,c+1,a,c+2 c+2}, M_{c+3,a,c+2,c+1 c+3}, M_{c+2,c+3,a,c+1 c+3},$ $M_{c+1,c+3,c+2,a c+3}, M_{c+3,c+2,a,c+1 c+3}, M_{c+3,c+1,c+2,a c+3}, M_{c+2,c+3,c+1,a c+3},$ $M_{c+3,c+2,c+1,a c+3}, M_{c+3,a,c+1,c+2 c+3}, M_{c+1,c+3,a,c+2 c+3}, M_{c+3,c+1,a,c+2 c+3}$
$P_{c+2,a,c+1,c+1 c+1}$		$M_{c+2,a,c+2,c+1 c+2}, M_{c+2,a,c+1,c+2 c+2}, M_{c+3,a,c+2,c+1 c+3}, M_{c+3,a,c+1,c+2 c+3},$ $M_{c+2,c+1,c+2,a c+2}, M_{c+2,c+1,a,c+2 c+2}, M_{c+3,c+2,a,c+1 c+3}, M_{c+3,c+1,c+2,a c+3},$ $M_{c+3,c+2,c+1,a c+3}, M_{c+3,c+1,a,c+2 c+3}$
$P_{c+1,c+1,a,c+2 c+1}$		$M_{c+2,c+1,a,c+2 c+2}, M_{c+1,c+2,a,c+2 c+2}, M_{c+2,c+1,c+2,a c+2}, M_{c+1,c+2,c+2,a c+2}$
$P_{c+1,c+1,c+2,a c+1}$		$M_{c+2,c+1,c+2,a c+2}, M_{c+1,c+2,c+2,a c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

情况 1: $d > c + 3$

令 $\nu = (a, c + 3, c + 1, d|c + 2)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$, 结合Theorem 2.1发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1) = P_\nu = 2M_{c+3,d,a,c+1|c+2} + 2M_{d,c+3,a,c+1|c+2} + 2M_{c+3,d,c+1,a|c+2} + 2M_{d,c+3,c+1,a|c+2}$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$ 极小并且重数为 1, 且 ν 只能由 $(a, c + 2, c + 1, d|c + 2) + (0, 1, 0, 0|0)$ 得到. 由Proposition 2.0.1, $M_{(a,c+2,c+1,d|c+2)} \in \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(c + 3, d, a, c + 1|c + 2)$ 只能由 $(c + 2, d, a, c + 1|c + 2) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c+2,d,a,c+1|c+2}$, 否则 $pr_\nu(P_\lambda \otimes V_6)$ 不可能包含完整的 $\text{Filt}(P_\nu)$. 同理, $M_{d,c+2,a,c+1|c+2}$, $M_{c+2,d,c+1,a|c+2}$, $M_{d,c+2,a,c+1|c+2}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{c+1,d,a,c|c}, M_{c+1,d,a,c+1|c+1}, M_{d,c+1,c,a|c}, M_{d,c+1,c,a,c+1|c+1}\}$. 我们发现, 但是Proposition 2.0.1表明 $M_{d,c+2,c+1,a|c+2} \in \text{Filt}(P_{c+1,d,a,c|c})$, $\text{Filt}(P_{c+1,d,c,a|c})$, $\text{Filt}(P_{d,c+1,a,c|c})$, $\text{Filt}(P_{d,c+1,c,a|c})$, 这说明 U' 不能包含 $\text{Filt}(P_{c+1,d,a,c|c})$, $\text{Filt}(P_{c+1,d,c,a|c})$, $\text{Filt}(P_{d,c+1,a,c|c})$, $\text{Filt}(P_{d,c+1,c,a|c})$, 进一步说明除了 P_λ , $Pr_\lambda(P_\mu \otimes V_6)$ 没有其他不可分解的直和项了.

情况 1: $d = c + 3$ 令 $\nu = (a, c + 3, c + 1, c + 3|c + 2)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$, 结合Theorem 2.1发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1) = 2P_\nu = 4M_{c+3,c+3,a,c+1|c+2} + 4M_{c+3,c+3,c+1,a|c+2} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$ 极小并且重数为 2, 且 ν 只能由 $(a, c + 2, c + 1, c + 3|c + 2) + (0, 1, 0, 0|0)$ 和 $(a, c + 3, c + 1, c + 2|c + 2) + (0, 0, 0, 1|0)$ 得到. 由Proposition 2.0.1, $M_{a,c+2,c+1,c+3|c+2}$, $M_{a,c+3,c+1,c+2|c+2} \in \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_1) = 2P_\nu$. 又由于 $(c + 3, c + 3, a, c + 1|c + 2)$ 只能由 $(c + 2, c + 3, a, c + 1|c + 2) + (1, 0, 0, 0|0)$ 和 $(c + 2, c + 3, a, c + 1|c + 2) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{c+3,c+2,a,c+1|c+2}$ 和两个 $M_{c+2,c+3,a,c+1|c+2}$, 否则 $pr_\nu(P_\lambda \otimes V_6)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{c+3,c+2,c+1,a|c+2}$, $M_{c+2,c+3,c+1,a|c+2}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

此时, 未确定的 Verma 模的多重集变成了 $U' = \{M_{c+1,d,a,c|c}, M_{c+1,d,a,c+1|c+1}, M_{d,c+1,c,a|c}, M_{d,c+1,c,a,c+1|c+1}\}$. 我们发现, 但是Proposition 2.0.1表明 $M_{d,c+2,c+1,a|c+2} \in \text{Filt}(P_{c+1,d,a,c|c})$, $\text{Filt}(P_{c+1,d,c,a|c})$, $\text{Filt}(P_{d,c+1,a,c|c})$, $\text{Filt}(P_{d,c+1,c,a|c})$, 这说明 U' 不能包含 $\text{Filt}(P_{c+1,d,a,c|c})$, $\text{Filt}(P_{c+1,d,c,a|c})$, $\text{Filt}(P_{d,c+1,a,c|c})$, $\text{Filt}(P_{d,c+1,c,a|c})$, 进一步说明除了 P_λ , $Pr_\lambda(P_\mu \otimes V_6)$ 没有其他不可分解的直和项了.

综上 $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$.

$$(1.3) \quad a = c + 1, b \neq c + 1$$

$$(1.3.1) \quad d \leq c + 1$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 0|0) = (c + 2, b, c + 1, c + 1|c).$$

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{c+1,b,c,d|c} + \sum M_{c+1,b,c+1,d|c+1} + \sum M_{c+2,b,c+1,d|c+2}$$

All items belong to P_λ by Proposition 2.0.1.

$$(1.3.2) \quad d > c + 1$$

$$(1.3.2.1) \quad b > d > c + 2$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 0|0).$$

$$P_\mu = M_{d,b,c+1,c+2|c} + M_{b,d,c+2,c+1|c} + M_{b,c+2,c+1,d|c} + M_{c+2,b,d,c+1|c} + M_{b,d,c+1,c+2|c} + M_{d,b,c+2,c+1|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d,b,c,c+1|c} + 2M_{d,b,c+1,c+1|c+1} + M_{d,b,c+1,c+2|c+2} + M_{b,d,c+1,c|c} + 2M_{b,d,c+1,c+1|c+1} \\ &= \sum M_{c+1,b,c,d|c} + \sum M_{c+1,b,c+1,d|c+1} + \sum M_{d,b,c+1,c+1|c+1} + \sum M_{c+2,b,c+1,d|c+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d,c+1,c+1|c+1}, M_{d,b,c+1,c+1|c+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 68.

$$(1.3.2.2) \quad b > d = c + 2$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 0|0).$$

$$P_\mu = M_{b,c+2,c+1,c+2|c} + M_{c+2,b,c+2,c+1|c} + M_{c+2,b,c+1,c+2|c} + M_{b,c+2,c+2,c+1|c}$$

Table 67: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
	$P_{c+1,d,a,c c}$	$\emptyset$
	$P_{c+1,d,a,c+1 c+1}$	$M_{c+1,d,a,c+2 c+2}, M_{d,c+1,a,c+2 c+2}, M_{c+1,d,c+2,a c+2}, M_{d,c+1,c+2,a c+2}$
	$P_{c+2,d,a,c+1 c+2}$	$M_{c+3,d,a,c+1 c+3}, M_{d,c+3,a,c+1 c+3}, M_{c+3,d,c+1,a c+3}, M_{d,c+3,c+1,a c+3}$
	$P_{d,c+1,c,a c}$	$\emptyset$
	$P_{d,c+1,c+1,a c+1}$	$M_{d,c+1,c+2,a c+2}$
	$P_{d,c+2,c+1,a c+2}$	$M_{d,c+3,c+1,a c+3}$
	$P_{d,c+1,a,c+1 c+1}$	$M_{d,c+1,a,c+2 c+2}, M_{d,c+1,c+2,a c+2}$
	$P_{d,c+1,a,c c}$	$\emptyset$
	$P_{d,c+2,a,c+1 c+2}$	$M_{d,c+3,a,c+1 c+3}, M_{d,c+3,c+1,a c+3}$
	$P_{c+1,d,c+1,a c+1}$	$M_{c+1,d,c+2,a c+2}, M_{d,c+1,c+2,a c+2}$
	$P_{c+1,d,c,a c}$	$\emptyset$
	$P_{c+2,d,c+1,a c+2}$	$M_{c+3,d,c+1,a c+3}, M_{d,c+3,c+1,a c+3}$
	$P_{c+1,c+1,a,d c+1}$	$M_{c+2,c+1,a,d c+2}, M_{c+1,c+2,a,d c+2}, M_{d,c+1,a,c+2 c+2}, M_{c+2,c+1,d,a c+2},$ $M_{d,c+1,c+2,a c+2}, M_{c+1,d,a,c+2 c+2}, M_{c+1,c+2,d,a c+2}, M_{c+1,d,c+2,a c+2}$
	$P_{d,a,c+1,c+1 c+1}$	$M_{d,a,c+2,c+1 c+2}, M_{d,a,c+1,c+2 c+2}, M_{d,c+1,c+2,a c+2}, M_{d,c+1,a,c+2 c+2}$
	$P_{a,d,c+1,c+1 c+1}$	$M_{a,d,c+2,c+1 c+2}, M_{a,d,c+1,c+2 c+2}, M_{d,a,c+2,c+1 c+2}, M_{c+1,d,c+2,a c+2},$ $M_{d,c+1,c+2,a c+2}, M_{d,a,c+1,c+2 c+2}, M_{c+1,d,a,c+2 c+2}, M_{d,c+1,a,c+2 c+2}$
	$P_{c+1,c+1,d,a c+1}$	$M_{c+2,c+1,d,a c+2}, M_{c+1,c+2,d,a c+2}, M_{d,c+1,c+2,a c+2}, M_{c+1,d,c+2,a c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 68: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
	$P_{b,d,c+1,c+1 c+1}$	$M_{b,d,c+2,c+1 c+2}, M_{b,d,c+1,c+2 c+2}$
	$P_{d,b,c+1,c+1 c+1}$	$M_{d,b,c+2,c+1 c+2}, M_{d,b,c+1,c+2 c+2}, M_{b,d,c+2,c+1 c+2}, M_{b,d,c+1,c+2 c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b,c+1,c,c+2|c} + M_{b,c+2,c,c+1|c} + M_{b,c+1,c+1,c+2|c+1} + 2M_{b,c+2,c+1,c+1|c+1} + M_{b,c+2,c+1,c+1|c+1} \\ &= \sum M_{c+1,b,c,c+2|c} + \sum M_{c+1,b,c+1,c+2|c+1} + \sum M_{c+2,b,c+1,c+1|c+1} + \sum M_{c+2,b,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+2,b,c+1,c+1|c+1}, M_{b,c+2,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 69](#).

Table 69: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+2,b,c+1,c+1 c+1}$	$M_{c+2,b,c+2,c+1 c+2}, M_{c+2,b,c+1,c+2 c+2}, M_{c+3,b,c+2,c+1 c+3}, M_{c+3,b,c+1,c+2 c+3},$ $M_{b,c+2,c+2,c+1 c+2}, M_{b,c+2,c+1,c+2 c+2}, M_{b,c+3,c+2,c+1 c+3}, M_{b,c+3,c+1,c+2 c+3}$
$P_{b,c+2,c+1,c+1 c+1}$	$M_{b,c+2,c+2,c+1 c+2}, M_{b,c+2,c+1,c+2 c+2}, M_{b,c+3,c+2,c+1 c+3}, M_{b,c+3,c+1,c+2 c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.3.2.3) $b = d > c + 2$

Let $\mu := \lambda + (1, 0, 1, 0|0)$.

$$\begin{aligned} P_\mu &= M_{b,b,c+1,c+2|c} + M_{c+2,b,c+1,b|c} + M_{b,c+2,b,c+1|c} + M_{b,b,c+2,c+1|c} + M_{b,c+2,c+1,b|c} + M_{c+2,b,b,c+1|c} \\ Pr_\lambda(P_\mu \otimes V_6) &= M_{b,b,c,c+1|c} + 2M_{b,b,c+1,c+1|c+1} + M_{b,b,c+1,c+2|c+2} + M_{c+1,b,c,b|c} + M_{c+1,b,c+1,b|c+1} \\ &= \sum M_{c+1,b,c,b|c} + \sum M_{c+1,b,c+1,b|c+1} + \sum M_{c+2,b,c+1,b|c+2} + M_{b,b,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,b,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 70](#).

Table 70: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{b,b,c+1,c+1 c+1}$	$M_{b,b,c+2,c+1 c+2}, M_{b,b,c+1,c+2 c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.3.2.4) $b = d = c + 2$

Let $\mu := \lambda + (1, 0, 1, 0|0)$.

$$\begin{aligned} P_\mu &= M_{c+2,c+2,c+1,c+2|c} + M_{c+2,c+2,c+2,c+1|c} \\ Pr_\lambda(P_\mu \otimes V_6) &= M_{c+1,c+2,c,c+2|c} + M_{c+2,c+1,c,c+2|c} + M_{c+2,c+2,c,c+1|c} + M_{c+1,c+2,c+1,c+2|c+1} + M_{c+1,c+2,c+1,c+2|c+1} \\ &= \sum M_{c+1,c+2,c,c+2|c} + \sum M_{c+1,c+2,c+1,c+2|c+1} + M_{c+2,c+2,c+1,c+1|c+1} + \sum M_{c+2,c+2,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+2,c+2,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 71](#).

(1.3.2.5) $d > b > c + 2$

Let $\mu := \lambda + (1, 0, 1, 0|0)$.

$$\begin{aligned} P_\mu &= M_{b,d,c+1,c+2|c} + M_{d,b,c+2,c+1|c} + M_{d,c+2,c+1,b|c} + M_{c+2,d,b,c+1|c} + M_{c+2,b,c+1,d|c} + M_{b,c+2,d,c+1|c} \\ Pr_\lambda(P_\mu \otimes V_6) &= M_{b,d,c,c+1|c} + 2M_{b,d,c+1,c+1|c+1} + M_{b,d,c+1,c+2|c+2} + M_{d,b,c+1,c|c} + 2M_{d,b,c+1,c+1|c+1} \\ &= \sum M_{c+1,b,c,d|c} + \sum M_{c+1,b,c+1,d|c+1} + \sum M_{c+2,b,c+1,d|c+2} + \sum M_{b,d,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d,b,c+1,c+1|c+1}, M_{b,d,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 72](#).

Table 71: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+2,c+2,c+1,c+1 c+1}$	$M_{c+2,c+2,c+2,c+1 c+2}, M_{c+2,c+2,c+1,c+2 c+2}, M_{c+3,c+2,c+2,c+1 c+3}, M_{c+2,c+3,c+2,c+1 c+3},$ $M_{c+3,c+2,c+1,c+2 c+3}, M_{c+2,c+3,c+1,c+2 c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 72: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d,b,c+1,c+1 c+1}$	$M_{d,b,c+2,c+1 c+2}, M_{d,b,c+1,c+2 c+2}$
$P_{b,d,c+1,c+1 c+1}$	$M_{b,d,c+2,c+1 c+2}, M_{b,d,c+1,c+2 c+2}, M_{d,b,c+2,c+1 c+2}, M_{d,b,c+1,c+2 c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.3.2.6) $d > b = c + 2$

Let $\mu := \lambda + (1, 0, 1, 0|0)$.

$$P_\mu = M_{d,c+2,c+1,c+2|c} + M_{c+2,d,c+2,c+1|c} + M_{c+2,c+2,c+1,d|c} + M_{c+2,d,c+1,c+2|c} + M_{d,c+2,c+2,c+1|c} + \dots$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d,c+1,c,c+2|c} + M_{d,c+2,c,c+1|c} + M_{d,c+1,c+1,c+2|c+1} + 2M_{d,c+2,c+1,c+1|c+1} + M_{d,c+1,c+1,c+2|c+1} \\ &= \sum M_{c+1,c+2,c,d|c} + \sum M_{c+1,c+2,c+1,d|c+1} + \sum M_{c+2,c+2,c+1,d|c+2} + \sum M_{c+2,d,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+2,d,c+1,c+1|c+1}, M_{d,c+2,c+1,c+1|c+1}, \dots\}$

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 73](#).

Table 73: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+2,d,c+1,c+1 c+1}$	$M_{c+2,d,c+2,c+1 c+2}, M_{c+2,d,c+1,c+2 c+2}, M_{c+3,d,c+2,c+1 c+3}, M_{c+3,d,c+1,c+2 c+3},$ $M_{d,c+2,c+2,c+1 c+2}, M_{d,c+2,c+1,c+2 c+2}, M_{d,c+3,c+2,c+1 c+3}, M_{d,c+3,c+1,c+2 c+3}$
$P_{d,c+2,c+1,c+1 c+1}$	$M_{d,c+2,c+2,c+1 c+2}, M_{d,c+2,c+1,c+2 c+2}, M_{d,c+3,c+2,c+1 c+3}, M_{d,c+3,c+1,c+2 c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(1.3.2.7) $d > c + 2, b < c$

Let $\mu := \lambda + (1, 0, 1, 0|0)$.

$$P_\mu = M_{c+2,d,b,c+1|c} + M_{d,c+2,c+1,b|c} + M_{c+2,b,c+1,d|c} + M_{d,c+1,b,c+2|c} + M_{d,b,c+2,c+1|c} + M_{c+2,c+1,d,b|c} + \dots$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c+1,d,b,c|c} + M_{c+1,d,b,c+1|c+1} + M_{c+2,d,b,c+1|c+2} + M_{d,c+1,b,c|c} + 2M_{d,c+1,c+1,b|c+1} \\ &= \sum M_{c+1,b,c,d|c} + \sum M_{c+1,b,c+1,d|c+1} + \sum M_{c+2,b,c+1,d|c+2} + \sum M_{d,b,c+1,c+1|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d,c+1,b,c+1|c+1}, M_{d,c+1,c+1,b|c+1}, \dots\}$

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 74](#).

(1.3.2.8) $d = c + 2, b < c$

Table 74: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d,c+1,b,c+1 c+1}$	$M_{d,c+2,b,c+1 c+2}, M_{d,c+1,b,c+2 c+2}, M_{d,c+2,c+1,b c+2}, M_{d,c+1,c+2,b c+2}$
$P_{d,c+1,c+1,b c+1}$	$M_{d,c+2,c+1,b c+2}, M_{d,c+1,c+2,b c+2}$
$P_{d,b,c+1,c+1 c+1}$	$M_{d,b,c+2,c+1 c+2}, M_{d,b,c+1,c+2 c+2}, M_{d,c+2,b,c+1 c+2}, M_{d,c+1,c+2,b c+2},$ $M_{d,c+2,c+1,b c+2}, M_{d,c+1,b,c+2 c+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Let $\mu := \lambda + (1, 0, 1, 0|0)$.

$$P_\mu = M_{c+2,c+2,b,c+1|c} + M_{c+2,b,c+1,c+2|c} + M_{c+2,c+1,c+2,b|c} + M_{c+2,c+2,c+1,b|c} + M_{c+2,b,c+2,c+1|c} + M_{c+2,c+1,b,c+2|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{c+1,c+2,b,c|c} + M_{c+2,c+1,b,c|c} + M_{c+1,c+2,b,c+1|c+1} + 2M_{c+2,c+1,b,c+1|c+1} + M_{c+2,c+1,b,c+2|c+1} \\ &= \sum M_{c+1,b,c,c+2|c} + \sum M_{c+1,b,c+1,c+2|c+1} + \sum M_{c+2,b,c+1,c+1|c+1} + \sum M_{c+2,b,c+1,c+2|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+2,c+1,c+1,b|c+1}, M_{c+2,b,c+1,c+1|c+1}, M_{c+2,c+1,b,c+2|c+1}, M_{c+2,b,c+1,c+2|c+1}\}$.
We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 75](#).

Table 75: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c+2,c+1,c+1,b c+1}$	$M_{c+2,c+2,c+1,b c+2}, M_{c+2,c+1,c+2,b c+2}, M_{c+3,c+2,c+1,b c+3}, M_{c+3,c+1,c+2,b c+3}$
$P_{c+2,b,c+1,c+1 c+1}$	$M_{c+2,b,c+2,c+1 c+2}, M_{c+2,b,c+1,c+2 c+2}, M_{c+3,b,c+2,c+1 c+3}, M_{c+3,b,c+1,c+2 c+3},$ $M_{c+2,c+2,b,c+1 c+2}, M_{c+2,c+1,c+2,b c+2}, M_{c+2,c+2,c+1,b c+2}, M_{c+2,c+1,b,c+2 c+2},$ $M_{c+3,c+2,b,c+1 c+3}, M_{c+3,c+1,c+2,b c+3}, M_{c+3,c+2,c+1,b c+3}, M_{c+3,c+1,b,c+2 c+3}$
$P_{c+2,c+1,b,c+1 c+1}$	$M_{c+2,c+2,b,c+1 c+2}, M_{c+2,c+1,b,c+2 c+2}, M_{c+3,c+2,b,c+1 c+3}, M_{c+3,c+1,b,c+2 c+3},$ $M_{c+2,c+2,c+1,b c+2}, M_{c+2,c+1,c+2,b c+2}, M_{c+3,c+2,c+1,b c+3}, M_{c+3,c+1,c+2,b c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.4) \quad a = c + 1, b = c + 1$$

$$(1.4.1) \quad d > c + 2$$

Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{d,c+2,c+1,c+2|c} + M_{c+2,d,c+2,c+1|c} + M_{c+2,c+2,c+1,d|c} + M_{c+2,d,c+1,c+2|c} + M_{d,c+2,c+2,c+1|c} + M_{c+2,c+1,d,c+2|c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d,c+1,c,c+1|c} + 2M_{d,c+1,c+1,c+1|c+1} + M_{d,c+1,c+1,c+2|c+2} + 2M_{d,c+2,c+1,c+1|c+2} + M_{d,c+1,c+1,c+2|c+2} \\ &= \sum M_{c+1,c+1,c,d|c} + \sum M_{c+1,c+1,c+1,d|c+1} + \sum M_{c+1,c+2,c+1,d|c+2} + \sum M_{c+1,d,c+1,c+2|c+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,d,c+1,c+1|c+1}, M_{c+2,d,c+1,c+1|c+2}, M_{c+1,c+1,d,c+2|c+2}, M_{c+1,c+1,c+2,d|c+2}\}$.

We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 76](#).

$$(1.4.2) \quad d = c + 2$$

Let $\mu := \lambda + (1, 1, 1, 0|0)$.

$$P_\mu = M_{c+2,c+2,c+1,c+2|c} + M_{c+2,c+2,c+2,c+1|c}$$

Table 76: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+1,d,c+1,c+1 c+1}$	$M_{c+1,d,c+2,c+1 c+2}, M_{c+1,d,c+1,c+2 c+2}, M_{d,c+1,c+2,c+1 c+2}, M_{d,c+1,c+1,c+2 c+2}$
$P_{c+2,d,c+1,c+1 c+2}$	$M_{c+3,d,c+1,c+1 c+3}, M_{d,c+3,c+1,c+1 c+3}$
$P_{d,c+1,c+1,c+1 c+1}$	$M_{d,c+1,c+2,c+1 c+2}, M_{d,c+1,c+1,c+2 c+2}$
$P_{d,c+2,c+1,c+1 c+2}$	$M_{d,c+3,c+1,c+1 c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$Pr_\lambda(P_\mu \otimes V_{10}) = M_{c+1,c+1,c,c+2|c} + M_{c+1,c+2,c,c+1|c} + M_{c+2,c+1,c,c+1|c} + M_{c+1,c+1,c+1,c+2|c+1} + 2M_{c+1,c+1,c+1,c+1|c+1} \\ = \sum M_{c+1,c+1,c,c+2|c} + \sum M_{c+1,c+1,c+1,c+2|c+1} + \sum M_{c+1,c+2,c+1,c+1|c+1} + \sum M_{c+1,c+1,c+1,c+1|c+1}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+2,c+1,c+1|c+1}, M_{c+2,c+1,c+1,c+1|c+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 77](#).

Table 77: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c+1,c+2,c+1,c+1 c+1}$	$M_{c+1,c+2,c+2,c+1 c+2}, M_{c+1,c+2,c+1,c+2 c+2}, M_{c+1,c+3,c+2,c+1 c+3}, M_{c+1,c+3,c+1,c+2 c+3},$ $M_{c+2,c+1,c+2,c+1 c+2}, M_{c+2,c+1,c+1,c+2 c+2}, M_{c+3,c+1,c+2,c+1 c+3}, M_{c+2,c+3,c+1,c+1 c+3},$ $M_{c+3,c+2,c+1,c+1 c+3}, M_{c+3,c+1,c+1,c+2 c+3}$
$P_{c+2,c+1,c+1,c+1 c+1}$	$M_{c+2,c+1,c+2,c+1 c+2}, M_{c+2,c+1,c+1,c+2 c+2}, M_{c+3,c+2,c+1,c+1 c+3}, M_{c+3,c+1,c+2,c+1 c+3},$ $M_{c+3,c+1,c+1,c+2 c+3}$
$P_{c+2,c+2,c+1,c+1 c+2}$	$M_{c+3,c+2,c+1,c+1 c+3}, M_{c+2,c+3,c+1,c+1 c+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(1.4.3) \quad d \leq c+1$$

$$\text{Let } \mu := \lambda + (1, 1, 1, 0|0) = (c+2, c+2, c+1, d|c).$$

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{a,c+1,c,d|c} + M_{a,c+1,c+1,d|c+1} + \sum M_{a,c+2,c+1,d|c+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2) \quad d = c$$

$$(2.1) \quad a \neq c+1, b \neq c+1$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|1) = (a, b, c, c|c+1).$$

$$Pr_\lambda(P_\mu \otimes V_1) = \sum M_{a,b,c,c|c} + \sum M_{a,b,c,c+1|c+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2.2) \quad a = c+1, b \neq c+1$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 1|0) = (c+2, b, c+1, c+1|c).$$

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{c+1,b,c,c|c} + \sum M_{c+1,b,c,c+1|c+1} + \sum M_{c+2,b,c,c+1|c+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2.3) \quad a \neq c + 1, b = c + 1$$

$$(2.3.1) \quad a > c + 2, b = c + 1$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|2) = (a, c + 1, c, c|c + 2).$$

$$Pr_\lambda(P_\mu \otimes V_5) = \sum M_{a,c+1,c,c|c} + \sum M_{a,c+1,c,c+1|c+1} + \sum M_{a,c+2,c,c+1|c+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(2.3.2) \quad a < c, b = c + 1$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|2).$$

$$P_\mu = M_{c+1,c,a,c|c+2} + M_{c,c+1,c,a|c+2} + M_{a,c+1,c,c|c+2} + M_{c,c+1,a,c|c+2} + M_{c+1,c,c,a|c+2} + M_{c+1,a,c,c|c+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{c+2,c+1,a,c|c+2} + M_{c+2,c,a,c+1|c+2} + 2M_{c+1,c+1,a,c|c+1} + M_{c+1,c,a,c+1|c+1} + M_{c+1,c,a,c|c+1} \\ &= \sum M_{a,c+1,c,c|c} + \sum M_{a,c+1,c,c+1|c+1} + \sum M_{a,c+2,c,c+1|c+2} + \sum M_{c+1,c+1,a,c|c+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+1,c+1,a,c|c+1}, M_{c+1,c+1,c,a|c+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_5) = P_\lambda$ by [table 78](#).

Table 78: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_5))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_5))$ (via Prop. ~ 1)
$P_{c+1,c+1,a,c c+1}$	$M_{c+2,c+1,a,c c+2}, M_{c+1,c+2,a,c c+2}, M_{c+2,c+1,c,a c+2}, M_{c+1,c+2,c,a c+2}$	
$P_{c+1,c+1,c,a c+1}$	$M_{c+2,c+1,c,a c+2}, M_{c+1,c+2,c,a c+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.3.3) \quad a = c + 2, b = c + 1$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|3).$$

$$P_\mu = M_{c+2,c+1,c,c|c+3}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_9) &= M_{c+3,c+2,c+1,c|c+3} + M_{c+3,c+2,c,c+1|c+3} + M_{c+2,c+2,c+1,c|c+2} + M_{c+2,c+2,c,c+1|c+2} + M_{c+2,c+1,c,c|c} \\ &= M_{c+2,c+1,c,c|c} + \sum M_{c+2,c+1,c,c+1|c+1} + \sum M_{c+2,c+2,c,c+1|c+2} + \sum M_{c+3,c+2,c,c+1|c+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c+3,c+2,c+1,c|c+3}, M_{c+3,c+2,c,c+1|c+3}\}$.

We have $Pr_\lambda(P_\mu \otimes V_9) = P_\lambda$ by [table 79](#).

Table 79: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_9))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_9))$ (via Prop. ~ 1)
$P_{c+3,c+2,c+1,c c+3}$	$M_{c+4,c+2,c+1,c c+4}$	
$P_{c+3,c+2,c,c+1 c+3}$	$M_{c+4,c+2,c,c+1 c+4}, M_{c+4,c+2,c+1,c c+4}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

□

Theorem 3.4. Let $a, b, c, d \in \mathbb{Z}$ and $\lambda = (a, b, c, d|d)$ with $d \notin \{a, b, c\}$. The projective objects P_λ in $\mathfrak{B}_{a,b,c}$ have the following Verma flag formulae.

Proof. The Cases are as follows:

(1) $a \neq d+1, b \neq d+1, c \neq d+1$

(1.1) The generic case : none of the special configurations below occur. $\text{not}(a \leq c < b < d), (b < a < d, c > d+1)$

Let $\mu := \lambda + (0, 0, 0, 0|1) = (a, b, c, d|d+1)$.

$$Pr_\lambda(P_\mu \otimes V_1) = \sum M_{a,b,c,d|d} + \sum M_{a,b,c,d+1|d+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(1.2) $a < c < b < d$

Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$P_\mu = 2M_{b,d,a,c|d+1} + 2M_{d,b,c,a|d+1} + M_{b,a,c,d|d+1} + M_{d,c,a,b|d+1} + M_{a,b,d,c|d+1} + M_{c,d,b,a|d+1} + M_{c,b,a,d|d+1} +$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b,d+1,a,c|d+1} + 2M_{b,d,a,c|d} + 2M_{d+1,b,c,a|d+1} + 2M_{d,b,c,a|d} + M_{b,a,c,d+1|d+1} + M_{b,a,c,d|d} + \\ &= \sum M_{a,b,c,d|d} + \sum M_{b,d,a,c|d} + \sum M_{a,b,c,d+1|d+1} + \sum M_{b,d+1,a,c|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d+1,a,c|d+1}, M_{b,d,a,c|d}, M_{d+1,b,c,a|d+1}, M_{d,b,c,a|d}\}$. 根据 [table 80](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{b,d,a,c|d}$ 或 $P_{d,b,a,c|d}$ 或 $P_{b,d,c,a|d}$ 或 $P_{d,b,c,a|d}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,c|d}, M_{d,b,a,c|d}, M_{b,d,c,a|d}, M_{d,b,c,a|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, c, d|d+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$, 结合 [Theorem 2.1](#) 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2) = 2P_\nu = 2\nu + 4M_{b,d,a,c|d+1} + 4M_{d,b,a,c|d+1} + 4M_{b,d,c,a|d+1} + 4M_{d,b,c,a|d+1} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(a, b, c, d|d) + (0, 0, 0, 0|1)$ 和 $(a, b, c, d+1|d+1) + (0, 0, 0, -1|0)$ 得到. 由 [Proposition 2.0.1](#), $M_{(a,b,c,d+1|d+1)} \subset \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$. 又由于 $(b, d, a, c|d+1)$ 只能由 $(b, d, a, c|d) + (0, 0, 0, 0|1)$ 和 $(b, d+1, a, c|d+1) + (0, -1, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,c|d}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,c|d}, M_{b,d,c,a|d}, M_{d,b,c,a|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

(1.3) $a = c < b < d$

Let $\mu := \lambda + (0, 0, 0, 0|1)$.

$$P_\mu = 2M_{b,d,a,a|d+1} + M_{b,a,a,d|d+1} + M_{a,b,d,a|d+1} + M_{a,d,a,b|d+1} + M_{d,a,b,a|d+1} + 2M_{d,b,a,a|d+1} + M_{a,b,a,d|d+1} +$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= 2M_{b,d+1,a,a|d+1} + 2M_{b,d,a,a|d} + M_{b,a,a,d+1|d+1} + M_{b,a,a,d|d} + M_{a,b,d+1,a|d+1} + M_{a,b,d,a|d} + \\ &= \sum M_{a,b,a,d|d} + \sum M_{b,d,a,a|d} + \sum M_{a,b,a,d+1|d+1} + \sum M_{b,d+1,a,a|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d+1,a,a|d+1}, M_{b,d,a,a|d}, M_{d+1,b,a,a|d+1}, M_{d,b,a,a|d}\}$. 根据 [table 81](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{b,d,a,a|d}$ 或 $P_{d,b,a,a|d}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,a|d}, M_{d,b,a,a|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a, d|d+1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$, 结合 [Theorem 2.1](#) 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2) = 2P_\nu = 2\nu + 4M_{b,d,a,a|d+1} + 4M_{d,b,a,a|d+1} \dots$$

Table 80: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b,d+1,a,c d+1}$	$M_{b,c+2,a,c c+2}, M_{c+2,b,a,c c+2}, M_{b,c+2,c,a c+2}, M_{c+2,b,c,a c+2}$
$P_{b,d,a,c d}$	$\emptyset$
$P_{d+1,b,c,a d+1}$	$M_{c+2,b,c,a c+2}$
$P_{d,b,c,a d}$	$\emptyset$
$P_{d+1,b,a,c d+1}$	$M_{c+2,b,a,c c+2}, M_{c+2,b,c,a c+2}$
$P_{d,b,a,c d}$	$\emptyset$
$P_{b,d+1,c,a d+1}$	$M_{b,c+2,c,a c+2}, M_{c+2,b,c,a c+2}$
$P_{b,d,c,a d}$	$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 81: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{b,d+1,a,a d+1}$	$M_{b,c+2,a,a c+2}, M_{c+2,b,a,a c+2}$
$P_{b,d,a,a d}$	$\emptyset$
$P_{d+1,b,a,a d+1}$	$M_{c+2,b,a,a c+2}$
$P_{d,b,a,a d}$	$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(a, b, c, b|c) + (0, 0, 0, 0|1)$ 和 $(a, b, c + 1, b|c + 1) + (0, 0, -1, 0|0)$ 得到. 由 [Proposition 2.0.1](#), $M_{(a,b,a,d+1|d+1)} \subset \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$. 又由于 $(b, d, a, a|d + 1)$ 只能由 $(b, d, a, a|d) + (0, 0, 0, 0|1)$ 和 $(b, d + 1, a, a|d + 1) + (0, -1, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,a|d}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,a|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2. 综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

$$(1.4) \quad b < a < d, c > d + 1$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|1).$$

$$P_\mu = M_{a,b,c,d|d+1} + M_{d,c,b,a|d+1} + 2M_{c,d,a,b|d+1} + M_{c,a,b,d|d+1} + M_{a,c,d,b|d+1} + M_{c,b,d,a|d+1} + M_{d,a,c,b|d+1} + M_{a,d,c,b|d+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a,b,c,d+1|d+1} + M_{a,b,c,d|d} + M_{d+1,c,b,a|d+1} + M_{d,c,b,a|d} + 2M_{c,d+1,a,b|d+1} + 2M_{c,d,a,b|d} + M_{c,d+1,b,a|d+1} \\ &= \sum M_{a,b,c,d|d} + \sum M_{a,b,c,d+1|d+1} + \sum M_{c,a,d,b|d} + \sum M_{c,a,d+1,b|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,d+1,a,b|d+1}, M_{c,d,a,b|d}, M_{c,a,d+1,b|d+1}, M_{c,a,d,b|d}\}$. 根据 [table 82](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{b,d,a,a|d}$ 或 $P_{d,b,a,a|d}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_1)$ 的直和分解中.

为了排除这种可能, 我们仅需证明 $M_{b,d,a,a|d}, M_{d,b,a,a|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可.

令 $\nu = (a, b, a, d|d + 1)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$, 结合 [Theorem 2.1](#) 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2) = 2P_\nu = 2\nu + 4M_{b,d,a,a|d+1} + 4M_{d,b,a,a|d+1} \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_1)) \otimes V_2)$ 极小并且重数为 2, 且 ν 只能由 $(a, b, c, b|c) + (0, 0, 0, 0|1)$ 和 $(a, b, c + 1, b|c + 1) + (0, 0, -1, 0|0)$ 得到. 由 [Proposition 2.0.1](#), $M_{(a,b,a,d+1|d+1)} \subset \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_2) = 2P_\nu$. 又由于 $(b, d, a, a|d + 1)$ 只能由 $(b, d, a, a|d) + (0, 0, 0, 0|1)$ 和 $(b, d + 1, a, a|d + 1) + (0, -1, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{b,d,a,a|d}$, 否则 $pr_\nu(P_\lambda \otimes V_2)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. 同理, $M_{d,b,a,a|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数也是 2. 综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 82: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_1))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_1))$ (via Prop. ~ 1)
$P_{c,d+1,a,b d+1}$		$M_{c,d+2,a,b d+2}$
$P_{c,d,a,b d}$		$\emptyset$
$P_{c,a,d+1,b d+1}$		$M_{c,a,d+2,b d+2}, M_{c,d+2,a,b d+2}$
$P_{c,a,d,b d}$		$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2) \quad a = d + 1, b \neq d + 1, c \neq d + 1$$

$$(2.1) \quad b < d, c > d + 2$$

$$\text{Let } \mu := \lambda + (1, 0, 0, 1|0).$$

$$P_\mu = M_{d+2,c,b,d+1|d} + M_{c,d+2,d+1,b|d} + M_{c,b,d+2,d+1|d} + M_{d+2,d+1,c,b|d} + M_{c,d+2,b,d+1|d} + M_{d+2,c,d+1,b|d} + M_{d+2,b,c,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d+1,c,b,d|d} + M_{d+1,c,b,d+1|d+1} + M_{d+2,c,b,d+1|d+2} + M_{c,d+1,d,b|d} + 2M_{c,d+1,d+1,b|d+1} + M_{c,d+1,b,d|d} \\ &= \sum M_{d+1,b,c,d|d} + \sum M_{d+1,b,c,d+1|d+1} + \sum M_{d+2,b,c,d+1|d+2} + M_{c,d+1,d+1,b|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,d+1,d+1,b|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 83](#).

Table 83: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{c,d+1,d+1,b d+1}$	$M_{c,d+2,d+1,b d+2}$	$M_{c,d+1,d+2,b d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.2) \quad b < d, c = d + 2$$

$$\text{Let } \mu := \lambda + (1, 0, 0, 1|0).$$

$$P_\mu = M_{d+2,d+2,b,d+1|d} + M_{d+2,d+1,d+2,b|d} + M_{d+2,d+2,d+1,b|d} + M_{d+2,b,d+2,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d+1,d+2,b,d|d} + M_{d+2,d+1,b,d|d} + M_{d+1,d+2,b,d+1|d+1} + M_{d+2,d+1,b,d+1|d+1} + M_{d+2,d+2,b,d+1|d+1} \\ &= \sum M_{d+1,b,d+2,d|d} + \sum M_{d+1,b,d+2,d+1|d+1} + M_{d+2,d+1,d+1,b|d+1} + \sum M_{d+2,b,d+2,d+1|d+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+2,d+1,d+1,b|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 84](#).

Table 84: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+2,d+1,d+1,b d+1}$	$M_{d+2,d+2,d+1,b d+2}$	$M_{d+2,d+1,d+2,b d+2}, M_{d+3,d+2,d+1,b d+3}, M_{d+3,d+1,d+2,b d+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(2.3) \quad \text{Others not}(b < d, c > d + 1)$$

$$\text{Let } \mu := \lambda + (1, 0, 0, 1|0) = (d + 2, b, c, d + 1|d).$$

$$Pr_\lambda(P_\mu \otimes V_1) = \sum M_{d+1,b,c,d|d} + \sum M_{d+1,b,c,d+1|d+1} + \sum M_{d+2,b,c,d+1|d+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(3) \quad a \neq d + 1, b = d + 1, c \neq d + 1$$

$$(3.1) \quad a \neq d + 2, a > d + 2$$

$$\text{Let } \mu := \lambda + (0, 1, 0, 1|0) = (a, d + 2, c, d + 1|d).$$

$$Pr_\lambda(P_\mu \otimes V_1) = \sum M_{a,d+1,c,d|d} + \sum M_{a,d+1,c,d+1|d+1} + \sum M_{a,d+2,c,d+1|d+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(3.2) \quad a \neq d + 2, a < d$$

$$(3.2.1) \quad c > d + 2$$

$$\text{Let } \mu := \lambda + (0, 1, 0, 1|0).$$

$$P_\mu = M_{d+2,c,a,d+1|d} + M_{c,d+2,d+1,a|d} + M_{a,d+2,c,d+1|d} + M_{d+1,c,d+2,a|d} + M_{c,a,d+2,d+1|d} + M_{d+2,d+1,c,a|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d+1,c,a,d|d} + M_{d+1,c,a,d+1|d+1} + M_{d+2,c,a,d+1|d+2} + M_{c,d+1,d,a|d} + 2M_{c,d+1,d+1,a|d+1} + \\ &= \sum M_{a,d+1,c,d|d} + \sum M_{a,d+1,c,d+1|d+1} + \sum M_{a,d+2,c,d+1|d+2} + \sum M_{d+1,d+1,c,a|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,c,d+1,a|d+1}, M_{c,d+1,d+1,a|d+1}, M_{c,d+1,d+1,c,a|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 85](#).

Table 85: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+1,c,d+1,a d+1}$	$M_{d+2,c,d+1,a d+2}, M_{d+1,c,d+2,a d+2}, M_{c,d+2,d+1,a d+2}, M_{c,d+1,d+2,a d+2}$
$P_{c,d+1,d+1,a d+1}$	$M_{c,d+2,d+1,a d+2}, M_{c,d+1,d+2,a d+2}$
$P_{d+1,d+1,c,a d+1}$	$M_{d+2,d+1,c,a d+2}, M_{d+1,d+2,c,a d+2}, M_{c,d+1,d+2,a d+2}, M_{d+2,c,d+1,a d+2},$ $M_{c,d+2,d+1,a d+2}, M_{d+1,c,d+2,a d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.2.2) \quad c = d + 2$$

$$\text{Let } \mu := \lambda + (0, 1, 0, 1|0).$$

$$P_\mu = M_{d+2,d+2,a,d+1|d} + M_{a,d+2,d+2,d+1|d} + M_{d+2,d+1,d+2,a|d} + M_{d+2,d+2,d+1,a|d} + M_{d+2,a,d+2,d+1|d} + M_{d+2,d+1,a,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d+1,d+2,a,d|d} + M_{d+2,d+1,a,d|d} + M_{d+1,d+2,a,d+1|d+1} + M_{d+2,d+1,a,d+1|d+1} + M_{d+2,d+2,a,d+1|d+1} \\ &= \sum M_{a,d+1,d+2,d|d} + \sum M_{a,d+1,d+2,d+1|d+1} + \sum M_{a,d+2,d+2,d+1|d+2} + \sum M_{d+1,d+1,a,d+1|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+2,d+1,d+1,a|d+1}, M_{d+1,d+1,d+2,a|d+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 86](#).

Table 86: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+2,d+1,d+1,a d+1}$	$M_{d+2,d+2,d+1,a d+2}, M_{d+2,d+1,d+2,a d+2}, M_{d+3,d+2,d+1,a d+3}, M_{d+3,d+1,d+2,a d+3}$
$P_{d+1,d+1,d+2,a d+1}$	$M_{d+2,d+1,d+2,a d+2}, M_{d+1,d+2,d+2,a d+2}, M_{d+2,d+2,d+1,a d+2}$
$P_{d+1,d+2,d+1,a d+1}$	$M_{d+2,d+2,d+1,a d+2}, M_{d+1,d+2,d+2,a d+2}, M_{d+1,d+3,d+2,a d+3}, M_{d+2,d+1,d+2,a d+2},$ $M_{d+3,d+1,d+2,a d+3}, M_{d+2,d+3,d+1,a d+3}, M_{d+3,d+2,d+1,a d+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.2.3) \quad a > c > d$$

$$\text{Let } \mu := \lambda + (0, 1, 0, 1|0).$$

$$P_\mu = M_{d+1,d+2,a,c|d} + M_{d+2,d+1,c,a|d} + M_{d+2,c,a,d+1|d} + M_{c,d+2,d+1,a|d} + M_{a,d+2,c,d+1|d} + M_{d+2,a,d+1,c|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d,d+1,a,c|d} + 2M_{d+1,d+1,a,c|d+1} + M_{d+1,d+2,a,c|d+2} + M_{d+1,d,c,a|d} + 2M_{d+1,d+1,c,a|d+1} \\ &= \sum M_{a,d+1,c,d|d} + \sum M_{a,d+1,c,d+1|d+1} + \sum M_{a,d+2,c,d+1|d+2} + \sum M_{d+1,d+1,a,c|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,a,c|d+1}, M_{d+1,d+1,c,a|d+1}\}$.

We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 87](#).

$$(3.2.4) \quad a = c > d$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 0|0).$$

$$P_\mu = M_{d+1,d+2,a,a|d} + M_{a,d+2,a,d+1|d} + M_{d+2,a,d+1,a|d} + M_{d+2,d+1,a,a|d} + M_{d+2,a,a,d+1|d} + M_{a,d+2,d+1,a|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d,d+1,a,a|d} + 2M_{d+1,d+1,a,a|d+1} + M_{d+1,d+2,a,a|d+2} + M_{a,d+1,a,d|d} + M_{a,d+1,a,d+1|d+1} \\ &= \sum M_{a,d+1,a,d|d} + \sum M_{a,d+1,a,d+1|d+1} + \sum M_{a,d+2,a,d+1|d+2} + M_{d+1,d+1,a,a|d+1} \end{aligned}$$

Table 87: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)			
$P_{d+1,d+1,a,c d+1}$	$M_{d+2,d+1,a,c d+2}, M_{d+1,d+2,a,c d+2}, M_{d+2,d+1,c,a d+2}, M_{d+1,d+2,c,a d+2}$				
$P_{d+1,d+1,c,a d+1}$	$M_{d+2,d+1,c,a d+2}, M_{d+1,d+2,c,a d+2}$				

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 88: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)	
$P_{d+1,d+1,a,a d+1}$	$M_{d+2,d+1,a,a d+2}, M_{d+1,d+2,a,a d+2}$		

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,a,a|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 88.

$$(3.2.5) \quad c > a > d$$

Let $\mu := \lambda + (0, 1, 0, 1|0)$.

$$P_\mu = M_{d+1,d+2,c,a|d} + M_{d+2,d+1,a,c|d} + M_{d+2,a,c,d+1|d} + M_{a,d+2,d+1,c|d} + M_{d+2,d+1,c,a|d} + M_{d+1,d+2,a,c|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d,d+1,c,a|d} + 2M_{d+1,d+1,c,a|d+1} + M_{d+1,d+2,c,a|d+2} + M_{d+1,d,a,c|d} + 2M_{d+1,d+1,a,c|d+1} \\ &= \sum M_{a,d+1,c,d|d} + \sum M_{a,d+1,c,d+1|d+1} + \sum M_{d+1,d+1,c,a|d+1} + \sum M_{a,d+2,c,d+1|d+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,c,a|d+1}, M_{d+1,d+1,a,c|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 89.

Table 89: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)			
$P_{d+1,d+1,c,a d+1}$	$M_{d+2,d+1,c,a d+2}, M_{d+1,d+2,c,a d+2}, M_{d+2,d+1,a,c d+2}, M_{d+1,d+2,a,c d+2}$				
$P_{d+1,d+1,a,c d+1}$	$M_{d+2,d+1,a,c d+2}, M_{d+1,d+2,a,c d+2}$				

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.3) \quad a = d + 2$$

$$(3.3.1) \quad c < d$$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{d+3,d+2,d+1,c|d} + M_{d+3,d+2,c,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d+2,d+1,d,c|d} + M_{d+2,d+1,d+1,c|d+1} + M_{d+2,d+2,d+1,c|d+2} + M_{d+3,d+2,d+1,c|d+3} + M_{d+3,d+2,c,d+1|d+3} \\ &= \sum M_{d+2,d+1,c,d|d} + \sum M_{d+2,d+1,c,d+1|d+1} + \sum M_{d+2,d+2,c,d+1|d+2} + \sum M_{d+3,d+2,c,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+2,d+1,c|d+3}, M_{d+3,d+2,c,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by table 90.

Table 90: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d+3,d+2,d+1,c d+3}$	$M_{d+4,d+2,d+1,c d+4}$
$P_{d+3,d+2,c,d+1 d+3}$	$M_{d+4,d+2,c,d+1 d+4}, M_{d+4,d+2,d+1,c d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(3.3.2) $c > d + 3$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{c,d+3,d+2,d+1|d} + M_{d+3,d+2,c,d+1|d} + M_{d+3,c,d+2,d+1|d} + M_{c,d+2,d+3,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{c,d+2,d+1,d|d} + M_{c,d+2,d+1,d+1|d+1} + 2M_{c,d+2,d+2,d+1|d+2} + M_{c,d+3,d+2,d+1|d+3} + M_{d+3,d+2,c,d+1|d+3} \\ &= \sum M_{d+2,d+1,c,d|d} + \sum M_{d+2,d+1,c,d+1|d+1} + \sum M_{d+2,d+2,c,d+1|d+2} + \sum M_{d+3,d+2,c,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{c,d+3,d+2,d+1|d+3}, M_{d+3,d+2,c,d+1|d+3}, M_{d+3,d+2,d+1,c|d+4}, M_{c,d+4,d+2,d+1,c|d+4}, M_{d+4,d+2,c,d+1|d+4}, M_{d+4,d+2,d+1,c|d+4}\}$. 根据 table 91, 仅仅由 Proposition 2.0.1 不足以判断 $P_{c,d+2,d+2,d+1|d+2}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_{10})$ 的直和分解中. 因此由 Theorem 3.3, 我们引入它们完整的滤过结构如下.

$$P_{c,d+2,d+2,d+1|d+2} = M_{c,d+3,d+2,d+1|d+3} + M_{c,d+2,d+3,d+1|d+3} + M_{c,d+2,d+2,d+1|d+2} = \sum M_{c,d+2,d+2,d+1|d+2} \quad (15)$$

可以看出 $\text{Filt}(P_{c,d+2,d+2,d+1|d+2})$ 所有的 Verma 模都在多重集 U 中. 且其重数未超过 U 中对应项的重数. 因此因此仅通过滤过比较仍无法排除它们中某一个作为 $pr_\lambda(P_\mu \otimes V_{10})$ 直和项的可能性. 我们反证法排除这个可能性.

首先, 假设 $P_{c,d+2,d+2,d+1|d+2}$ 是 $pr_\lambda(P_\lambda \otimes V_{10})$ 的一个直和项, 那么, 存在一个投射模 Q 使得

$$pr_\lambda(P_\lambda \otimes V_{10}) = Q + P_{c,d+2,d+2,d+1|d+2}$$

令 $\nu = (d+2, d+2, c, d+1|d+3)$, 那么 ν 是典型的权. 我们就有

$$pr_\nu(Q \otimes V_2) = 2M_{d+2,d+2,c,d+1|d+3} + 2M_{d+2,c,d+2,d+1|d+3} + M_{c,d+2,d+2,d+1|d+3} \quad (16)$$

观察到 equation (16) 中, 权 ν 在滤过中是极小的, 且重数为 2. Q 是投射模, 因此 $pr_\nu(Q \otimes V_2)$ 也是投射模. 根据 Lemma 2.0.1, 这表明 $pr_\nu(Q \otimes V_1)$ 必然有两个 P_ν 作为直和项. 但是 Theorem 2.1 给出 P_ν 的完整滤过的结构如下

$$\begin{aligned} P_\nu &= 2M_{d+2,d+2,c,d+1|d+3} + 2M_{d+2,c,d+2,d+1|d+3} + M_{c,d+2,d+2,d+1|d+3} \\ &= \sum M_{d+2,d+2,c,d+1|d+3} \end{aligned} \quad (17)$$

比较 equation (16) 和 equation (17), 我们发现 $pr_\nu(Q \otimes V_1)$ 不可能有两个 P_ν 作为直和项. 矛盾, 因此排除了 $P_{c,d+2,d+2,d+1|d+2}$ 作为 $pr_\lambda(P_\lambda \otimes V_{10})$ 直和项的可能.

综上 $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$.

(3.3.3) $c = d + 3$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{d+3,d+2,d+3,d+1|d} + M_{d+3,d+3,d+2,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d+2,d+1,d+3,d|d} + M_{d+3,d+1,d+2,d|d} + M_{d+2,d+1,d+3,d+1|d+1} + M_{d+3,d+1,d+2,d+1|d+1} + \\ &= \sum M_{d+2,d+1,d+3,d|d} + \sum M_{d+2,d+1,d+3,d+1|d+1} + \sum M_{d+2,d+2,d+3,d+1|d+2} + M_{d+3,d+2,d+3,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+2,d+3,d+1|d+3}, M_{d+3,d+2,d+2,d+1|d+3}, M_{d+3,d+2,d+3,d+1|d+3}, M_{d+3,d+2,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by table 92.

Table 91: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{c,d+3,d+2,d+1 d+3}$	$M_{c,d+4,d+2,d+1 d+4}$
$P_{d+3,d+2,c,d+1 d+3}$	$M_{d+4,d+2,c,d+1 d+4}, M_{c,d+2,d+4,d+1 d+4}, M_{d+4,c,d+2,d+1 d+4}, M_{c,d+4,d+2,d+1 d+4}$
$P_{d+3,c,d+2,d+1 d+3}$	$M_{d+4,c,d+2,d+1 d+4}, M_{c,d+4,d+2,d+1 d+4}$
$P_{c,d+2,d+2,d+1 d+2}$	$\emptyset$
$P_{c,d+2,d+3,d+1 d+3}$	$M_{c,d+2,d+4,d+1 d+4}, M_{c,d+4,d+2,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Table 92: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d+3,d+2,d+3,d+1 d+3}$	$M_{d+4,d+2,d+3,d+1 d+4}, M_{d+3,d+2,d+4,d+1 d+4}, M_{d+4,d+3,d+2,d+1 d+4}, M_{d+3,d+4,d+2,d+1 d+4}$
$P_{d+3,d+2,d+2,d+1 d+2}$	$M_{d+4,d+3,d+2,d+1 d+4}, M_{d+4,d+2,d+3,d+1 d+4}$
$P_{d+3,d+3,d+2,d+1 d+3}$	$M_{d+4,d+3,d+2,d+1 d+4}, M_{d+3,d+4,d+2,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(3.3.4) \quad c = d + 2$$

Let $\mu := \lambda + (1, 1, 0, 1|0)$.

$$P_\mu = M_{d+3,d+2,d+2,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d+2,d+1,d+2,d|d} + M_{d+2,d+2,d+1,d|d} + M_{d+2,d+1,d+2,d+1|d+1} + M_{d+2,d+2,d+1,d+1|d+1} + \\ &= \sum M_{d+2,d+1,d+2,d|d} + \sum M_{d+2,d+1,d+2,d+1|d+1} + M_{d+2,d+2,d+2,d+1|d+2} + M_{d+3,d+2,d+2,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+2,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 93](#).

Table 93: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d+3,d+2,d+2,d+1 d+3}$		$M_{d+4,d+2,d+2,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4) \quad c = d + 1, a \neq d + 1, b \neq d + 1$$

$$(4.1) \quad a \neq d + 2, b \neq d + 2$$

$$(4.1.1) \quad a < d, b > d + 2$$

Let $\mu := \lambda + (0, 0, 1, 1|0)$.

$$P_\mu = M_{d+2,b,a,d+1|d} + M_{b,d+2,d+1,a|d} + M_{d+1,b,d+2,a|d} + M_{b,a,d+2,d+1|d} + M_{b,d+2,a,d+1|d} + M_{d+2,b,d+1,a|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d+1,b,a,d|d} + M_{d+1,b,a,d+1|d+1} + M_{d+2,b,a,d+1|d+2} + M_{b,d+1,d,a|d} + 2M_{b,d+1,d+1,a|d+1} + \\ &= \sum M_{a,b,d+1,d|d} + \sum M_{a,b,d+1,d+1|d+1} + \sum M_{d+1,b,d+1,a|d+1} + \sum M_{a,b,d+2,d+1|d+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,b,d+1,a|d+1}, M_{b,d+1,d+1,a|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 94](#).

Table 94: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+1,b,d+1,a d+1}$	$M_{d+2,b,d+1,a d+2}, M_{d+1,b,d+2,a d+2}, M_{b,d+2,d+1,a d+2}, M_{b,d+1,d+2,a d+2}$	
$P_{b,d+1,d+1,a d+1}$	$M_{b,d+2,d+1,a d+2}, M_{b,d+1,d+2,a d+2}$	

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.1.2) \quad a < d, b < a$$

Let $\mu := \lambda + (0, 0, 1, 1|0)$.

$$P_\mu = M_{a,b,d+2,d+1|d} + M_{d+1,d+2,b,a|d} + 2M_{d+2,d+1,a,b|d} + M_{d+2,a,b,d+1|d} + M_{a,d+2,d+1,b|d} + M_{d+2,b,d+1,a|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{a,b,d+1,d|d} + M_{a,b,d+1,d+1|d+1} + M_{a,b,d+2,d+1|d+2} + M_{d,d+1,b,a|d} + 2M_{d+1,d+1,b,a|d+1} + \\ &= \sum M_{a,b,d+1,d|d} + \sum M_{a,b,d+1,d+1|d+1} + \sum M_{a,b,d+2,d+1|d+2} + \sum M_{d+1,b,d+1,a|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d,a,b|d}, 2M_{d+1,d+1,a,b|d+1}, M_{d+2,d+1,d+1,a,b|d+2}\}$. 根据 [table 95](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{d+1,d,a,b|d}$ 或 $P_{d+1,a,d,b|d}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_6)$ 的直和分解中. 因此由 [Theorem 3.2](#) 和 [Theorem 3.3](#), 我们引入它们完整的滤过结构如下。

$$P_{d+1,d,a,b|d} = M_{d+1,d,a,b|d} + M_{d+1,d+1,a,b|d+1} + M_{d+2,d+1,a,b|d+2} \quad (18)$$

$$P_{d+1,a,d,b|d} = M_{d+1,d,a,b|d} + M_{d+1,d+1,a,b|d+1} + M_{d+2,d+1,a,b|d+2} + M_{d+1,a,d,b|d} + M_{d+1,a,d+1,b|d+1} + M_{d+1,a,d+2,b|d+2} + \dots \quad (19)$$

可以看出 $\text{Filt}(P_{d+1,d,a,b|d})$ 和 $\text{Filt}(P_{d+1,a,d,b|d})$ 所有的 Verma 模都在多重集 U 中. 且其重数未超过 U 中对应项的重数. 因此因此仅通过滤过比较仍无法排除它们中某一个作为 $pr_\lambda(P_\mu \otimes V_6)$ 直和项的可能性.

为了排除这种可能, 我们仅需证明 $M_{d+1,d,a,b|d}, M_{d+1,a,d,b|d}$ 在 $\text{Filt}(P_\lambda)$ 中的重数是 2 即可. 令 $\nu = (a, b, d+3, d+1|d+2)$, 我们计算 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1)$, 结合 Theorem 2.1 发现

$$pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_1) = P_\nu = \nu + 2M_{d+3,d+1,a,b|d+2} + \dots$$

注意到 M_ν 在 $pr_\nu((pr_\lambda(P_\mu \otimes V_6)) \otimes V_2)$ 极小并且重数为 1, 且 ν 只能由 $(a, b, d+2, d+1|d+2) + (0, 0, 1, 0|0)$ 得到. 由 Proposition 2.0.1, $M_{(a,b,c,d+1|d+1)} \subset \text{Filt}(P_\lambda)$. 因此 $pr_\nu(P_\lambda \otimes V_1) = P_\nu$. 又由于 $(d+3, d+1, a, b|d+2)$ 只能由 $(d+2, d+1, a, b|d+2) + (1, 0, 0, 0|0)$ 得到, 因此我们可以得到 $\text{Filt}(P_\lambda)$ 中必然有两个 $M_{d+2,d+1,a,b|d+2}$, 否则 $pr_\nu(P_\lambda \otimes V_1)$ 不可能包含完整的两个 $\text{Filt}(P_\nu)$. equation (18) 表明 $M_{d+2,d+1,a,b|d+2} \in \text{Filt}(P_{d+1,d,a,b|d})$, 这说明 $pr_\lambda(P_\mu \otimes V_6)$ 直和分解中不可能同时包含 P_λ 和 $P_{d+1,d,a,b|d}$. 同理, equation (19) 表明 $pr_\lambda(P_\mu \otimes V_6)$ 直和分解中不可能同时包含 P_λ 和 $P_{d+1,a,d,b|d}$.

综上 $Pr_\lambda(P_\mu \otimes V_1) = P_\lambda$.

Table 95: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+1,d,a,b d}$	$\emptyset$
$2M_{d+1,d+1,a,b d+1}$	$M_{d+1,d+2,a,b d+2}$
$P_{d+2,d+1,a,b d+2}$	$M_{d+3,d+1,a,b d+3}$
$P_{d+1,d+1,b,a d+1}$	$M_{d+2,d+1,b,a d+2}, M_{d+1,d+2,b,a d+2}, M_{d+1,d+2,a,b d+2}$
$P_{d+1,b,d+1,a d+1}$	$M_{d+2,b,d+1,a d+2}, M_{d+1,b,d+2,a d+2}, M_{d+2,d+1,b,a d+2}, M_{d+1,d+2,b,a d+2},$ $M_{d+1,a,d+2,b d+2}, M_{d+1,d+2,a,b d+2}$
$2M_{d+1,a,d+1,b d+1}$	$M_{d+1,a,d+2,b d+2}, M_{d+1,d+2,a,b d+2}$
$P_{d+1,a,d,b d}$	$\emptyset$
$P_{d+2,a,d+1,b d+2}$	$M_{d+3,a,d+1,b d+3}, M_{d+3,d+1,a,b d+3}$
$P_{a,d+1,d+1,b d+1}$	$M_{a,d+2,d+1,b d+2}, M_{a,d+1,d+2,b d+2}, M_{d+1,d+2,a,b d+2}, M_{d+1,a,d+2,b d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.1.3) \quad a < d, a < b < d$$

Let $\mu := \lambda + (0, 0, 1, 1|0)$.

$$P_\mu = M_{b,a,d+2,d+1|d} + M_{d+1,d+2,a,b|d} + M_{d+2,d+1,b,a|d} + M_{d+2,b,a,d+1|d} + M_{a,d+1,d+2,b|d} + M_{b,d+2,d+1,a|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b,a,d+1,d|d} + M_{b,a,d+1,d+1|d+1} + M_{b,a,d+2,d+1|d+2} + M_{d,d+1,a,b|d} + 2M_{d+1,d+1,a,b|d+1} + \\ &= \sum M_{a,b,d+1,d|d} + \sum M_{a,b,d+1,d+1|d+1} + \sum M_{a,b,d+2,d+1|d+2} + \sum M_{a,d+1,d+1,b|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,a,b|d+1}, M_{d+1,d+1,b,a|d+1}, M_{d+1,d+1,a,b|d+2}, M_{d+1,d+1,b,a|d+2}, M_{d+2,d+1,a,b|d+2}, M_{d+2,d+1,b,a|d+2}, M_{d+1,d+2,a,b|d+2}, M_{d+1,d+2,b,a|d+2}, M_{d+1,d+2,a,b|d+3}, M_{d+1,d+2,b,a|d+3}, M_{d+2,d+2,a,b|d+2}, M_{d+2,d+2,b,a|d+2}, M_{d+3,d+1,a,b|d+3}, M_{d+3,d+1,b,a|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 96.

Table 96: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+1,d+1,a,b d+1}$	$M_{d+2,d+1,a,b d+2}, M_{d+1,d+2,a,b d+2}, M_{d+2,d+1,b,a d+2}, M_{d+1,d+2,b,a d+2}$
$P_{d+1,d+1,b,a d+1}$	$M_{d+2,d+1,b,a d+2}, M_{d+1,d+2,b,a d+2}$
$P_{d+1,a,d+1,b d+1}$	$M_{d+2,a,d+1,b d+2}, M_{d+1,a,d+2,b d+2}, M_{d+2,d+1,a,b d+2}, M_{d+2,b,d+1,a d+2},$ $M_{d+2,d+1,b,a d+2}, M_{d+1,d+2,a,b d+2}, M_{d+1,b,d+2,a d+2}, M_{d+1,d+2,b,a d+2}$
$P_{d+1,b,d+1,a d+1}$	$M_{d+2,b,d+1,a d+2}, M_{d+1,b,d+2,a d+2}, M_{d+2,d+1,b,a d+2}, M_{d+1,d+2,b,a d+2}$
$P_{a,d+1,d+1,b d+1}$	$M_{a,d+2,d+1,b d+2}, M_{a,d+1,d+2,b d+2}, M_{d+2,a,d+1,b d+2}, M_{d+1,d+2,a,b d+2},$ $M_{b,d+2,d+1,a d+2}, M_{d+2,d+1,a,b d+2}, M_{d+2,b,d+1,a d+2}, M_{d+1,d+2,b,a d+2},$ $M_{d+2,d+1,b,a d+2}, M_{d+1,a,d+2,b d+2}, M_{b,d+1,d+2,a d+2}, M_{d+1,b,d+2,a d+2}$
$P_{b,d+1,d+1,a d+1}$	$M_{b,d+2,d+1,a d+2}, M_{b,d+1,d+2,a d+2}, M_{d+2,b,d+1,a d+2}, M_{d+1,d+2,b,a d+2},$ $M_{d+2,d+1,b,a d+2}, M_{d+1,b,d+2,a d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.1.4) \quad a < d, a = b$$

$$\text{Let } \mu := \lambda + (0, 0, 1, 1|0).$$

$$P_\mu = M_{d+1,d+2,a,a|d} + M_{a,d+1,d+2,a|d} + M_{a,d+2,a,d+1|d} + M_{d+2,a,d+1,a|d} + M_{a,a,d+2,d+1|d} + M_{d+2,d+1,a,a|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{d,d+1,a,a|d} + 2M_{d+1,d+1,a,a|d+1} + M_{d+1,d+2,a,a|d+2} + M_{a,d,d+1,a|d} + 2M_{a,d+1,d+1,a|d+1} \\ &= \sum M_{a,a,d+1,d|d} + \sum M_{a,a,d+1,d+1|d+1} + \sum M_{a,a,d+2,d+1|d+2} + \sum M_{a,d+1,d+1,a|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,a,a|d+1}, M_{d+1,a,d+1,a|d+1}, M_{d+1,d+1,a,a|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by table 97.

Table 97: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{d+1,d+1,a,a d+1}$	$M_{d+2,d+1,a,a d+2}, M_{d+1,d+2,a,a d+2}$
$P_{d+1,a,d+1,a d+1}$	$M_{d+2,a,d+1,a d+2}, M_{d+1,a,d+2,a d+2}, M_{d+2,d+1,a,a d+2}, M_{d+1,d+2,a,a d+2}$
$P_{a,d+1,d+1,a d+1}$	$M_{a,d+2,d+1,a d+2}, M_{a,d+1,d+2,a d+2}, M_{d+2,a,d+1,a d+2}, M_{d+1,d+2,a,a d+2},$ $M_{d+2,d+1,a,a d+2}, M_{d+1,a,d+2,a d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.1.5) \quad a > d + 2, b < d$$

$$\text{Let } \mu := \lambda + (0, 0, 1, 1|0).$$

$$P_\mu = M_{a,d+2,d+1,b|d} + M_{a,b,d+2,d+1|d} + M_{a,d+2,b,d+1|d} + M_{a,d+1,d+2,b|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{a,d+1,d,b|d} + 2M_{a,d+1,d+1,b|d+1} + M_{a,d+2,d+1,b|d+2} + M_{a,b,d+1,d|d} + M_{a,b,d+1,d+1|d+1} + \\ &= \sum M_{a,b,d+1,d|d} + \sum M_{a,b,d+1,d+1|d+1} + \sum M_{a,b,d+2,d+1|d+2} + M_{a,d+1,d+1,b|d+1} \end{aligned}$$

Table 98: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_6))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_6))$ (via Prop. ~ 1)
$P_{a,d+1,d+1,b d+1}$	$M_{a,d+2,d+1,b d+2}$	$M_{a,d+1,d+2,b d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

Similarly, the set of indeterminate composition factors is $U = \{M_{a,d+1,d+1,b|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_6) = P_\lambda$ by [table 98](#).

$$(4.1.6) \text{ Others not}(a < d), (a > d+2, b < d)$$

Let $\mu := \lambda + (0, 0, 0, 1|1) = (a, b, d+2, d+1|d)$.

$$Pr_\lambda(P_\mu \otimes V_6) = \sum M_{a,b,d+1,d|d} + \sum M_{a,b,d+1,d+1|d+1} + \sum M_{a,b,d+2,d+1|d+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

$$(4.2) \ a = d+2, b = d+2$$

Let $\mu := \lambda + (0, 0, 0, 0|3)$.

$$P_\mu = M_{d+2,d+2,d+1,d|d+3}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_9) &= M_{d+3,d+2,d+2,d+1|d+3} + M_{d+2,d+3,d+2,d+1|d+3} + M_{d+2,d+2,d+2,d+1|d+2} + M_{d+2,d+2,d+1,d+1|d+2} \\ &= M_{d+2,d+2,d+1,d|d} + M_{d+2,d+2,d+1,d+1|d+1} + M_{d+2,d+2,d+2,d+1|d+2} + \sum M_{d+2,d+3,d+2,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+2,d+2,d+1|d+3}, M_{d+2,d+3,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_9) = P_\lambda$ by [table 19](#).

Table 99: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_9))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_9))$ (via Prop. ~ 1)
$P_{d+3,d+2,d+2,d+1 d+3}$	$M_{d+4,d+2,d+2,d+1 d+4}$	
$P_{d+2,d+3,d+2,d+1 d+3}$	$M_{d+2,d+4,d+2,d+1 d+4}$	$M_{d+4,d+2,d+2,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.3) \ a \neq d+2, b = d+2$$

$$(4.3.1) \ a > d+3$$

Let $\mu := \lambda + (0, 0, 0, 0|3)$.

$$P_\mu = M_{a,d+2,d+1,d|d+3}$$

$$Pr_\lambda(P_\mu \otimes V_9) = M_{a,d+3,d+2,d+1|d+3} + M_{a,d+2,d+2,d+1|d+2} + M_{a,d+2,d+1,d+1|d+1} + M_{a,d+2,d+1,d|d}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{a,d+3,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_9) = P_\lambda$ by [table 100](#).

$$(4.3.2) \ a = d+3$$

Let $\mu := \lambda + (1, 1, 1, 1|0)$.

$$P_\mu = M_{d+4,d+3,d+2,d+1|d}$$

$$Pr_\lambda(P_\mu \otimes V_{14}) = M_{d+3,d+2,d+1,d|d} + M_{d+3,d+2,d+1,d+1|d+1} + M_{d+3,d+2,d+2,d+1|d+2} + M_{d+3,d+3,d+2,d+1|d+3}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+3,d+2,d+1|d+3}, M_{d+4,d+3,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{14}) = P_\lambda$ by [table 101](#).

Table 102: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj.	Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)				
$P_{d+2,d+3,a,d+1 d+3}$		$M_{d+2,d+4,a,d+1 d+4},$	$M_{d+4,d+2,a,d+1 d+4},$	$M_{d+2,d+4,d+1,a d+4},$	$M_{d+4,d+2,d+1,a d+4}$	
$P_{d+3,d+2,d+1,a d+3}$		$M_{d+4,d+2,d+1,a d+4}$				
$P_{d+1,d+3,d+2,a d+3}$		$M_{d+1,d+4,d+2,a d+4},$	$M_{d+4,d+1,d+2,a d+4},$	$M_{d+2,d+4,d+1,a d+4},$	$M_{d+4,d+2,d+1,a d+4}$	
$P_{d+3,a,d+2,d+1 d+3}$		$M_{d+4,a,d+2,d+1 d+4},$	$M_{d+4,d+2,a,d+1 d+4},$	$M_{d+4,d+1,d+2,a d+4},$	$M_{d+4,d+2,d+1,a d+4}$	
$P_{d+2,d+2,a,d+1 d+2}$		$\emptyset$				
$P_{d+3,d+2,a,d+1 d+3}$		$M_{d+4,d+2,a,d+1 d+4}, M_{d+4,d+2,d+1,a d+4}$				
$P_{d+1,d+2,d+1,a d+1}$		$M_{d+1,d+2,d+2,a d+2}, M_{d+2,d+1,d+2,a d+2}$				
$P_{d+2,d+2,d+1,a d+2}$		$\emptyset$				
$P_{d+2,d+3,d+1,a d+3}$		$M_{d+2,d+4,d+1,a d+4}, M_{d+4,d+2,d+1,a d+4}$				
$P_{d+2,d+1,d+1,a d+1}$		$M_{d+2,d+1,d+2,a d+2}$				
$P_{d+3,d+1,d+2,a d+3}$		$M_{d+4,d+1,d+2,a d+4}, M_{d+4,d+2,d+1,a d+4}$				
$P_{a,d+3,d+2,d+1 d+3}$		$M_{a,d+4,d+2,d+1 d+4}, M_{d+4,a,d+2,d+1 d+4}, M_{d+2,d+4,a,d+1 d+4}, M_{d+1,d+4,d+2,a d+4},$ $M_{d+4,d+2,a,d+1 d+4}, M_{d+4,d+1,d+2,a d+4}, M_{d+2,d+4,d+1,a d+4}, M_{d+4,d+2,d+1,a d+4}$				

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.4) \quad a = d + 2, b \neq d + 2$$

$$(4.4.1) \quad b > d + 3$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 1|0).$$

$$P_\mu = M_{b,d+3,d+2,d+1|d} + M_{d+3,b,d+2,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{b,d+2,d+1,d|d} + M_{b,d+2,d+1,d+1|d+1} + M_{b,d+2,d+2,d+1|d+2} + M_{b,d+3,d+2,d+1|d+3} + M_{d+3,b,d+2,d+1|d+3} \\ &= \sum M_{d+2,b,d+1,d|d} + \sum M_{d+2,b,d+1,d+1|d+1} + \sum M_{d+2,b,d+2,d+1|d+2} + \sum M_{d+3,b,d+2,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{b,d+3,d+2,d+1|d+3}, M_{d+3,b,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 103](#).

Table 103: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{b,d+3,d+2,d+1 d+3}$	$M_{b,d+4,d+2,d+1 d+4}$
$P_{d+3,b,d+2,d+1 d+3}$	$M_{d+4,b,d+2,d+1 d+4}, M_{b,d+4,d+2,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.4.2) \quad b = d + 3$$

$$\text{Let } \mu := \lambda + (1, 0, 1, 1|0).$$

$$P_\mu = M_{d+3,d+3,d+2,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d+2,d+3,d+1,d|d} + M_{d+3,d+2,d+1,d|d} + M_{d+2,d+3,d+1,d+1|d+1} + M_{d+3,d+2,d+1,d+1|d+1} + \\ &= \sum M_{d+2,d+3,d+1,d|d} + \sum M_{d+2,d+3,d+1,d+1|d+1} + \sum M_{d+2,d+3,d+2,d+1|d+2} + M_{d+3,d+2,d+1,d+1|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+3,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 104](#).

Table 104: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d+3,d+3,d+2,d+1 d+3}$	$M_{d+4,d+3,d+2,d+1 d+4}, M_{d+3,d+4,d+2,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

$$(4.4.3) \quad b < d$$

$$\text{Let } \mu := \lambda + (0, 0, 0, 0|0).$$

$$P_\mu = M_{d+3,d+2,d+1,b|d} + M_{d+3,b,d+2,d+1|d} + M_{d+3,d+2,b,d+1|d} + M_{d+3,d+1,d+2,b|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d+2,d+1,d,b|d} + 2M_{d+2,d+1,d+1,b|d+1} + M_{d+2,d+2,d+1,b|d+2} + M_{d+3,d+2,d+1,b|d+3} + M_{d+3,b,d+2,d+1|d+3} \\ &= \sum M_{d+2,b,d+1,d|d} + \sum M_{d+2,b,d+1,d+1|d+1} + \sum M_{d+2,b,d+2,d+1|d+2} + \sum M_{d+3,b,d+2,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+2,d+1,b|d+3}, M_{d+3,b,d+2,d+1|d+3}\}$. We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 105](#).

$$(5) \quad a = d + 1, b = d + 1, c \neq d + 1 \text{ Let } \mu := \lambda + (1, 1, 0, 1|0) = (b + 1, b + 2, c, d|d).$$

$$Pr_\lambda(P_\mu \otimes V_{10}) = \sum M_{d+1,d+1,c,d|d} + \sum M_{d+1,d+1,c,d+1|d+1} + \sum M_{d+1,d+2,c,d|d+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

Table 105: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d+3,d+2,d+1,b d+3}$	$M_{d+4,d+2,d+1,b d+4}$
$P_{d+3,b,d+2,d+1 d+3}$	$M_{d+4,b,d+2,d+1 d+4}, M_{d+4,d+2,b,d+1 d+4}, M_{d+4,d+1,d+2,b d+4}, M_{d+4,d+2,d+1,b d+4}$
$P_{d+3,d+2,b,d+1 d+3}$	$M_{d+4,d+2,b,d+1 d+4}, M_{d+4,d+2,d+1,b d+4}$
$P_{d+2,d+1,d+1,b d+1}$	$M_{d+2,d+2,d+1,b d+2}, M_{d+2,d+1,d+2,b d+2}$
$P_{d+3,d+1,d+2,b d+3}$	$M_{d+4,d+1,d+2,b d+4}, M_{d+4,d+2,d+1,b d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(6) $b = d + 1, c = d + 1, a \neq d + 1$

(6.1) $a > d + 2$

Let $\mu := \lambda + (0, 0, 0, 0|2) = (a, b + 2, d + 1, d|d + 2)$.

$$Pr_\lambda(P_\mu \otimes V_{10}) = M_{a,d+2,d+1,d+1|d+2} + M_{a,d+1,d+2,d+1|d+2} + M_{a,d+1,d+1,d+1|d+1} + M_{a,d+1,d+1,d|d} = M_{a,d+1,d+1,d+1|d+1}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(6.2) $a = d + 2$

Let $\mu := \lambda + (0, 0, 0, 0|3)$.

$$P_\mu = M_{d+2,d+1,d+1,d|d+3}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_9) &= M_{d+3,d+2,d+1,d+1|d+3} + M_{d+3,d+1,d+2,d+1|d+3} + M_{d+2,d+2,d+1,d+1|d+2} + M_{d+2,d+1,d+2,d+1|d+2} \\ &= M_{d+2,d+1,d+1,d|d} + M_{d+2,d+1,d+1,d+1|d+1} + \sum M_{d+2,d+1,d+2,d+1|d+2} + \sum M_{d+3,d+1,d+2,d+1|d+3} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+3,d+2,d+1,d+1|d+3}, M_{d+3,d+1,d+2,d+1|d+3}, M_{d+2,d+2,d+1,d+1|d+2}, M_{d+2,d+1,d+2,d+1|d+2}, M_{d+2,d+1,d+1,d|d}, M_{d+2,d+1,d+1,d+1|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_9) = P_\lambda$ by [table 106](#).

Table 106: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_9))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_9))$ (via Prop. ~ 1)
$P_{d+3,d+2,d+1,d+1 d+3}$	$M_{d+4,d+2,d+1,d+1 d+4}$
$P_{d+3,d+1,d+2,d+1 d+3}$	$M_{d+4,d+1,d+2,d+1 d+4}, M_{d+4,d+2,d+1,d+1 d+4}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(6.3) $a < d$

Let $\mu := \lambda + (1, 0, 0, 0|0)$. 若 $d = a - 1$, P_μ 结构可参考[Theorem 3.1](#), 进而其他情况可以依此归纳得到.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{d+2,a,d+1,d+1|d+2} + M_{d+2,d+1,a,d+1|d+2} + M_{d+2,d+1,d+1,a|d+2} + M_{a,d+2,d+1,d+1|d+2} + M_{d+1,d+1,d+1,d+1|d+1} \\ &= \sum M_{a,d+1,d+1,d|d} + \sum M_{a,d+1,d+1,d+1|d+1} + \sum M_{a,d+1,d+2,d+1|d+2} + M_{d+1,d+1,d+1,a|d+1} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,d+1,a|d+1}\}$. We have $Pr_\lambda(P_\mu \otimes V_2) = P_\lambda$ by [table 107](#).

Table 107: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_2))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_2))$ (via Prop. ~ 1)			
	$P_{d+1,d+1,d+1,a d+1}$	$M_{d+2,d+1,d+1,a d+2}$	$M_{d+1,d+2,d+1,a d+2}$	$M_{d+1,d+1,d+2,a d+2}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(7) $c = d + 1, a = d + 1, b \neq d + 1$

(7.1) $b > d + 2$

Let $\mu := \lambda + (0, 0, 0, 0|2) = (d + 1, b, d + 1, d|d + 2)$.

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{d+2,b,d+1,d+1|d+2} + M_{d+1,b,d+2,d+1|d+2} + M_{d+1,b,d+1,d+1|d+1} + M_{d+1,b,d+1,d|d} + M_{b,d+2,d+1,d+1|d+2} \\ &= \sum M_{d+1,b,d+1,d|d} + \sum M_{d+1,b,d+1,d+1|d+1} + \sum M_{d+1,b,d+2,d+1|d+2} \end{aligned}$$

All items belong to P_λ by [Proposition 2.0.1](#).

(7.2) $b = d + 2$

Let $\mu := \lambda + (0, 0, 0, 0|3)$.

$$P_\mu = M_{d+1,d+2,d+1,d|d+3} + M_{d+2,d+1,d+1,d|d+3}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_9) &= M_{d+2,d+3,d+1,d+1|d+3} + M_{d+1,d+3,d+2,d+1|d+3} + 2M_{d+2,d+2,d+1,d+1|d+2} + M_{d+1,d+2,d+2,d+1|d+2} \\ &= \sum M_{d+1,d+2,d+1,d|d} + \sum M_{d+1,d+2,d+1,d+1|d+1} + \sum M_{d+1,d+2,d+2,d+1|d+2} + \sum M_{d+1,d+2,d+2,d+1|d+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+2,d+3,d+1,d+1|d+3}, M_{d+1,d+3,d+2,d+1|d+3}, M_{d+2,d+2,d+1,d+1|d+2}, M_{d+1,d+2,d+2,d+1|d+2}, M_{d+1,d+2,d+1,d+1|d+1}, M_{d+1,d+2,d+1,d|d}\}$.

根据 [table 108](#), 仅仅由 [Proposition 2.0.1](#) 不足以判断 $P_{d+2,d+2,d+1,d+1|d+2}$ 是否出现在 $pr_\lambda(P_\lambda \otimes V_9)$ 的直和分解中. 因此由 [Theorem 3.1](#), 我们引入它们完整的滤过结构如下.

$$P_{d+2,d+2,d+1,d+1|d+2} = M_{d+2,d+2,d+1,d+1|d+2} + M_{d+2,d+3,d+1,d+1|d+3} + M_{d+3,d+2,d+1,d+1|d+3} = M_{d+2,d+2,d+1,d+1|d+2} + M_{d+2,d+3,d+1,d+1|d+3} + M_{d+3,d+2,d+1,d+1|d+3} \quad (24)$$

可以看出 $\text{Filt}(P_{d+2,d+2,d+1,d+1|d+2})$ 所有的 Verma 模都在多重集 U 中. 且其重数未超过 U 中对应项的重数. 因此因此仅通过滤过比较仍无法排除它们中某一个作为 $pr_\lambda(P_\mu \otimes V_9)$ 直和项的可能性. 我们反证法排除这个可能性.

首先, 假设 $P_{d+2,d+2,d+1,d+1|d+2}$ 是 $pr_\lambda(P_\lambda \otimes V_9)$ 的一个直和项, 那么, 存在一个投射模 Q 使得

$$pr_\lambda(P_\lambda \otimes V_{10}) = Q + P_{d+2,d+2,d+1,d+1|d+2}$$

令 $\nu = (d + 1, d + 4, d + 2, d + 1|d + 3)$, 那么 ν 是典型的权. 我们就有

$$pr_\nu(Q \otimes V_1) = M_{d+1,d+4,d+2,d+1|d+3} + M_{d+4,d+1,d+2,d+1|d+3} \quad (25)$$

观察到 [equation \(25\)](#) 中, 权 ν 在滤过中是极小的, 且重数为 1. Q 是投射模, 因此 $pr_\nu(Q \otimes V_1)$ 也是投射模. 根据 [Lemma 2.0.1](#), 这表明 $pr_\nu(Q \otimes V_1)$ 必然有个 P_ν 作为直和项. 但是 [Theorem 2.1](#) 给出 P_ν 的完整滤过的结构如下

$$\begin{aligned} P_\nu &= M_{d+1,d+4,d+2,d+1|d+3} + M_{d+4,d+1,d+2,d+1|d+3} + M_{d+2,d+4,d+1,d+1|d+3} + M_{d+4,d+2,d+1,d+1|d+3} \\ &= \sum M_{d+1,d+4,d+2,d+1|d+3} \end{aligned} \quad (26)$$

比较 [equation \(25\)](#) 和 [equation \(26\)](#), 我们发现 $pr_\nu(Q \otimes V_1)$ 不可能有个 P_ν 作为直和项. 矛盾, 因此排除了 $P_{d+2,d+2,d+1,d+1|d+2}$ 作为 $pr_\lambda(P_\lambda \otimes V_9)$ 直和项的可能.

综上 $Pr_\lambda(P_\mu \otimes V_9) = P_\lambda$.

Table 108: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_9))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_9))$ (via Prop. ~ 1)
$P_{d+2,d+3,d+1,d+1 d+3}$	$M_{d+2,d+4,d+1,d+1 d+4}, M_{d+4,d+2,d+1,d+1 d+4}$
$P_{d+1,d+3,d+2,d+1 d+3}$	$M_{d+1,d+4,d+2,d+1 d+4}, M_{d+4,d+1,d+2,d+1 d+4}, M_{d+2,d+4,d+1,d+1 d+4}, M_{d+4,d+2,d+1,d+1 d+4}$
$P_{d+3,d+2,d+1,d+1 d+3}$	$M_{d+4,d+2,d+1,d+1 d+4}$
$P_{d+3,d+1,d+2,d+1 d+3}$	$M_{d+4,d+1,d+2,d+1 d+4}, M_{d+4,d+2,d+1,d+1 d+4}$
$P_{d+2,d+2,d+1,d+1 d+2}$	$\emptyset$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(7.3) $b < d$

Let $\mu := \lambda + (1, 0, 1, 1|0)$.

$$P_\mu = M_{d+2,d+2,b,d+1|d} + M_{d+2,d+1,d+2,b|d} + M_{d+2,d+2,d+1,b|d} + M_{d+2,b,d+2,d+1|d}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_{10}) &= M_{d+1,d+1,b,d|d} + M_{d+1,d+1,b,d+1|d+1} + M_{d+1,d+2,b,d+1|d+2} + M_{d+2,d+1,b,d+1|d+2} + M_{d+1,d,d+1|d+1} \\ &= \sum M_{d+1,b,d+1,d|d} + \sum M_{d+1,b,d+1,d+1|d+1} + M_{d+1,d+1,d+1,b|d+1} + \sum M_{d+1,b,d+2,d+1|d+2} \end{aligned}$$

Similarly, the set of indeterminate composition factors is $U = \{M_{d+1,d+1,d+1,b|d+1}, M_{d+2,d+1,d+1,b|d+2}\}$.

We have $Pr_\lambda(P_\mu \otimes V_{10}) = P_\lambda$ by [table 109](#).

Table 109: Calculation of Verma modules in $\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\lambda \otimes V_{10}))$

Proj. Cover	$\text{Filt}(P_*) \setminus \text{Filt}(pr_\lambda(P_\mu \otimes V_{10}))$ (via Prop. ~ 1)
$P_{d+1,d+1,d+1,b d+1}$	$M_{d+1,d+2,d+1,b d+2}, M_{d+1,d+1,d+2,b d+2}$
$P_{d+2,d+1,d+1,b d+2}$	$M_{d+3,d+1,d+1,b d+3}$

Interpretation: This table displays only selected computational results. This partial presentation does not affect our conclusion.

(8) $c = d + 1, a = d + 1, b = d + 1$ Let $\mu := \lambda + (0, 0, 0, 0|2) = (d + 1, d + 1, d + 1, d|d + 2)$.

$$Pr_\lambda(P_\mu \otimes V_5) = M_{d+1,d+1,d+1,d|d} + M_{d+1,d+1,d+1,d+1|d+1} \sum M_{d+1,d+1,d+2,d+1|d+2}$$

All items belong to P_λ by [Proposition 2.0.1](#).

□

References

- [1] Maria Gorelik. Annihilation theorem and separation theorem for basic classical lie superalgebras. *Journal of the American Mathematical Society*, 15(1):113–165, 2002.